

ECMA 30760

Introduction to Economic Design

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The big question for this course is how to design economic institutions. These include:

- Voting,
- Auctions,
- Matching, and
- Resource allocation.

Our role varies:

- Sometimes we will be a benevolent planner (from outside the market).
- Other times, we will try to extract surplus.

Challenges include:

- People are strategic (and reoptimize choices when the rules change).¹
- Information and power are decentralized.

We will formalize assumptions into mathematical axioms and use basic techniques like calculus, logic, linear programming, and linear algebra to answer our questions.

I refer to agents 1 and i with female pronouns and agents 2 and j with male pronouns.

¹Refer to the Boston school choice mechanism.

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1.1 Introduction

Notation 1.1.1.

Let A be a finite set of outcomes and let N be a finite set of agents (also referred to as voters).

We take each agent to have strict preferences $\succ_i$ over A . $x \succ_i y$ if agent i strictly prefers outcome x to outcome y . We write $x \succeq_i y$ if either $x \succ_i y$ or $x = y$.

Formally, $\succ_i$ is a *binary relation*; i.e., it is a subset of $A \times A$ with the *infix* notation of $x \succ_i y$ instead of $(x, y) \in \succ_i$.

Remark 1.1.2. We have the following properties of $\succ_i$:

- Irreflexive: For every x , it is not the case that $x \succ_i x$.
- Complete: For every x, y , if $x \neq y$, then either $x \succ_i y$ or $y \succ_i x$.
- Transitive: If $x \succ_i y$ and $y \succ_i z$, then $x \succ_i z$.
- Asymmetric: For every $x \neq y$, if $x \succ_i y$, then we do not have $y \succ_i x$.

A binary relation satisfying these properties is called a *strict total order* or *strict linear order*. 

Let P be the set of strict orders on A and let P^N be the set of *preference profiles*:

$$P^N = \{(\succ_1, \dots, \succ_{|N|}) : \succ_i \in P, \forall i\}.$$

Definition 1.1.3 (Social choice function).

A social choice function is a mapping $f : P^N \rightarrow A$.

Informally, a social choice function is a decision rule for each preference profile.

We explore the following social choice functions, which are also referred to as voting rules:

Definition 1.1.4 (Common Voting Rules).

- Plurality: Choose whichever outcome is top-ranked by the most people. (Along with some tiebreaking rule.)
- Instant runoff: All outcomes but the two top-ranked outcomes^a are dropped, then

the winner is decided by plurality.

- **Elimination:** The outcome that is top-ranked by the fewest people is removed and the process repeats.
- **Borda count:** Let $n = |A|$. Give n points to a candidate for each agent that top-ranks them, $n - 1$ points for each agent that second-ranks them, and so on. The outcome with the most points wins.
- **Summation social choice functions:** Assign points $\phi_i(m)$ for agent i 's m th ranked outcome, and the outcome with the most points wins.
- **Pairwise comparison graph:** We arrange the outcomes into a directed graph with all possible edges labeled by the number of people who prefer the tail node to the head node. A special case is the Copeland social choice function, which counts for each outcome how many arrows point out, and the winner is the node with the highest count. See Figure (1.1).

^aThe two outcomes that have the highest number of top-rank votes.

Example 1.1.5.

For instance, if we have

$$A = \{w, x, y, z\}$$

with the preference profile of:

- 6 people with $w \succ_i x \succ_i y \succ_i z$
- 5 people with $x \succ_i y \succ_i w \succ_i z$
- 4 people with $y \succ_i x \succ_i w \succ_i z$
- 3 people with $z \succ_i y \succ_i x \succ_i w$,

then the plurality rule chooses w , the instant runoff chooses x , elimination chooses y , the Borda count chooses x , and the Copeland rule chooses x . 

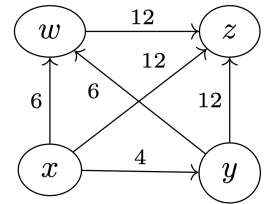
The first three voting rules are common, while the Borda count and summation social choice functions are mostly seen in sports.

Definition 1.1.6 (Condorcet winner, Condorcet property).

An outcome x is a Condorcet winner if it beats all other outcomes in a head-to-head election. A social choice function satisfies the Condorcet property if it picks a Condorcet winner, when one exists.

In the above profile, x is a Condorcet winner. We may want to always select a Condorcet winner, but it may not exist.

Figure 1.1. Copeland voting rule



Example 1.1.7.

Consider

$$\begin{aligned} x &\succ_1 y \succ_1 z \\ y &\succ_2 z \succ_2 x \\ z &\succ_3 x \succ_3 y. \end{aligned}$$

Then in head-to-heads, we have $x \leftarrow y$, $y \leftarrow z$, but $x \rightarrow z$. Thus there is no Condorcet winner. 

Definition 1.1.8 (Dominating set of outcomes, Smith set, Smith property).

A set of outcomes $A' \subseteq A$ is dominating if it is nonempty and for all $x \in A'$, $y \in A \setminus A'$, x beats y in a head-to-head election.

The Smith set is the smallest dominating set. A social choice function f satisfies the Smith property if it always chooses something in the Smith set.

A dominating set always exists because we can take $A' = A$. Furthermore, dominating sets must be “nested.”

Lemma 1.1.9.

If X and Y are dominating sets, then $X \subseteq Y$ or $Y \subseteq X$.

Proof of Lemma 1.1.9. Suppose not. Then any $x \in X \setminus Y$ and $y \in Y \setminus X$ have to both beat each other in head-to-head elections, which cannot happen under strict preferences. 

This lemma tells us that the Smith set is the intersection of all dominating sets. If a Condorcet winner exists, then it is in the Smith set.

1.2 Gibbard–Satterthwaite**Definition 1.2.1 (Strategy-proof).**

A social choice function is strategy-proof if there is no preference profile $(\succ_1, \dots, \succ_{|N|})$, agent i , and preference $\succ'_i$ such that

$$f(\succ_1, \dots, \succ_{i-1}, \succ'_i, \succ_{i+1}, \dots, \succ_{|N|}) \succ_i f(\succ_1, \dots, \succ_{i-1}, \succ_i, \succ_{i+1}, \dots, \succ_{|N|}).$$

A social choice function is not strategy-proof if there is some preference profile and agent where the agent would prefer to misrepresent her preferences. Strategy-proofness only restricts unilateral deviations and does not consider the effects of collusion.

We need some definitions for the next theorem.

Definition 1.2.2 (Lower contour set).

Given a preference $\succ_i$ on A and $a \in A$, define the lower contour set as $L_{\succ_i}(a) = \{b : a \succ_i b\}$, which is the set of outcomes dispreferred to a .

Definition 1.2.3 (Monotonic social choice function).

f is monotonic if for any preference profiles $\succ, \succ'$ such that for every i , $L_{\succ'_i}(f(\succ)) \supseteq L_{\succ_i}(f(\succ))$, we have $f(\succ') = f(\succ)$.

Under a monotonic social choice function, changing preferences such that the lower contour set of the outcome doesn't shrink preserves the original outcome. In particular, if the lower contour set of the outcome grows, then the original outcome is still chosen. If we shuffle around alternatives ranked below the outcome, the outcome is preserved.

Definition 1.2.4 (Pareto efficient social choice function).

f is Pareto efficient if for all preference profiles $\succ$, $a \succ_i b$ for all i implies that $f(\succ) \neq b$.

A Pareto efficient social choice function does not pick outcomes that all agents disprefer to another (single) outcome.

Theorem 1.2.5 (Gibbard–Satterthwaite).

If $|A| > 2$, then every surjective and strategy-proof social choice function is dictatorial, in that there is some agent i such that

$$f(\succ) = \arg \max_{\succ_i} A$$

for all $\succ$. (Here, the $\arg \max$ over $\succ_i$ means that the ordering of the right hand side is determined by $\succ_i$.)

Importantly, i does *not* depend on $\succ$. The dictatorial property says that i 's top choice is chosen for all preference profiles.

Proof of Theorem 1.2.5. We first describe the proof structure:

- (i) Deduce that surjectivity and strategy-proofness imply monotonicity and Pareto efficiency.
- (ii) Assume that $|N| = 2$, then show monotonicity and Pareto efficiency imply the social choice function is dictatorial.
- (iii) Induct.

Now we prove the theorem. Suppose f is strategy-proof and surjective.

Step 1: Monotonicity and Pareto efficiency. Let $\succ, \succ'$ be as in the definition of monotonicity, i.e.

$$L_{\succ'_i}(f(\succ)) \supseteq L_{\succ_i}(f(\succ)).$$

To show monotonicity, we want to show that $f(\succ) = f(\succ')$. By strategy-proofness,

$$f(\succ_1, \succ_2, \dots, \succ_{|N|}) \succsim_1 f(\succ'_1, \succ_2, \dots, \succ_{|N|}).$$

Can it be the case that

$$f(\succ) \succ_1 f(\succ'_1, \succ_{-1})$$

strictly? Suppose this held. Using our assumption of $L_{\succ'_i}(f(\succ)) \supseteq L_{\succ_i}(f(\succ))$, since $f(\succ'_1, \succ_{-1})$ is still below $f(\succ)$ when switching from $\succ_1$ to $\succ'_1$, we get

$$f(\succ) \succ'_1 f(\succ'_1, \succ_{-1}).$$

Thus, under $\succ'_1$, agent 1 strictly prefers $f(\succ)$ to $f(\succ'_1, \succ_{-1})$, which violates strategy-proofness of f . Therefore

$$f(\succ) = f(\succ'_1, \succ_{-1}).$$

We can iterate to arrive that $f(\succ) = f(\succ')$, so f is monotonic.

We claim that monotonicity and surjectivity imply Pareto efficiency. Suppose not, so for some $\succ$, we have that for every i , $a \succ_i b$ but $f(\succ) = b$. Construct a new profile $\succ'$ from $\succ$ by moving a to the top of everyone's list. Then by the lower contour sets of b staying the same and monotonicity, $f(\succ') = b$.

At the same time, by surjectivity, there exists a profile $\succ^*$ such that $f(\succ^*) = a$. Modify $\succ^*$ to $\succ'^*$ by moving a to the top of all agents' list. By the same argument, $f(\succ'^*) = a$ as well. Comparing $\succ'$ to $\succ'^*$, we have that a is at the top of everyone's orderings in both preference profiles, but $f(\succ') = b \neq a = f(\succ'^*)$, which violates f 's monotonicity,¹ a contradiction. Therefore, f must be Pareto efficient.

Step 2: 2-agent case. Suppose

$$a \succ_1 b \succ_1 \dots, \quad b \succ_2 a \succ_2 \dots. \quad (1.1)$$

Suppose further that f is strategy-proof and surjective, so it is monotonic and Pareto efficient.

We know that $f(\succ_1, \succ_2) \in \{a, b\}$ by Pareto efficiency, as $a \succ_i c$ for $i = 1, 2$ and any $c \neq a, b$. Assume without loss that $f(\succ_1, \succ_2) = a$. Consider a new profile $\succ'_2$ such that $b \succ'_2 \dots \succ'_2 a$. We see that $f(\succ_1, \succ'_2) = b$ cannot hold, as by monotonicity, the original profile would end with b , as the lower contour set of b does not shrink from $\succ_2$ to $\succ'_2$.² (This would also violate strategy-proofness.) By Pareto efficiency, $f(\succ_1, \succ'_2)$ is also not any $c \neq a, b$. Therefore, $f(\succ_1, \succ'_2) = a$. By monotonicity, $f(\succ''_1, \succ''_2) = a$ whenever $a = \arg \max_{\succ''_1} A$, as the lower contour sets of a stay the same from $\succ_1$ to $\succ''_1$ and grow from $\succ'_2$ to $\succ''_2$. Thus, it makes sense to say that agent 1 is the dictator over a .

Redefine the original preferences, fix $c \neq a, b$ ³ and consider

$$c \succ_1 b \succ_1 \dots, \quad b \succ_2 c \succ_2 \dots.$$

Is $f(\succ_1, \succ_2) = b$ possible? No, because by the same logic, 2 would be the dictator over b , which breaks down when 1 top-ranks a and 2 top-ranks b . This implies that $f(\succ_1, \succ_2) = c$, so 1 is the dictator over c . Since c was arbitrary other than a, b , we know that 1 is the dictator over all outcomes $A \setminus \{b\}$.

Finally, consider another new profile

$$b \succ_1 a \succ_1 \dots, \quad a \succ_2 b \succ_2 \dots.$$

We know from Pareto efficiency that $f(\succ_1, \succ_2) \in \{a, b\}$. Suppose $f(\succ_1, \succ_2) = a$, so 2 is a dictator over a .⁴ By our assumption that $|A| > 2$ and our previous work, 1 is the dictator over some $c \neq a, b$. However, because 2 could top-rank a while 1 top-ranks c , this leads to a contradiction. Therefore $f(\succ_1, \succ_2) = b$ and 1 is the dictator at b , so 1 is the dictator over everything.

After deciding that agent 1 wins in the tie of Equation (1.1), we:

- (i) Show that 1 is the dictator over a .
- (ii) Show that 1 is the dictator over all $c \neq a, b$.
- (iii) Show that 1 is the dictator over b .

To show that 1 is the dictator over a , we create a strategy profile where 2 top-ranks b , bottom-ranks a , show that the SCF still picks a , and use monotonicity to upgrade to all other strategy profiles where 1 top-ranks a .

Given an outcome $a \in A$, we say that agent i is the dictator over a if $f(\succ) = a$ whenever $a = \arg \max_{\succ_i} A$. Two agents can be dictators over an outcome at the same time, but it is impossible for two agents to be dictators over different outcomes, as both of them top-ranking their respective dictatorial outcomes leads to a contradiction.

¹Monotonicity is violated because the lower contour sets of $a = f(\succ'^*)$ are the same, yet $f(\succ') = b \neq a$.

²We can shuffle the outcomes below b in $\succ'_2$ to arrive at $\succ_2$, which monotonicity says should lead to $\succ_2$ having the same outcome as $\succ'_2$, which is b .

³We know that there exists $c \neq a, b$ because $|A| > 2$.

⁴This does not contradict 1 being a dictator over a . If we have $A = \{a, b\}$, and b is chosen only when both agents rank it first, then both are dictators over a . This is why we need $|A| > 2$.

Whoever we side with in Equation (1.1) is the dictator.

Step 3: Induct. Assume that the Gibbard–Satterthwaite theorem holds with $|N| = n$ agents. Let f be a strategy-proof and surjective mechanism for $n + 1$ agents.

We want to reduce the domain of f . Define

$$g(\succ_1, \succ_2) = f(\succ_1, \succ_2, \dots, \succ_2).$$

We first want to show that g is dictatorial. g is surjective because f is Pareto efficient (for every outcome, we can have everyone top-rank that outcome). To show that g is strategy-proof, consider $\succ_2$ and $\succ'_2$, and fix $\succ_1$. Then by strategy-proofness of f ,

$$\begin{aligned} g(\succ_1, \succ_2) &= f(\succ_1, \succ_2, \dots, \succ_2) \\ &\succsim_2 f(\succ_1, \succ'_2, \succ_2, \dots, \succ_2) \\ &\succsim_2 f(\succ_1, \succ'_2, \succ'_2, \succ_2, \dots, \succ_2) \\ &\vdots \\ &\succsim_2 f(\succ_1, \succ'_2, \dots, \succ'_2) = g(\succ_1, \succ'_2). \end{aligned}$$

So $g(\succ_1, \succ_2) \succsim_2 g(\succ_1, \succ'_2)$ and g is strategy-proof for agent 2.

Strategy-proofness of f also (more directly) gives strategy-proofness of g for agent 1. By the 2-agent case, g is dictatorial with dictator 1.

If 1 is the dictator, we have that f is dictatorial after applying monotonicity.⁵ Now consider 2 being the dictator. Fix $\succ_1^*$ and define the n -agent social choice function

$$h(\succ_2, \dots, \succ_{n+1}) = f(\succ_1^*, \succ_2, \dots, \succ_{n+1}).$$

h is surjective because for any $a \in A$, agents $2, \dots, |N|$ can all report a as their top choices and recover g , which 2 is the dictator of. h is strategy-proof by f 's strategy-proofness. By the induction assumption, for this particular profile $\succ_1^*$, there exists a dictator K of h .

Fix $\succ_2^*, \dots, \succ_{K-1}^*, \succ_{K+1}^*, \dots, \succ_{n+1}^*$ and define

$$q(\succ_1, \succ_K) = f(\succ_1, \succ_2^*, \dots, \succ_{K-1}^*, \succ_K, \succ_{K+1}^*, \dots, \succ_{n+1}^*).$$

q inherits strategy-proofness from f .

If $\succ_1 = \succ_1^*$, then we know from h that K is the dictator, so q is surjective. By the 2-agent case, q is dictatorial; thus, since K is the dictator over everything when agent 1 reports $\succ_1^*$, K must be the dictator over everything no matter what agent 1 reports. Since $\succ_2^*, \dots, \succ_{K-1}^*, \succ_{K+1}^*, \dots, \succ_{n+1}^*$ were arbitrary and K is the dictator of f no matter what 1 reports, K must be the dictator of f in all cases.

Through g , we showed that if 1 is the dictator of g , then 1 is the dictator of f . If 2 is the dictator of g , then there exists agent K who is the dictator of f , where K does not depend on the $\succ_1^*$ we fixed. 

Remark 1.2.6. There are some ways around Theorem (1.2.5):

- (i) We may not care if we have to restrict the image of the social choice function to cardinality 2.
- (ii) Strategy-proofness is a strong restriction. Once we fix a social choice function, a game is induced. Strategy-proofness is a dominant strategy equilibrium in the game.
 - For each agent i , we have a type space Θ_i .

⁵Exercise: Show the details.

- For each type $\theta_i \in \Theta_i$, we have a utility function $u_i(\theta_i, a)$, which is the utility for agent i of type θ_i if a is chosen. We also assume that there is a common prior (belief) over the distribution of types people may have: $\mu \in \Delta\left(\prod_{i=1}^{|N|} \Theta_i\right)$

Everything, including the existence of dominant strategies, depends on μ .

- (iii) Some may say that P is too large or that there is more information in particular contexts. For example, we may expect single-peaked preferences, under which we show there exist strategy-proof social choice functions that are not dictatorial (Problem 3 of the notes). Furthermore, we may want to expand P to allow for indifference between outcomes. In a resource allocation problem, agents only care about their own allocations.

Using information in cardinal utilities does not allow for nondictatorial social choice functions, as the restriction of strategy-proofness becomes stronger.

- (iv) We could allow $f : P^N \rightarrow \Delta(A)$. This is explored in Problem 4 of the notes, which shows that there is always a random Condorcet winner. However, this does not get us out of the theorem. Gibbard (1977) showed that any strategy-proof mechanism $f : P^N \rightarrow \Delta(A)$ is a randomization over mechanisms that are either dictatorial or “duple” (restricted to two outcomes).

Criticism (ii) is the strongest.



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How should a benevolent social planner decide the outcome if people report their individual preferences to her?

2.1 Ordinal Aggregation

Notation 2.1.1.

Let R denote the set of weak preferences on A ; i.e., the set of orderings, allowing ties.

R is the set of complete, transitive, and reflexive binary relations $\succsim$ on A .

Definition 2.1.2 (Strict, weak social welfare function).

A strict social welfare function is a function $F : P^N \rightarrow P$. A weak social welfare function is a function $F : P^N \rightarrow R$.

Note that even weak social welfare functions are defined on the set of *strict* preference profiles.

Definition 2.1.3 (Kendall–Tau distance).

For any two strict rankings $\succsim_i, \succsim'_i$ on A , define the distance $d(\succsim_i, \succsim'_i)$ between the two rankings as the number of pairs $a, b \in A$ such that $a \prec_i b$ and $a \succ'_i b$ or $a \succ_i b$ and $a \prec'_i b$.

Note that the outcome of social welfare functions is a preference itself. Examples include:

- (i) Dictatorship: $F(\succsim_1, \dots, \succsim_{|N|}) = \succsim_1$ for every $(\succsim_1, \dots, \succsim_{|N|})$.
- (ii) Elimination: rank in increasing order of round left.
- (iii) Borda count.
- (iv) Copeland method.
- (v) Kemeny–Young: Choose $F(\succsim)$ to minimize

$$\sum_{i=1}^{|N|} d(F(\succsim), \succsim_i).$$

- (vi) Dodgson: For any outcome a and preference profile $\succsim$, let $\succsim^a$ be the profile such that

- (i) a is a Condorcet winner,
- (ii) $\sum_{i=1}^{|N|} d(\succ_i, \succ_i^a)$ is minimized.

Then define $z(a) = \sum_{i=1}^{|N|} d(\succ_i, \succ_i^a)$. Rank a above b iff $z(a) \leq z(b)$.

The elimination SWF is weak, as in a given round, the same number of people could rank the bottom two outcomes as their top choices. The Borda count is also weak.

Definition 2.1.4 (Paretian property).

A social welfare function F satisfies the Paretian property if for any $\succ$ and $a, b \in A$ is such that if for all i , $a \succ_i b$, then

$$aF(\succ)b.$$

Definition 2.1.5 (Independence of irrelevant alternatives).

A social welfare function F satisfies independence of irrelevant alternatives (IIA) if for any two profiles $\succ, \succ'$ such that for all i , $\succ_i$ and $\succ'_i$ agree when restricted to $\{a, b\}$ for $a, b \in A$, then

$$aF(\succ)b \iff aF(\succ')b.$$

Example 2.1.6 (IIA assumption).

Consider $A = \{a, b, c\}$ where $a \succ_1 b \succ_1 c$ and $b \succ_2 c \succ_2 a$. Without IIA, we might want to rank $b \succ_S a \succ_S c$, which seems sensible. Why would we want IIA? It is because in the preference $\succ_S$, we inferred cardinal rankings from ordinal rankings. To see why this is problematic, consider adding outcomes $d_1, \dots, d_k$, where d_ℓ means that outcome a is selected and you lose ℓ cents. Then

$$\begin{aligned} a \succ'_1 d_1 \succ'_1 \dots \succ'_1 b \succ'_1 c \\ b \succ'_2 c \succ'_2 a \succ'_2 d_1 \succ'_2 \dots \succ'_2 d_k, \end{aligned}$$

in which case IIA prohibits us from picking a . 

The Borda count infers cardinal information when it shouldn't (and does not satisfy IIA); this is why in the Kendall–Tau distance, we just use the number of outcomes that the two preferences disagree on, and not anything relating to the ordering.

Theorem 2.1.7 (Arrow's Impossibility).

If $|A| > 2$, then the only (weak or strict) social welfare functions that satisfy IIA and the Paretian property are dictatorial.

2.2 Cardinal Aggregation

Assume that each agent i has a utility function $u_i : A \rightarrow \mathbb{R}^+$, where we have a class $\mathcal{U}$ so $u_i \in \mathcal{U}$. We want to provide axioms on a function F mapping $\mathcal{U}^N \rightarrow R$ or $\mathcal{U}^N \rightarrow U$. For each $a \in A$, we can stack the utilities into a vector $(u_1(a), u_2(a), \dots, u_{|N|}(a))'$. Instead of defining axioms on F , we could provide axioms on a binary relation $\succ_s$ on $\mathbb{R}_+^{|N|}$. This would allow us to compare outcomes $a, b \in A$ by comparing their vectors $(u_1(a), \dots, u_{|N|}(a))$ and

$(u_1(b), \dots, u_{|N|}(b))$. We could use this to form a function $F : \mathcal{U}^{|N|} \rightarrow R$, where

$$a \succsim b \iff (u_1(a), \dots, u_{|N|}(a)) \succsim (u_1(b), \dots, u_{|N|}(b)).$$

Definition 2.2.1 (Cardinal social welfare ranking).

A cardinal social welfare ranking is a complete, transitive and reflexive binary relation $\succsim_s$ on $\mathbb{R}_+^{|N|}$.

Example 2.2.2 (Cardinal social welfare rankings).

We have some examples:^a

- (i) Utilitarian: $u \succsim_s u'$ iff $\sum_i u_i \geq \sum_i u'_i$
- (ii) Rawlsian: $u \succsim_s u'$ iff $\min_i u_i \geq \min_i u'_i$
- (iii) Leximin: $u \succ_s v \iff u_k^* \succ v_k^*$, where u^*, v^* are u, v rearranged in increasing order, and k is the first index where u, v differ.

^aThe leximin ranking is discussed further later in this section.



Now we define properties we want cardinal rankings to satisfy.

Definition 2.2.3 (Continuous cardinal social welfare ranking).

We say that $\succsim_s$ is continuous if for any $u \in \mathbb{R}_+^{|N|}$, the sets

$$\{u' : u' \succsim_s u\}, \quad \{u' : u \succsim_s u'\}$$

are closed. In other words, if $\{u^n\} \rightarrow u^*$ is such that $u^n \succsim_s u'$ for each $n \in \mathbb{N}$, then $u^* \succsim_s u'$. Similarly, if $u' \succsim_s u^n$ for each $n \in \mathbb{N}$, then $u' \succsim_s u^*$.

Definition 2.2.4 (Independence of unconcerned agents).

$\succsim_s$ is independent of unconcerned agents if for any $x, y \in \mathbb{R}_+$ and any agent i , we have

$$(x, u_{-i}) \succsim_s (x, u'_{-i}) \iff (y, u_{-i}) \succsim_s (y, u'_{-i})$$

for any u_{-i}, u'_{-i} .

We can represent a well-behaved ranking by a function that is additively separable across agents.

Theorem 2.2.5 (Debreu).

For any $\succsim_s$ satisfying continuity and independence of unconcerned agents, there exist functions $\{g_i\}_{i \in N}$ such that

$$u \succsim_s u' \iff \sum_i g_i(u_i) \geq \sum_i g_i(u'_i).$$

We have the following analogue of the Paretian property.

Definition 2.2.6 (Strict monotonicity).

We say that $\succsim_s$ is strictly monotonic if whenever $u_i \geq u'_i$ for all i and there is some j with $u_j > u'_j$, then $u \succ_s u'$.

Definition 2.2.7 (Scale invariance).

We say $\succsim_s$ is scale invariant if

$$u \succsim_s u' \iff \lambda u \succsim_s \lambda u', \quad \forall \lambda \in \mathbb{R}_{>0}.$$

Theorem 2.2.8.

If $\succsim_s$ satisfies continuity, independence of unconcerned agents, strict monotonicity, and scale invariance, and if $|N| \geq 2$, then there are positive weights $w_1, \dots, w_{|N|}$ which sum to 1 and some $\gamma \in \mathbb{R}$ such that

$$u \succsim_s v \iff \sum_i w_i \frac{u_i^{1-\gamma} - 1}{1-\gamma} \geq \sum_i w_i \frac{v_i^{1-\gamma} - 1}{1-\gamma}.$$

We now explore properties of this CRRA transformation of the utilities. We see that when $\gamma = 0$, we have

$$u \succsim_s v \iff \sum_i w_i u_i \geq \sum_i w_i v_i,$$

which is a weighted utilitarian preference.

Furthermore,

$$\lim_{\gamma \uparrow 1} \frac{u^{1-\gamma} - 1}{1-\gamma} = \ln(u),$$

so for γ close to 1,

$$u \succsim_s v \iff \sum_i w_i \ln(u_i) \geq \sum_i w_i \ln(v_i) \iff \prod_i u_i^{w_i} \geq \prod_i v_i^{w_i},$$

which is called *Nash welfare*.

As $\gamma \rightarrow \infty$, we get the leximin¹ order, in that

$$u \succ_s v \iff u_k^* \succ v_k^*,$$

where u^*, v^* are u, v rearranged in increasing order, and k is the first index where u, v differ. The leximin ordering is a refinement (in the English, not mathematical sense) of the Rawlsian ordering.

Note that Theorem (2.2.8) does not say that we can pick γ ; instead, after fixing some ranking $\succsim_s$ of the planner, then we know there is *some* γ that represents the planner's rankings in this additively separable, CRRA way.

¹A portmanteau of lexicographic and min.

2.3 Aggregation Under Uncertainty

Suppose there is a finite number of states $\Theta = \{\theta_1, \dots, \theta_m\}$. Each agent i has utility $u_i : \Theta \rightarrow \mathbb{R}$ and evaluates a lottery p over Θ by the Von Neumann–Morgenstern expected utility function

$$U_i(p) = \sum_{k=1}^m p_k \underbrace{u_i(\theta_k)}_{\text{Bernoulli index}}.$$

(We use U to denote utility over $\Delta(\Theta)$, including degenerate lotteries, and u to denote utility over Θ .)

There are good normative reasons for people to be VNM maximizers, but in an experimental setting, people violate this utility structure. We assume that the planner is also VNM rational with utility

$$U_s(p) = \sum_{k=1}^m p_k u_s(\theta_k).$$

Definition 2.3.1 (Paretian property).

The planner's utility function $U_s(\cdot)$ satisfies the Paretian property if $U_i(p) \geq U_i(q)$ for all i implies $U_s(p) \geq U_s(q)$.

Theorem 2.3.2 (Harsanyi).

The planner's utility U_s is Paretian iff there is a constant β and weights $(w_i)_{i \in N}$ which are nonnegative such that

$$u_s(\theta) = \beta + \sum_i w_i u_i(\theta), \quad \forall \theta \in \Theta.$$

Equivalently, we have a necessary and sufficient condition for the Paretian property of

$$U_s(p) = \beta + \sum_i w_i U_i(p).$$

To prove this theorem, we first recall Lemma (A.3.3).

Proof of Theorem 2.3.2. We can write this as a linear programming problem, where the theorem holds if there exist β and $(w_i)_{i \in N}$ such that

$$\begin{pmatrix} 1 & u_1(\theta_1) & u_2(\theta_1) & \cdots & u_{|N|}(\theta_1) \\ 1 & u_1(\theta_2) & u_2(\theta_2) & \cdots & u_{|N|}(\theta_2) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & u_1(\theta_m) & u_2(\theta_m) & \cdots & u_{|N|}(\theta_m) \end{pmatrix} \begin{pmatrix} \beta \\ w_1 \\ w_2 \\ \vdots \\ w_{|N|} \end{pmatrix} = \begin{pmatrix} u_s(\theta_1) \\ u_s(\theta_2) \\ \vdots \\ u_s(\theta_m) \end{pmatrix}$$

and

$$-\begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ 0 & & & \ddots & 0 \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix} \begin{pmatrix} \beta \\ w_1 \\ w_2 \\ \vdots \\ w_{|N|} \end{pmatrix} \leq \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

If the feasibility alternative in Farkas' Lemma holds, then we are done. So, assume that we do not have satisfactory β and $(w_i)_{i \in N}$. To apply Lemma (A.3.3), take the vector of unknowns to be

$$x = (\beta, w_1, \dots, w_{|N|})'.$$

The first displayed system above is the equality system $Bx = c$, with one equality for each state θ_k . The second displayed system is the inequality system $Ax \leq 0$, with one inequality $-w_j \leq 0$ for each agent j . Hence the Farkas certificate has two kinds of multipliers. The equality constraints receive unrestricted multipliers $q_1, \dots, q_m$, while the inequalities $-w_j \leq 0$ receive nonnegative multipliers $p_1, \dots, p_{|N|}$. Writing p and q as row vectors, the certificate alternative in Lemma (A.3.3) says that, if no such β, w exist, then these multipliers must satisfy

$$pA + qB = 0, \quad 0 > p \cdot b + q \cdot c = p \cdot 0 + q \cdot c = q \cdot c.$$

Here, the inequality matrix A has rows

$$(0, -e_j), \quad j \in N,$$

where the 0 is the coefficient on β and $-e_j$ selects the coefficient on w_j . Hence

$$pA = (0, -p_1, \dots, -p_{|N|}).$$

The equality matrix B has one row for each state:

$$(1, u_1(\theta_k), \dots, u_{|N|}(\theta_k)), \quad k = 1, \dots, m.$$

Therefore

$$qB = \left(\sum_{k=1}^m q_k, \sum_{k=1}^m q_k u_1(\theta_k), \dots, \sum_{k=1}^m q_k u_{|N|}(\theta_k) \right).$$

The condition $pA + qB = 0$ means that each component of this row vector is zero. From the first component, which is the coefficient on β , we have

$$\sum_{k=1}^m q_k = 0, \tag{2.1}$$

while the component corresponding to w_j is

$$-p_j + \sum_{k=1}^m q_k u_j(\theta_k),$$

so the coefficients on the w_j imply

$$\sum_{k=1}^m q_k u_j(\theta_k) - p_j = 0, \quad \forall j. \quad (2.2)$$

Lastly, c is the right-hand side of the equality system, so

$$c = (u_s(\theta_1), \dots, u_s(\theta_m))'.$$

Thus the strict inequality $q \cdot c < 0$ is

$$\sum_{k=1}^m q_k u_s(\theta_k) < 0. \quad (2.3)$$

Because $p_j \geq 0$, Equation (2.2) implies

$$\sum_{k=1}^m q_k u_j(\theta_k) \geq 0, \quad \forall j.$$

Notice that if q solves the above three constraints, then for every $\lambda > 0$, λq is a solution as well. Rescale q such that

$$|q_k| \leq \frac{1}{m}$$

for each k and define $r_k = q_k + \frac{1}{m}$ for all k .

Returning to the three constraints with r replacing q ,

$$\begin{aligned} \sum_{k=1}^m r_k &= 1, \quad r_k \geq 0, \quad \forall k \\ \sum_{k=1}^m r_k u_j(\theta_k) - \sum_{k=1}^m \frac{1}{m} u_j(\theta_k) &\geq 0, \quad \forall j \\ \sum_{k=1}^m r_k u_s(\theta_k) - \sum_{k=1}^m \frac{1}{m} u_s(\theta_k) &< 0. \end{aligned}$$

The second line says that every agent j weakly prefers the lottery defined by (r_k) to the uniform lottery, while the planner strictly prefers the uniform lottery. This is a violation of the Paretian property. 

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Suppose we have one good and two agents. If we ask the agents to report their values and assign the good to the person with the higher value, we run into an obvious problem.

3.1 Introduction

We assume the utility for agent i getting the good and being charged the transfer price p follows *quasilinear preferences*:

$$u(\text{getting the good, being charged } p) = v_i - p.$$

The notation is poor: there are separate v_i objects for the reported value and for the actual value.

We can try implementing a second-price auction: the agent with the higher reported value wins the object and pays the lower reported value.

Proposition 3.1.1.

The second-price transfer payment mechanism is strategy-proof.

Proof of Proposition 3.1.1. Consider true values v_1, v_2 and the misreport v'_1 : We have the following cases for the effect from agent 1's perspective, where $v_1 < v_2$ and agent 1 does not win the good:

Case 1 : $v'_1 < 1$ induces no change.

Case 2 : $v'_1 \in (v_1, v_2)$ also has no change.

Case 3 : $v'_1 = v_2$ leads to agent 1 winning with probability $\frac{1}{2}$, in which case she gets utility $v_1 - v_2 < 0$.

Case 4 : $v'_1 > v_2$ leads to agent 1 winning and getting utility $v_1 - v_2 < 0$.

When $v_2 < v_1$, agent 1 wins the good with utility $v_1 - v_2$. Misreporting v'_1 :

Case 1 : $v'_1 < v_2$ means agent 1 loses the good and has utility 0.

Case 2 : $v'_1 = v_2$ means agent 1 wins with probability $\frac{1}{2}$, in which case she gets utility $v_1 - v_2$.

Case 3 : $v'_1 > v_2$ leads to no change.

If $v_1 = v_2$, then under truthful reporting, agent 1 wins with probability $\frac{1}{2}$ and has utility 0. With $v'_1 < v_2$, she does not win the good and has utility 0. With $v'_1 > v_2$, she wins the good and gets utility $v_1 - v_2 = 0$.

In other words, bidding our true value allows us to win when we want to win and lose when we want to lose. 🐘

This mechanism implements the desired (efficient) outcome, but comes with the expense of the mechanism designer now collecting (a lot of) money.

3.2 Quasilinear Preferences

Consider outcomes in A and transfers in $\mathbb{R}$. We want to define preferences on $A \times \mathbb{R}$ and make statements like $(a, p) \succsim (a', p')$.

Definition 3.2.1 (Quasilinear Preferences).

A quasilinear preference v on $A \times \mathbb{R}$ can be written

$$v(a, p) = u(a) - p.$$

Definition 3.2.2 (Represents).

Suppose $\succsim$ is a weak preference on $A \times \mathbb{R}$; i.e., it is complete, transitive, and reflexive.

Suppose A is a second-countable^a topological space.

We say that $v : A \times \mathbb{R} \rightarrow \mathbb{R}$ represents $\succsim$ if

$$(a, p) \succsim (a', p') \iff v(a, p) \geq v(a', p').$$

^aIn other words, A 's topology is separable/has a countable base.

We take the following assumptions (axioms):

Assumption 3.2.3 (Axioms of preferences).

- (i) $(a, p) \succsim (a', p')$ iff $(a, p + \delta) \succsim (a', p' + \delta)$ for every δ .
- (ii) For any $a, a' \in A$, there exists p such that $(a, p) \sim (a', 0)$
- (iii) For any a, p and $\delta > 0$, we have $(a, p) \succ (a, p + \delta)$
- (iv) $\succsim$ is continuous.

Remark 3.2.4. Assumption (i) imposes that preferences over A are independent of transfer payments. We may not want this if a represents luxury goods.

Assumption (ii) assumes that we can compensate people enough money to take outcome a' over a . This may not be a good idea if a relates to health outcomes or property with personal value. 📖

Proposition 3.2.5 (Quasilinear representation of preferences).

If $\succsim$ satisfies the axioms in (3.2.3), there exists a utility function $u : A \rightarrow \mathbb{R}$ such that

$$(a, p) \succsim (a', p') \iff u(a) - p \geq u(a') - p'.$$

Proof of Proposition 3.2.5. See the course notes. 🐘

We assume from now on that people have quasilinear preferences.

3.3 Implementability, Cyclical Monotonicity, and Farkas' Lemma

We want to implement some social choice function $g : \mathcal{U}^{|N|} \rightarrow A$, where $\mathcal{U} := \{u : A \rightarrow \mathbb{R}\}$. We will do so using the transfer payments $p : \mathcal{U}^{|N|} \rightarrow \mathbb{R}^n$, where if the utilities are $(u_1, \dots, u_{|N|})$, agent i pays $p_i(u_1, \dots, u_{|N|})$ to the planner.

The main question we want to answer is the following: For which g is there a p such that truthful reporting is a dominant strategy?

Definition 3.3.1 (Implementability).

We say that a social choice function $g : \mathcal{U}^{|N|} \rightarrow A$ is implementable in dominant strategies if there exists $p : \mathcal{U}^{|N|} \rightarrow \mathbb{R}^n$ such that for every i, u_i, u_{-i} and u'_i ,

$$u_i(g(u_i, u_{-i})) - p_i(u_i, u_{-i}) \geq u_i(g(u'_i, u_{-i})) - p_i(u'_i, u_{-i}).$$

Equivalently,

$$p_i(u_i, u_{-i}) - p_i(u'_i, u_{-i}) \leq u_i(g(u_i, u_{-i})) - u_i(g(u'_i, u_{-i})).$$

To answer our question, we want to use Farkas' lemma. Note that the $u_i(g(\cdot))$ are fixed, as we take u and g as given. The p_i s are what we are free to choose, but $u_i \in \mathcal{U}$ can take infinitely many forms. First fix the set of quasilinear preferences $V \subseteq \mathcal{U}$, and assume $u_i \in V$ for each i . Fix agent i and $u_{-i} \in V^{N \setminus \{i\}}$ (which is fine since the transfers can depend on the agent). Let $x(v) := p_i(v, u_{-i})$, which is the transfer that i pays when she reports v (for an already-fixed u_{-i}). With the truthful value v_i and misreport v'_i , our condition is that

$$x(v_i) - x(v'_i) \leq v_i(g(v_i, u_{-i})) - v_i(g(v'_i, u_{-i})).$$

Definition 3.3.2 (Cycles and cyclical monotonicity).

Fix a player $i \in N$. A *cycle* in player i 's type space is a finite sequence

$$v^1, v^2, \dots, v^{q-1}, v^q = v^1,$$

where each $v^k \in V$. Given $v_{-i} \in V^{N \setminus \{i\}}$, this means that the reports of all players other than i are fixed at v_{-i} , while player i 's report moves around the cycle.

A social choice function $g : V^N \rightarrow A$ satisfies *cyclical monotonicity* if for every i , every v_{-i} , and every cycle in player i 's type space,

$$\sum_{k=1}^{q-1} v^k(g(v^k, v_{-i})) - v^k(g(v^{k+1}, v_{-i})) \geq 0.$$

Root keeps saying that a social choice function is not cyclically monotone if agents can pretend to be the next agent in the cycle and on average be better off. But it would be more accurate to say that there are q games, all in which agents $-i$ have values v_{-i} . In the k th game, agent i has value v^k and reports v^{k+1} . The social choice function is cyclically monotone if on average this is *not* profitable for agent i . a

Proposition 3.3.3 (Equivalence of cyclical monotonicity and implementability).

A social choice function $g : V^N \rightarrow A$ is implementable in dominant strategies iff g satisfies cyclical monotonicity.

Remark 3.3.4. The proposition still holds with $\mathcal{U}$ instead of V . This version of the proposition is now notable enough to be called a theorem, named after Rochet. 

Proof of Proposition 3.3.3. We first show that cyclical monotonicity implies implementability.

We want to use Farkas' lemma. We want to show existence of transfer payments — a vector with a component for each agent's reported value — that satisfy the inequality

$$x(v_i) - x(v'_i) \leq v_i(g(v_i, u_{-i})) - v_i(g(v'_i, u_{-i}))$$

for each (v_i, v'_i) pair of true value and misreport. We drop the subscripts and call this (v, v') .

Consider a matrix A where the rows are indexed by a pair (v, v') . There are $|V|$ possibilities for what i 's value can be and $|V| - 1$ possibilities for misreports. Thus, A should have $|V|(|V| - 1)$ rows. Furthermore, A should have $|V|$ columns, as our choice variables are

$$x := \begin{bmatrix} x(v_1) \\ \vdots \\ x(v_{|V|}) \end{bmatrix}.$$

The constraint vector should be

$$b := \begin{bmatrix} v(g(v, u_{-i})) - v(g(v', u_{-i})) \\ \vdots \end{bmatrix}.$$

The large matrix has entry in the row with index (v, v') and column v''

$$A_{(v,v'),v''} = \begin{cases} 1, & v'' = v \\ -1, & v'' = v' \\ 0, & \text{otherwise} \end{cases}.$$

Hence, on this finite space V of quasilinear preferences on the set of outcomes A , g is implementable in dominant strategies if

$$Ax \leq b.$$

By Farkas' Lemma, there exists no solution $x \in \mathbb{R}^{|V|}$ (transfer payment when u_{-i} is fixed) iff there exists $y \in \mathbb{R}^{|V|(|V|-1)}$ with $y \geq 0$ such that $yA = 0$ and $y \cdot b < 0$. Suppose this is the case and index y by (v, v') , where v can take on $|V|$ values and for each v , v' can take on $|V| - 1$ values.

$y = 0$ is not possible because we require $y \cdot b < 0$. By $y \geq 0$, we are guaranteed a particular v, v' such that $y_{(v,v')} > 0$. This element of y gets multiplied against everything in the (v, v') row of A ; in particular, it is multiplied against $A_{(v,v'),v'} = -1$. Since $yA = 0$, there must be some entry in the v' column of A with a value of 1, which is exactly rows in the form (v', v'') , that is multiplied against a positive component of y . This is the (v', v'') component, so we have found v'' such that $y_{(v',v'')} > 0$.

Iterating, there exists a sequence $v^1, v^2, \dots$ such that $y_{(v^k, v^{k+1})} > 0$ for all k . Since $|V|$ is finite, the sequence must eventually repeat an index. Let q be the first time that this occurs. Without loss, we can assume $v^q = v^1$.

Now we have

$$y_{(v_1, v_2)} > 0, y_{(v_2, v_3)} > 0, \dots, y_{(v^{q-2}, v^{q-1})} > 0, y_{(v^{q-1}, v^1)} > 0.$$

Let $y^* \in \mathbb{R}^{|V|(|V|-1)}$ be a vector¹ of indicators such that

$$y_{(v,v')}^* = \mathbb{1} \{ (v, v') = (v^k, v^{k+1}) \text{ for some } k \}.$$

Essentially, it is just y with all positive weights made to be 1.

Note that $y^*A = 0$ by construction. Let $\alpha := \min_k \{y_{(v^k, v^{k+1})}^*\}$. By distributing, $(y - \alpha y^*)A = 0$. Furthermore, $(y - \alpha y^*)$ has at least one more 0 than y does. By definition of α , $y - \alpha y^* \geq 0$. We can repeat this process of finding indices where the residual vector is positive until the residual is 0, so we get

$$y = \alpha_1 y_1^* + \cdots + \alpha_p y_p^*$$

for some p and each y_ℓ^* is a cycle like y^* .

With this representation of y and recalling $y \cdot b < 0$, we know that $y_\ell^* \cdot b < 0$ for at least one ℓ .

We have shown that if there exist no transfers to implement g in dominant strategies, then there exists a sequence of utilities $v^1, \dots, v^r$ (coming from y_ℓ^*) such that

$$0 > \sum_{j=1}^{r-1} b_{(v^j, v^{j+1})} = \sum_{j=1}^{r-1} v^j (g(v^j, u_{-i})) - v^j (g(v^{j+1}, u_{-i})).$$

The net change from the $r-1$ games where agent i misreports v^j as v^{j+1} is negative. Thus, if g is cyclically monotone, there cannot be such a y with $y \geq 0$, $yA = 0$, and $y \cdot b < 0$. By Farkas' Lemma, $Ax \leq b$ has a solution. Taking $p_i(v, u_{-i}) = x(v)$ for this fixed i and u_{-i} , and repeating for every i and u_{-i} , gives transfer payments that implement g .

Conversely, suppose g is implementable in dominant strategies by the transfer rule p . Fix i , v_{-i} , and a cycle $v^1, \dots, v^{q-1}, v^q = v^1$. For each $k = 1, \dots, q-1$, incentive compatibility gives

$$p_i(v^k, v_{-i}) - p_i(v^{k+1}, v_{-i}) \leq v^k (g(v^k, v_{-i})) - v^k (g(v^{k+1}, v_{-i})).$$

Summing over k , the left-hand side telescopes to

$$p_i(v^1, v_{-i}) - p_i(v^q, v_{-i}) = 0,$$

since $v^q = v^1$. Hence

$$0 \leq \sum_{k=1}^{q-1} v^k (g(v^k, v_{-i})) - v^k (g(v^{k+1}, v_{-i})),$$

so g satisfies cyclical monotonicity. □

Proposition 3.3.5.

Suppose $g : \mathcal{U}^{|N|} \rightarrow A$ is such that

$$g(u_1, u_2, \dots, u_{|N|}) \in \arg \max_{a \in A} \sum_{i \in N} u_i(a).$$

Then g is cyclically monotone.

¹ y^* is a finite sequence of types in the support of y .

Proof of Proposition 3.3.5. By definition of g as utilitarian, we have that for every (u_i, u_{-i}) ,

$$u_i(g(u_i, u_{-i})) + \sum_{j \neq i} u_j(g(u_i, u_{-i})) \geq u_i(a) + \sum_{j \neq i} u_j(a), \quad \forall a \in A$$

Adding and subtracting,

$$u_i(g(u_i, u_{-i})) - u_i(a) \geq \sum_{j \neq i} u_j(a) - u_j(g(u_i, u_{-i})), \quad \forall a \in A.$$

Fix u_{-i} and consider the cycle $u^1, \dots, u^{q-1}, u^q = u^1$. To check cyclical monotonicity, we compute

$$\begin{aligned} & \sum_{\ell=1}^{q-1} u^\ell(g(u^\ell, u_{-i}) - u^\ell(g(u^{\ell+1}, u_{-i})) \\ & \geq \sum_{\ell=1}^{q-1} \sum_{j \neq i} u_j(g(u^{\ell+1}, u_{-i})) - u_j(g(u^\ell, u_{-i})) \text{ by the add-subtract result} \\ & = \sum_{j \neq i} \sum_{\ell=1}^{q-1} u_j(g(u^{\ell+1}, u_{-i})) - u_j(g(u^\ell, u_{-i})) = 0. \end{aligned}$$

$\underbrace{\hspace{10em}}_{=0 \text{ by telescoping around a cycle}}$



Corollary 3.3.6.

Pairing this result with Proposition (3.3.3) yields that any welfare-maximizing social choice function is implementable in dominant strategies.

3.4 Another Approach to Cyclical Monotonicity

The proof to Proposition (3.3.3) relied on Farkas' lemma and is not beautiful. We can refer to the course notes for a more instructive proof using convex functions and their subdifferentials.

3.5 The Vickrey–Clarke–Groves Mechanism

We need to determine the transfer payments p_i .

Proposition 3.5.1.

Let g be a welfare-maximizing utility function. Consider the rule

$$p_i(u_i, u_{-i}) := - \sum_{j \neq i} u_j(g(u_i, u_{-i})).$$

Under these transfer payments, truthful reporting is a dominant strategy; p implements g .

Implementability only says that some transfers support truthful reporting; it does not by itself guarantee budget balance, individual rationality, or any fairness property. The VCG construction below gives a family of transfers for welfare maximization by charging each agent the externality she imposes on others, though even VCG need not be budget-balanced or distributionally fair in general.

Proof of Proposition 3.5.1. Considering the deviation from reporting u_i to u'_i , we can show the following chain between agent i 's welfare under truthful and misreporting, where the inequality comes from g being welfare-maximizing.

$$\begin{aligned}
 u_i(g(u_i, u_{-i})) - p_i(u_i, u_{-i}) &= u_i(g(u_i, u_{-i})) - \left(- \sum_{j \neq i} u_j(g(u_i, u_{-i})) \right) \\
 &= \sum_k u_k(g(u_i, u_{-i})) \\
 &\geq \sum_k u_k(g(u'_i, u_{-i})) \\
 &= u_i(g(u'_i, u_{-i})) - \left(- \sum_{j \neq i} u_j(g(u'_i, u_{-i})) \right) \\
 &= u_i(g(u'_i, u_{-i})) - p_i(u'_i, u_{-i}).
 \end{aligned}$$



This payment plan is super expensive, though; the planner has to pay out $|N| - 1$ times total utility. What a stupid silly mechanism.

There is a better mechanism, which uses the idea that if your payment does not depend on your report, you cannot game a lower payment by misreporting.

Definition 3.5.2 (Vickrey–Clarke–Groves (VCG) mechanism).

The VCG mechanism has social choice function

$$g(u_1, \dots, u_{|N|}) \in \arg \max_{a \in A} \sum_i u_i(a)$$

and payments

$$p_i(u_i, u_{-i}) = \sum_{j \neq i} [u_j(a^*(u_{-i})) - u_j(g(u_i, u_{-i}))],$$

where $a^*(u_{-i}) \in \arg \max_{a \in A} \sum_{j \neq i} u_j(a)$.

Proposition 3.5.3.

If $h : \mathcal{U}^{|N|-1} \rightarrow \mathbb{R}$ and p implements g , then if

$$p'_i(u_i, u_{-i}) = p_i(u_i, u_{-i}) + h(u_{-i}),$$

$p' = (p'_i, p_{-i})$ also implements g .

If a payment rule p implements g (i.e. (g, p) is incentive-compatible), then adding to i 's payment any function of u_{-i} to form p'_i does not change incentive compatibility.

Corollary 3.5.4.

In particular, the payments in the VCG mechanism implement the VCG mechanism's social choice function.

Proof of Corollary 3.5.4. We take Proposition (3.5.3) as given. It implies Corollary (3.5.4), but we can show it directly as well. We make the same comparison as before:

$$\begin{aligned}
u_i(g(u_i, u_{-i})) - p_i(u_i, u_{-i}) &= u_i(g(u_i, u_{-i})) - \sum_{j \neq i} [u_j(a^*(u_{-i})) - u_j(g(u_i, u_{-i}))] \\
&= \sum_k u_k(g(u_i, u_{-i})) - \sum_{j \neq i} u_j(a^*(u_{-i})) \\
&\geq \sum_k u_k(g(u'_i, u_{-i})) - \sum_{j \neq i} u_j(a^*(u_{-i})) \\
&= u_i(g(u'_i, u_{-i})) - \sum_{j \neq i} [u_j(a^*(u_{-i})) - u_j(g(u'_i, u_{-i}))] \\
&= u_i(g(u'_i, u_{-i})) - p_i(u'_i, u_{-i}),
\end{aligned}$$

where the inequality follows from g being utilitarian under the true profile. 

It turns out that maximizing welfare is implementable in dominant strategies, and the only such social welfare functions that are implementable are those that are utilitarian (with weights $\{w_i\}$ and biases over outcomes $\{C_a\}$). Just like Gibbard–Satterthwaite, Roberts' theorem depends crucially on the domain of g being unrestricted.

Theorem 3.5.5 (Roberts').

Suppose that $|A| \geq 3$. $g : \mathcal{U}^{|N|} \rightarrow A$ is implementable in dominant strategies iff there are nonnegative weights $\{w_i\}_{i \in N}$ which are not identically zero and constants $\{C_a\}_{a \in A}$ for each $a \in A$ such that

$$g(u_1, \dots, u_{|N|}) \in \arg \max_{a \in A} \sum_{i \in N} w_i u_i(a) + C_a.$$

We call the social choice function g in Roberts' theorem an *affine maximizer*.

Proof of Theorem 3.5.5. We omit the proof. 

3.6 Domain Restrictions

The VCG mechanism is designed for the space $\mathcal{U}^{|N|}$. If we have some $\mathcal{V} \subseteq \mathcal{U}$ and a social choice function $g' : \mathcal{V}^{|N|} \rightarrow A$ that maximizes welfare, does the VCG mechanism restricted to $\mathcal{V}$ implement g' in dominant strategies? Yes, because restricting the space to optimize over makes the problem easier. If you didn't want to deviate from your truthful utility in $\mathcal{U}$, then you would not want to deviate in $\mathcal{V}$ either.

Example 3.6.1.

Consider a single good to be allocated to one agent out of n . We can identify A with the unit vectors in $\mathbb{R}^n$. Suppose each agent gets value b_i if she receives the good and utility 0 if she doesn't get the good. The agent with the highest value $b_{(1)}$ gets the good and she pays $b_{(2)}$. 

Example 3.6.2.

Suppose Google has K sponsored slots $1, 2, \dots, K$ for search results. Each slot ℓ has a clickthrough rate of α_ℓ . Assume that $\alpha_1 \geq \alpha_2 \geq \dots \geq \alpha_K$. If $K < |N|$, add many slots with clickthrough rate 0. Each advertising firm i has value v_i from a click, so if they are given slot ℓ , they have utility $v_i \alpha_\ell$.

Here, A is the set of permutations for which firm is assigned to what slot. We have a very large domain restriction; firm i 's utility is in a specific form that is invariant to other firms' utilities.

Order the agents so that $v_1 \geq v_2 \geq \dots \geq v_{|N|}$. We claim with great conviction that welfare is maximized by assigning slot ℓ to firm ℓ for $\ell = 1, \dots, |N|$.

The VCG payments are

$$p_i(v_i, v_{-i}) = \sum_{j>i} (\alpha_{j-1} - \alpha_j) v_j.$$



Cyclical monotonicity is the correct characterization (condition) for implementability in dominant strategies, as Rochet's theorem is not sensitive to domain restrictions, as seen in Proposition (3.3.3) and the subsequent remark.

However, the forward direction of Roberts' theorem no longer holds with domain restrictions, since implementability of g is easier to satisfy when the condition must hold over fewer utility functions. There are known mechanisms for domain restrictions that are not affine maximizers. Hence, we should try to better understand cyclical monotonicity. We can try weakening it by imposing the cycles are only of length 2.

Definition 3.6.3 (Weak monotonicity).

Let $\mathcal{V} \subseteq \mathcal{U}$. We say that $g : \mathcal{V}^{|N|} \rightarrow A$ satisfies weak monotonicity if it satisfies cyclical monotonicity for all cycles of length 2. That is, for all i , v_{-i} , and all v_i, v'_i , we have

$$v_i(g(v_i, v_{-i})) - v_i(g(v'_i, v_{-i})) + v'_i(g(v'_i, v_{-i})) - v'_i(g(v_i, v_{-i})) \geq 0.$$

More intuitively,

$$v_i(g(v_i, v_{-i})) - v_i(g(v'_i, v_{-i})) \geq v'_i(g(v_i, v_{-i})) - v'_i(g(v'_i, v_{-i})).$$

v_i 's change in utility from the misreport v'_i to the true value v_i must be bigger than v'_i 's change in utility from the true value v'_i to the misreport v_i .

Theorem 3.6.4 (Saks and Yu, 2005).

Suppose $\mathcal{V} \subseteq \mathcal{U}$ is convex. $g : \mathcal{V}^{|N|} \rightarrow A$ is implementable in dominant strategies iff it is weakly monotonic.

The theorem says that if our domain restriction is to a convex set, then we only need to check cycles of length 2, which is much easier to check than cycles of possibly infinite length.

Social Choice with Indifferences, Constrained Allocation

4

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In Gibbard–Satterthwaite, we assumed everyone had strict preferences, but now we can suppose that agents can be indifferent between allocations, e.g., if they are indifferent between allocations which assign them the same objects.

4.1 Introduction and the Equivalence of Constrained Allocation

We introduce seemingly different problems that will turn out to be equivalent.

Definition 4.1.1 (Social choice problem with indifferences).

For each agent $i \in N$, suppose we have a partition $A^i := \{A_1^i, \dots, A_{m_i}^i\}$ of A such that if $x, y \in A_\ell^i$, then i is indifferent between x and y . Let P^i be the set of strict preferences on A^i . Then any $\succsim_i \in P^i$ induces a weak ranking $\succsim_i^*$ on A , defined by the following relation for $x \in A_\ell^i, y \in A_k^i$:

$$\begin{cases} x \succsim_i^* y, & A_\ell^i \neq A_k^i \text{ and } A_\ell^i \succ_i A_k^i \\ x \sim_i^* y, & A_\ell^i = A_k^i \end{cases}.$$

A social choice function with indifferences is a function $f : \prod_{i \in N} P^i \rightarrow A$ and is defined by

$$(N, A, (A^i)_{i \in N}),$$

and the social choice problem is to select such a function given the indifference partitions.

Definition 4.1.2 (Constrained allocation problem).

Suppose for each agent $i \in N$, we have a set X_i of available objects. An allocation is a function $\mu : N \rightarrow \cup_{i \in N} X_i$ such that for every $i \in N$, $\mu(i) \in X_i$. An allocation specifies for each agent i a feasible object to be received. Denote $\mathcal{A} := \{\mu : N \rightarrow \cup_{i \in N} X_i\}$ to be the set of allocations.

A constraint is a set $C \subseteq \mathcal{A}$ that is nonempty. Allocations in C are called *feasible*, while allocations not in C are *infeasible*. Let $\tilde{P}_i$ be the set of strict preferences on X_i . A mechanism is a function $f : \prod_{i \in N} \tilde{P}_i \rightarrow C$.

The constrained allocation problem is defined by

$$(N, (X_i)_{i \in N}, C).$$

The constraint set is given to us in the allocation problem.

Proposition 4.1.3 (Equivalence of social choice with indifferences and constrained allocation).

Every constrained allocation problem can be written as a social choice problem with indifferences. Conversely, every social choice problem with indifferences can be written as a constrained allocation problem. Under these translations, mechanisms are the same up to outcomes that all agents regard as indifferent.

Proof of Proposition 4.1.3. ($\implies$) First start with a constrained allocation problem defined on

$$(N, (X_i)_{i \in N}, C).$$

Replace each X_i with $\{x \in X_i : \mu(i) = x \text{ for some } \mu \in C\}$, so every remaining available object is assigned to agent i in some feasible allocation. Then set $A := C$. For each $i \in N$, define a partition of A by

$$A^i := \{\{\mu \in C : \mu(i) = x\} : x \in X_i\}.$$

Thus two feasible allocations are in the same indifference class for agent i exactly when they give agent i the same object.¹

A strict preference $\succ_i$ on X_i induces a strict preference $\succ_i^*$ on A^i by

$$\{\mu \in C : \mu(i) = x\} \succ_i^* \{\mu \in C : \mu(i) = y\} \iff x \succ_i y.$$

This gives a canonical identification of the constrained allocation preference domain $\tilde{P}_i$ with the social choice preference domain P^i . Hence a constrained allocation mechanism $f : \prod_{i \in N} \tilde{P}_i \rightarrow C$ can be regarded as the social choice function with indifferences $f^* : \prod_{i \in N} P^i \rightarrow A$ given by

$$f^* (\succ_1^*, \dots, \succ_{|N|}^*) = f (\succ_1, \dots, \succ_{|N|}).$$

In this direction, the equivalence of mechanisms is immediate because the social outcomes are exactly the feasible allocations: $A = C$.

($\impliedby$) Conversely, start with a social choice problem with indifferences defined on $(N, A, (A^i)_{i \in N})$. Set $X_i := A^i$, so agent i 's available objects are her indifference classes. With this definition, the constrained allocation preference domain $\tilde{P}_i$ is exactly P^i , the set of strict preferences over A^i . Note that any allocation μ chooses one class $\mu(i) \in A^i$ for each agent i . Define the constraint as a subset of all possible allocations $\mathcal{A} := \{\mu : N \rightarrow \times_{i \in N} X_i\} = \{\mu : N \rightarrow \times_{i \in N} A_i\}$ by

$$C := \left\{ \mu \in \mathcal{A} : \bigcap_{i \in N} \mu(i) \neq \emptyset \right\}.$$

A tuple $(\mu(i))_i \in \prod_{i \in N} A^i$ of assigned indifference classes is feasible if there exists at least one original outcome $a \in A$ such that a is in every assigned indifference class.

Thus any social choice function $f : \prod_{i \in N} P^i \rightarrow A$ induces a constrained allocation

¹This is the right indifference class because in the constrained allocation problem, agent i only has a strict preference over her own set X_i of objects and does not care about others' objects.

mechanism $\hat{f} : \prod_{i \in N} \tilde{P}_i \rightarrow C$ by

$$\hat{f}(\succ)(i) = A_\ell^i \quad \text{where } f(\succ) \in A_\ell^i.$$

$\hat{f}(\succ)(i)$ is the indifference class A_ℓ^i in i 's partition of A , which is agent i 's object in the constructed constrained allocation problem.

To show that no preference-relevant information is lost in this translation, take a constrained allocation mechanism $g : \prod_{i \in N} \tilde{P}_i \rightarrow C$ in the constructed allocation problem. Since each feasible allocation $g(\succ)$ is a tuple of indifference classes with nonempty intersection, choose for each $\mu \in C$ some representative $r(\mu) \in \cap_{i \in N} \mu(i)$ and define the social choice function $\hat{g} : \prod_{i \in N} P^i \rightarrow A$ by

$$\hat{g}(\succ) := r(g(\succ)).$$

The choice of representative does not matter for preferences, since every representative lies in the same indifference class for every agent. 

Example 4.1.4 (Examples of Equivalence).

Consider:

- (i) House allocation: Suppose we have a set of possible houses H , and put $X_i := H$ for every i . The constrained set of allocations is

$$C = \{\mu : N \rightarrow H : \mu(i) \neq \mu(j), \text{ if } i \neq j\}.$$

- (ii) School choice: Denote the set of schools X , and put $X_i := X$ for all i . Each school $x \in X$ has a capacity $q_x \geq 0$, and the constrained set is

$$C = \{\mu : N \rightarrow X : |\mu^{-1}(x)| \leq q_x, \forall x\}.$$

- (iii) One-sided matching:^a Suppose $X_i := N$ for every i and we have the constrained set

$$C = \{\mu : N \rightarrow N : \mu \circ \mu(i) = i, \forall i\}.$$

C requires that if i is matched with j , j must be matched with i . But it allows for people to be matched with themselves, i.e. unmatched.

- (iv) Two-sided matching: Consider $N = M \amalg W$. Set

$$X_i = \begin{cases} W, & i \in M \\ M, & i \in W \end{cases},$$

and put

$$C = \{\mu : N \rightarrow N : \mu \circ \mu(i) = i \text{ and } \mu(i) \in M \text{ if } i \in W, \forall i\}.$$

- (v) Social choice: Put $X_i := A$ for every i and let

$$C = \{\mu : N \rightarrow A : \mu(i) = \mu(j), \forall i, j\}.$$

Thus the social choice problem we have been thinking of this whole time is a

special case of the constrained allocation problem.

^aMore commonly called the roommates problem.



We reintroduce two concepts in this new setting.

Definition 4.1.5 (Strategy-proof, Pareto efficient).

In the setting of a constrained allocation problem, a mechanism $f : \prod_{i \in N} \tilde{P}_i \rightarrow C$ is strategy-proof if there does not exist $i, \succ_i, \succ'_i, \succ_{-i}$ such that

$$f(\succ'_i, \succ_{-i})(i) \succ_i f(\succ_i, \succ_{-i})(i).$$

f is Pareto efficient if for all $\succ \in \prod_{i \in N} \tilde{P}_i$, there does not exist $\mu \in C$ such that:

- (i) For every $j \in N$, $\mu(j) \succeq_j f(\succ)(j)$ and
- (ii) For some $i \in N$, $\mu(i) \succ_i f(\succ)(i)$.

4.2 Two Agent Constrained Allocation

Now, assume $N = \{1, 2\}$. Fix the possible sets X_1, X_2 and a constraint C .

Definition 4.2.1 (Local dictator).

A local dictator at an infeasible allocation $(x, y) \notin C$ is an agent who gets to keep her top choice when x, y are top-ranked by agents 1 and 2 respectively.

We will describe all strategy-proof and Pareto efficient mechanisms for C .

Proposition 4.2.2 (Two-agent local dictatorships).

Let G be the graph whose nodes are infeasible allocations and whose edges connect two infeasible allocations that share a coordinate. After removing objects that can never be assigned to an agent in any feasible allocation, a mechanism f is strategy-proof and Pareto efficient if and only if it is a local dictatorship: each connected component of G is assigned a local dictator, and for every profile with top-ranked objects (x, y) , the mechanism gives both agents their top choices if $(x, y) \in C$; otherwise, the local dictator for the component containing (x, y) keeps that agent's top choice and the other agent receives her favorite object compatible with the dictator's top choice.

Lemma 4.2.3.

Suppose that there exists $x \in X_i$ for some i such that $\mu(i) \neq x$ for every $\mu \in C$. If f is strategy-proof and Pareto efficient, then for $\succ_i$ and $\succ'_i$ that differ only in the position of x , then

$$f(\succ_i, \succ_j)(i) = f(\succ'_i, \succ_j)(i).$$

Furthermore, j 's object is unchanged by i 's misreport:

$$f(\succ_i, \succ_j)(j) = f(\succ'_i, \succ_j)(j).$$

Proof of Lemma 4.2.3. If the objects are not the same, then i has a strict preference

between the two objects. Suppose

$$f(\succ_i, \succ_j)(i) \succ_i f(\succ'_i, \succ_j)(i).$$

Since neither side is x and $\succ_i, \succ'_i$ differ only in the position of x , we also have

$$f(\succ_i, \succ_j)(i) \succ'_i f(\succ'_i, \succ_j)(i),$$

which violates strategy-proofness.

If instead

$$f(\succ_i, \succ_j)(i) \prec_i f(\succ'_i, \succ_j)(i),$$

strategy-proofness is immediately violated.

For the second claim, suppose for contradiction (without loss) that

$$f(\succ_i, \succ_j)(j) \succ_j f(\succ'_i, \succ_j)(j).$$

Then $f(\succ'_i, \succ_j)$ is not efficient, as $f(\succ_i, \succ_j) \in C$ and $f(\succ_i, \succ_j) \sim_i f(\succ'_i, \succ_j)$ by the first claim. 

We can write an allocation as a pair (x, y) . By Lemma (4.2.3), for each i , we can get rid of all $x \in X_i$ that can never be chosen and assume that for every $(x, y) \notin C$, there exist x', y' such that $(x', y) \in C$ and $(x, y') \in C$.

Let $\bar{C} \subseteq C$ be the set of infeasible allocations.

This is true because if there does not exist such a y' , $\mu(i) \neq x$ for all μ , but we already got rid of all such x .

Lemma 4.2.4.

(9.1.10) For any strategy-proof and efficient mechanism f , each infeasible allocation $(x, y) \notin C$ is associated with a local dictator. In particular, there exists a function $D : \bar{C} \rightarrow \{1, 2\}$ that describes who is the local dictator at every infeasible allocation. Furthermore, if $(x, y'') \notin C$ as well, then

$$D((x, y)) = 1 \implies D((x, y'')) = 1.$$

Proof of Lemma 4.2.4. Suppose (x, y) is not feasible and there exist x', y' such that $(x', y) \in C$ and $(x, y') \in C$. Consider the preferences

$$\begin{aligned} \succ_1 &: x \succ_1 x' \succ_1 \cdots \\ \succ_2 &: y \succ_2 y' \succ_2 \cdots \end{aligned}$$

We know that $f(\succ_1, \succ_2) \in \{(x, y'), (x', y)\}$ by Pareto efficiency of f . Suppose without loss that we side with agent 1 and $f(\succ_1, \succ_2) = (x, y')$.

Consider another preference $\succ'_2$ where y is top-ranked. We know by strategy-proofness that $f(\succ_1, \succ'_2)(2) \neq y$. This implies $f(\succ_1, \succ'_2)(1) = x$ by Pareto efficiency, as otherwise, we could have put $f(\succ_1, \succ'_2) = (x', y)$, making agent 1 no worse off and agent 2 strictly better off. By one more application of strategy-proofness, $f(\succ'_1, \succ'_2)(1) = x$ for any $(\succ'_1, \succ'_2)$ such that 1 top-ranks x and 2 top-ranks y , as if agent 1 does not get x by ranking $x \succ'_1 x'' \succ'_1 \cdots$, she could deviate to reporting $x \succ_1 x' \succ_1 \cdots$ and get x , which is her top choice under $\succ'_1$. By Pareto efficiency, this implies that agent 2 gets his favorite object compatible with 1 getting x .

We have shown that agent 1 is a local dictator over (x, y) . We can repeat this procedure

for every infeasible allocation, put

$$D : \overline{C} \rightarrow \{1, 2\},$$

where the function indicates who we sided with in getting his or her top-ranked object.

For the second claim, suppose not, so $D((x, y'')) = 2$. Consider the preferences

$$\begin{aligned} \succ_1 &: x \succ_1 \cdots \\ \succ_2 &: y \succ_2 y'' \succ_2 \cdots \\ \succ'_2 &: y'' \succ'_2 \cdots \end{aligned}$$

Under $(\succ_1, \succ_2)$ then 2 gets something worse than y'' , since 1 is the local dictator at (x, y) and (x, y'') is infeasible. But 2 can misreport $\succ'_2$ to get y'' , as he is the dictator at (x, y'') , which violates strategy-proofness.

The same argument works for infeasible allocations that share the second coordinate instead of the first coordinate. Repeating this argument along a connected path shows that all infeasible allocations in the same connected component of G have the same local dictator. 

Proof of Proposition 4.2.2. The first lemma lets us ignore any object that is never assigned to an agent in a feasible allocation, since changing the position of such an object cannot affect either agent's allocation. Thus, for any infeasible (x, y) , there exist x' and y' such that $(x', y) \in C$ and $(x, y') \in C$.

The second lemma then assigns a local dictator to every infeasible allocation and shows that this assignment is constant across infeasible allocations that share a coordinate. Equivalently, it is constant on every horizontal or vertical edge of G , and therefore on every connected component of G . Hence any strategy-proof and Pareto efficient mechanism must be a local dictatorship.

Conversely, suppose f is a local dictatorship. It is Pareto efficient because if the top-ranked pair (x, y) is feasible, both agents receive their top choices; if (x, y) is infeasible and agent 1 is the local dictator, then 1 receives x and agent 2 receives her favorite object among the feasible allocations with first coordinate x , so no feasible allocation can improve 2 without making 1 worse off. The case where agent 2 is the local dictator is symmetric.

To see strategy-proofness, first suppose the reported top-ranked objects are feasible. Then each truthful agent receives that agent's favorite object and cannot gain by deviating. If the reported top-ranked pair is infeasible, the local dictator cannot gain because the dictator keeps the dictator's favorite object. The non-dictator receives the best object compatible with the dictator's reported top choice. A deviation by the non-dictator to another infeasible allocation means the dictator is the same by Lemma (9.1.10), so the non-dictator cannot improve further on his or her best compatible object. A deviation that makes the top-ranked pair feasible gives the dictator the dictator's top choice and therefore gives the deviating non-dictator an object compatible with that same top choice, again no better than what the non-dictator already received. Thus no unilateral deviation is profitable. 

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Why are economic theorists interested in auctions?

Example 5.0.1 (Real-world auctions).

There are several useful auctions used in everyday life:

- Art, used cars, etc.
- Electricity ⚡
- Google/Facebook ads
- Electromagnetic spectrum
- Government debt



We start with a simple example.

5.1 Optimal Posted Price With a Single Buyer

Suppose we have a single buyer for a single good, with the buyer's value as $v \in [0, \bar{v}]$. Let $v \sim F$, where $\mathbb{P}[v \leq w] = F(w)$.

Assumption 5.1.1.

We assume F is differentiable and thus admits a density f . We further assume that the seller has no value for the good.

Proposition 5.1.2.

In this single-buyer, single-good environment with a posted price, the optimal price is the p^* satisfying

$$p^* = \frac{1 - F(p^*)}{f(p^*)}.$$

- The optimal price leads to an inefficient outcome, and
- The optimal price depends on F .

Proof of Proposition 5.1.2. With price p , the seller's expected revenue is

$$\underbrace{p(1 - F(p))}_{\text{probability the buyer buys}}.$$

The first order condition is

$$1 - F(p) + p(-f(p)) = 0,$$

so the optimal posted price is

$$p^* = \frac{1 - F(p^*)}{f(p^*)}.$$

Who is the lowest type that purchases the good? It is the person with value $v = p^*$, yielding the *virtual value* characterization:

$$v - \frac{1 - F(v)}{f(v)} = 0.$$

We see a source of inefficiency; even though the seller has no value for the good, she does not sell the good to buyers with low types, even though post-revealing type, the sale could go through with a price between 0 and the realized value to make everyone better off. 🐘

For example, with a uniform value distribution $v \sim \text{Unif}(0, 1)$, the virtual value is negative for $v < 1/2$.

5.2 Overview of Auctions

Consider a set N of agents who are the possible buyers of the single good, where each agent i has value $v_i \in [0, 1]$. Let the set of outcomes be

$$A = \left\{ (x_1, \dots, x_{|N|}) : x_i \geq 0 \ \forall i, \sum_{i \in N} x_i \leq 1 \right\}.$$

Hence, an outcome is a lottery over who gets the good (including the seller herself).

Assumption 5.2.1.

We assume the utilities take the following form: If agent i has value v_i for the good, gets it with probability x_i , and pays p_i , then the payoff is

$$u_i = v_i x_i - p_i.$$

The price p_i is determined by the reported value, which may not be the same as the true value.

Since the agent has linear utility in money, it's equivalent to charge agent i if she wins vs. charging a smaller amount proportional to her winning probability, regardless if she wins.

Example 5.2.2 (Common auctions).

We have the following common auctions:

- (i) **First-price auction:** The seller asks each $i \in N$ to submit a bid $b_i \geq 0$. The player with the highest bid wins and pays their bid.
- (ii) **Second-price auction:** The player with the highest bid wins and pays the second highest bid.
- (iii) **All-pay auction:** The player with the highest bid wins and every player i pays

b_i .



We specialize to a two-player symmetric case.

Assumption 5.2.3.

Let $N = \{1, 2\}$ and $v_i \stackrel{iid}{\sim} F$. Assume F admits a density f that is strictly positive on $[0, 1]$ and 0 elsewhere.

Proposition 5.2.4 (Revenue of a second price auction).

The revenue of a second-price auction is

$$R_{\text{SPA}} = 1 - \int_0^1 F_{(2)}(w)dw = \int_0^1 (1 - F(w))^2 dw.$$

Proof of Proposition 5.2.4. We know that truthful reporting is a dominant strategy in a second-price auction. Assume that $b_i = v_i$ for both agents. To understand the expected revenue, we need to understand the distribution of the lower value. We see that

$$F_{(2)}(w) = 1 - (1 - F(w))(1 - F(w)) = 2F(w) - F(w)^2,$$

as both players must have value *above* w . Hence, the expected revenue is

$$R = \int_0^1 w f_{(2)}(w)dw = 1 - \int_0^1 F_{(2)}(w)dw,$$

where the second equality follows from integrating by parts. 

The first-price auction analysis is trickier — we intuit that people should shade bids, so the effect of taking the highest bid may be offset by lower bids overall. We need to first take a detour to Bayesian games, where agents are unsure of others' payoffs.

Definition 5.2.5 (Bayesian game).

A Bayesian game is defined by the following primitives:

- A set N of agents.
- For each $i \in N$, there is a set of types T_i which encodes her own type and her beliefs over others' types. We denote the type space $T := \prod_{i \in N} T_i$.
- For each $i \in N$, there is a set of actions A_i , and we denote the set of action profiles $A := \prod_{i \in N} A_i$.
- For each $i \in N$, there is a utility function

$$U_i : T_i \times A \rightarrow \mathbb{R}.$$

- (We assume there is) A common prior $\mu \in \Delta(T)$. Player i 's belief over the others' types is determined by conditioning μ on her realized type $t_i \in T_i$ and is notated $\mu(t_{-i} \mid t_i)$.

We can take the derivative of $F_{(2)}$ to get

$$f_{(2)}(w) = 2f(w) - 2F(w)f(w).$$

A strategy for player i is a function

$$\sigma_i : T_i \rightarrow \Delta(A_i).$$

The strategy profile $\sigma = (\sigma_i)_{i \in N}$ induces the payoff for agent i

$$U_i(\sigma, t_i) = \sum_{t_{-i}} \sum_{a \in A} u_i(t_i, a) \sigma_i(a | t_i) \sigma(a_{-i} | t_{-i}) \mu(t_{-i} | t_i).$$

The strategy profile σ is a Bayes-Nash equilibrium if for every agent i , σ , σ'_i , and t_i ,

$$U_i(\sigma, t_i) \geq U_i(\sigma'_i, \sigma_{-i}, t_i).$$

Now we are ready to analyze the first-price auction.

Proposition 5.2.6 (Revenue of a first-price auction).

An equilibrium bidding strategy in a symmetric first price auction with independent value distribution with CDF F for a bidder with value v is

$$b(v) = v - \frac{1}{F(v)} \int_0^v F(w) dw.$$

The expected revenue is

$$R_{\text{FPA}} = 1 - \int_0^1 F_{(2)}(w) dw = \int_0^1 (1 - F(w))^2 dw,$$

which is exactly the expected revenue of the second-price auction.

Note that all the primitives of the Bayesian game map neatly onto our auction. The common prior $\mu \in \Delta(T)$ is the distribution from which the values are drawn from.

Proof of Proposition 5.2.6. To prove the proposition, we begin with a conjecture that there is an equilibrium of the first-price auction in which there is a function $b : [0, 1] \rightarrow [0, 1]$ such that agent i bids $b(v)$ when she has value v . We also conjecture that this b is strictly increasing and differentiable.

A necessary condition for a proposed b to be a Bayes-Nash equilibrium is that when player i has value v_i , her utility bidding according to $b(\cdot)$ is maximized when bidding as if she has value v_i :¹

$$v_i = \arg \max_{v'_i \in [0, 1]} (v_i - b(v'_i)) F(v'_i).$$

The first-order condition is

$$-b'(v'_1)F(v'_1) + (v_i - b(v'_i)) f(v'_i)|_{v'_i=v_i} = 0,$$

¹The utility in this expression is the utility when both players bid according to $b(\cdot)$, player j bids according to his true value, and player i bids as if she had value v'_i .

which yields

$$\begin{aligned}
 b'(v_i)F(v_i) + b(v_i)f(v_i) &= v_i f(v_i) \\
 \frac{d}{dv_i} b(v_i)F(v_i) &= v_i f(v_i) \\
 \int_0^v \frac{d}{dv_i} b(v_i)F(v_i) dv &= \int_0^v v_i f(v_i) dv \\
 b(v_i)F(v_i)|_0^v &= \int_0^v v_i f(v_i) dv_i \\
 b(v)F(v) &= v_i F(v_i)|_0^v - \int_0^v F(v_i) dv_i \\
 b(v)F(v) &= vF(v) - \int_0^v F(v_i) dv_i \\
 b(v) &= v - \frac{1}{F(v)} \int_0^v F(w) dw.
 \end{aligned}$$

When we divide by $F(v)$ in the last line, there is a problem if $v = 0$; we can assume/show that an agent must bid 0 if she has value 0.

For later use, we compute that

$$b'(v) = 1 + \frac{f(v)}{F(v)^2} \int_0^v F(w) dw - \frac{1}{F(v)} F(v) = \frac{f(v)}{F(v)^2} \int_0^v F(w) dw.$$

If the conjectured restrictions on the equilibrium hold, then we have one candidate function. By the assumption that b is increasing, expected revenue depends on the distribution of the higher value, which we find to be characterized by

$$F_{(1)}(w) = F(w)^2, \quad f_{(1)}(w) = 2F(w)f(w).$$

Hence, expected revenue is

$$\begin{aligned}
 R_{\text{FPA}} &= \int_0^1 b(w)f_{(1)}(w) dw \\
 &= b(w)F_{(1)}(w)|_0^1 - \int_0^1 b'(w)F_{(1)}(w) dw \\
 &= b(1) - \int_0^1 \frac{f(w)}{F(w)^2} \left(\int_0^w F(z) dz \right) \underbrace{F_{(1)}(w)}_{=F(w)^2} dw \\
 &= \overbrace{1 - \int_0^1 F(w) dw}^{b(1)} - \int_0^1 f(w) \left(\int_0^w F(z) dz \right) dw \\
 &= 1 - \int_0^1 F(w) dw - \left[F(v) \int_0^1 F(w) dw \right]_{v=0}^{v=1} + \int_0^1 F(v)^2 \\
 &= 1 - 2 \int_0^1 F(w) dw + \int_0^1 F(w)^2 dw \\
 &= 1 - \int_0^1 F_{(2)}(w) dw = R_{\text{SPA}}.
 \end{aligned}$$



5.3 Revelation Principle

The Revelation Principle greatly simplifies analysis of equilibria of games.

Definition 5.3.1 (Direct mechanism).

A direct mechanism is a game where $\mathcal{A}_i = V$ so that agents report a value v .

Proposition 5.3.2 (Revelation Principle).

For any G and equilibrium s , there exists a direct mechanism in which it is a Bayes–Nash equilibrium for agents to truthfully report such that the outcome (set of allocation probabilities and transfer payments) is the same under G, s and under the direct mechanism.

Proof of Proposition 5.3.2. Let $G := ((\mathcal{A}_i)_{i \in N}, x, p)$, fix an equilibrium strategy profile $s := (s_i)_{i \in N}$.

Consider the direct mechanism $M := ((V)_{i \in N}, x', p')$ with

$$\begin{aligned} x'(v_1, \dots, v_{|N|}) &= x(s_1(v_1), \dots, s_{|N|}(v_{|N|})) \\ p'(v_1, \dots, v_{|N|}) &= p(s_1(v_1), \dots, s_{|N|}(v_{|N|})). \end{aligned}$$

Then M has an equilibrium in which all agents truthfully report — a unilateral deviation corresponds to misplaying $s_i(v_i)$ as $s_i(v'_i)$ in G , which was shown to not be optimal. 🐘

5.4 Revenue-Maximizing Auctions

Consider a seller who wants to design a game G to maximize revenue with the following structure. For each $i \in N$, there is an action set $\mathcal{A}_i$. We denote the action profile $\mathcal{A} := \prod_{i \in N} \mathcal{A}_i$. The components of the game the seller must design are

$$\begin{aligned} x : \mathcal{A} &\rightarrow A \\ p : \mathcal{A} &\rightarrow \mathbb{R}^{|N|}. \end{aligned}$$

Each agent $i \in N$ comes up with a strategy $s_i : V \rightarrow \Delta \mathcal{A}_i$. Note that we are still assuming agent i has value $v_i \stackrel{iid}{\sim} F$ in $[0, 1]$. Assume the profile $s := (s_i)_{i \in N}$ is a Bayes–Nash equilibrium of the game G .

By the Revelation Principle, for any equilibrium of an auction, there exists a direct mechanism with truthful reporting as a Bayes–Nash equilibrium such that the allocation probabilities and payments are the same under the two mechanisms. In particular, the expected revenues will be equal, so when trying to determine the maximum achievable revenue, it suffices to consider the maximum attainable revenue under truth-telling direct mechanisms; we can safely assume that the reported value s_i is exactly v_i .

Fixing $(s_i)_{i \in N} = (v_i)_{i \in N}$, let's focus on one agent i and drop subscripts. Let i 's interim allocation $X(v)$ be the probability that i gets the good when she reports value v . Let $P(v)$ be i 's expected payment when she reports value v . In this case, i 's expected utility is $vX(v) - P(v)$, so under our assumption of $(s_i)_{i \in N} = (v_i)_{i \in N}$ being an equilibrium,

$$\begin{aligned} vX(v) - P(v) &\geq vX(v') - P(v') \\ v'X(v') - P(v') &\geq v'X(v) - P(v). \end{aligned}$$

In the direct mechanism, a unilateral deviation is just a different report v'_i , which induces the original-game action $s_i(v'_i)$. The revelation principle does not compare against arbitrary alternative strategies s'_i in the direct mechanism; those deviations were already covered by the assumption that s is an equilibrium of the original game.

We only consider the strategic-normal form.

Here, just think of $\mathcal{A}_i := V_i = [0, 1]$ as the possible set of values for agent i to report. This is because we can appeal to the Revelation Principle, which follows in two paragraphs.

Capital X and P denote expectations of win probabilities and payments when an agent knows her value but not those of others. We call them *interim* functions because they are not ex-ante (before an individual learns her own value) nor ex-post (after an individual learns everyone's values).

for every potential misreport $v' \in V := [0, 1]$. Rearranging, these lead to

$$\begin{aligned} v(X(v) - X(v')) &\geq P(v) - P(v') \\ v'(X(v') - X(v)) &\geq P(v') - P(v) \\ \iff v'(X(v) - X(v')) &\leq P(v) - P(v'), \end{aligned}$$

which imply the *payment difference sandwich*:

$$v'(X(v) - X(v')) \leq P(v) - P(v') \leq v(X(v) - X(v')) \quad (5.1)$$

We will derive several properties from Equation (5.1).

We first characterize the interim assignment probabilities and payments of a Bayes–Nash equilibrium. These are *necessary* conditions for *all* equilibria of auctions, and not just revenue-maximizing ones (though these are really the ones we care about the most).

Thinking ahead, if we're hoping to find probability assignments and a payment structure that is revenue-maximizing in an equilibrium, it must at least satisfy these conditions. Corollary (5.4.3) adds a further restriction for when we specialize to an individually-rational auction that is actually revenue-maximizing.

Proposition 5.4.1.

If s is a Bayes–Nash equilibrium of an auction game G , then the induced interim allocation probabilities X_i and payments P_i satisfy the following:

- (i) The interim allocation probability $X(\cdot)$ is nondecreasing in the report.
- (ii) For any v ,

$$P(v) = P(0) + vX(v) - \int_0^v X(t)dt.$$

The only choices that the seller has are in controlling $P(0)$ and $X(v)$ for each $v \in [0, 1]$. The payment at any value is then uniquely determined.

Proof of Proposition 5.4.1. (i) We can rearrange Equation (5.1) to get

$$(v - v')(X(v) - X(v')) \geq 0.$$

(ii) Let

$$0 = v_0 < v_1 < \cdots < v_k < v_{k+1} = v.$$

Then

$$P(v) - P(0) = P(v) - P(v_k) + P(v_k) - P(v_{k-1}) + \cdots + P(v_1) - P(0).$$

Applying the payment difference sandwich to each term, we get that

$$\sum_{j=1}^k v_j (X(v_{j+1}) - X(v_j)) \leq P(v) - P(0) \leq \sum_{j=0}^k v_{j+1} (X(v_{j+1}) - X(v_j)).$$

Note that the sums look like approximations to the Riemann integral of the area to the left of the interim allocation function using lower and upper sums, respectively. However, if there is a jump discontinuity in X , then its inverse is not well-defined and the area of one of these rectangles may not exist if v_j is not in the image of X .

Equivalently, express the left-of-curve sums as the total rectangle $vX(v)$ minus upper

and lower sums for the area under $X(\cdot)$.

$$\begin{aligned}
 \sum_{j=0}^k v_j (X(v_{j+1}) - X(v_j)) &= vX(v) - vX(v) + \sum_{j=0}^k v_j (X(v_{j+1}) - X(v_j)) \\
 &= vX(v) - v_{k+1}(X(v_{k+1})) + \sum_{j=1}^k v_j X(v_{j+1}) - \underbrace{\sum_{j=0}^k v_j X(v_j)}_{=\sum_{j=0}^{k-1} v_{j+1} X(v_{j+1})} \\
 &= vX(v) - \sum_{j=0}^k (v_{j+1} - v_j) X(v_{j+1}).
 \end{aligned}$$

We can derive a symmetric expression for the other side of the payment difference sandwich, then combine both sides to get

$$vX(v) - \sum_{j=0}^k (v_{j+1} - v_j) X(v_{j+1}) \leq P(v) - P(0) \leq vX(v) - \sum_{j=0}^k (v_{j+1} - v_j) X(v_j).$$

Since our choice of the partition $\{v_i\}_{i=0}^{k+1}$ of $[0, 1]$ was arbitrary, we can send $k \rightarrow \infty$ to get

$$vX(v) - \int_0^v X(t)dt \leq P(v) - P(0) \leq vX(v) - \int_0^v X(t)dt,$$

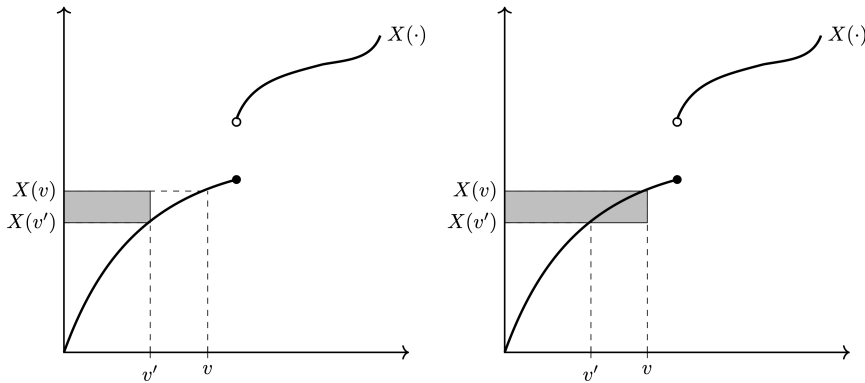
so

$$P(v) - P(0) = vX(v) - \int_0^v X(t)dt.$$



Intuition for the proof of property (ii) is displayed in Figure (5.1).

Figure 5.1: Payment difference sandwich.



Notes: The second approximation to the Riemann integral: summing along values of $X(\cdot)$.

Definition 5.4.2 (Individual rationality).

An auction game G with equilibrium s is individually rational if for each v ,

$$vX(v) - P(v) \geq 0.$$

(Interim) individual rationality is a condition that lies somewhere in between ex-ante and ex-post rationality.

We see immediately that $P(0) \leq 0$ is required for individual rationality, while result (ii) of Proposition (5.4.1) implies $P(0)$ should be as large as possible for revenue-maximization. Hence, among individually rational games, a revenue-maximizing payment schedule must have $P(0) = 0$.

Corollary 5.4.3 (A revenue-maximizing auction's payments).

A revenue-maximizing auction satisfying individual rationality has payment schedule

$$P(v) = vX(v) - \int_0^v X(t)dt.$$

This payment schedule and $X_i(\cdot)$ being nondecreasing (from Proposition 5.4.1) are necessary conditions for an auction to be revenue-maximizing. Note that we have not yet determined what exactly $X(\cdot)$ is in the revenue-maximizing auction.

We have now explained why the first-price, second-price, and all-pay auctions all raise the same expected revenue: they all share $X(v)$ as the probability of v being the highest value. So they must share the same payment schedule, and hence expected revenue as well.

Proposition 5.4.4.

For any G, G' and equilibria s, s' such that X_G and $X_{G'}$ are the same and $P_G(0) = P_{G'}(0)$, the revenue is the same.

We did *not* show that these auctions are revenue-maximizing. (They aren't!)

Proof of Proposition 5.4.4. Immediate from Proposition (5.4.1). 

So far, we fixed one agent at a time and derived necessary interim conditions for revenue maximization in an individually rational auction. We now need to construct an ex-post allocation rule and ex-post payment rule that take the entire value profile $v = (v_i)_{i \in N}$ as input and generate interim objects satisfying those necessary conditions. Under regularity, the virtual value auction provides such a construction.

Definition 5.4.5 (Regular value distribution, virtual value).

A distribution with CDF F and density f is regular if the virtual value

$$c(w) := w - \frac{1 - F(w)}{f(w)}$$

is strictly increasing.

Definition 5.4.6 (Virtual value auction).

In the virtual value auction, all buyers submit their values for the good. Using F , the seller converts these to virtual values. If no virtual value is positive, the seller keeps the good. Otherwise, she gives the good to the agent with the highest value. If there

are multiple such agents, then each gets the good with equal probability. Buyers pay the lowest value they could have submitted and still won the auction with, i.e. the second-highest value after excluding duplicates.

Theorem 5.4.7 (Revenue maximization of the virtual value auction).

Consider individually rational auctions with a single good and symmetric bidders whose values are drawn iid from a regular distribution F . One^a revenue-maximizing auction has assignment probabilities for each bidder i given by

$$x_i(v_i, v_{-i}) = \begin{cases} 0, & c(v_i) \leq 0 \text{ or } c(v_i) < c(v_j) \text{ for some } j \\ \frac{1}{m}, & m := |\{j : v_j = v_i\}| \end{cases}.$$

Moreover, the direct mechanism with this allocation rule and payments

$$p_i(v_i, v_{-i}) = v_i x_i(v_i, v_{-i}) - \int_0^{v_i} x_i(t, v_{-i}) dt$$

is a revenue-maximizing virtual value auction.

^aAny revenue-maximizing auction must have $x_i(v_i, v_{-i}) = 0$ for any i where $c(v_i) \leq 0$ or $c(v_i) < c(v_j)$ for some j .

Proof of Theorem 5.4.7. Without loss, use a game with $\mathcal{V}_i = [0, 1]$. Fix functions $x : V^{|N|} \rightarrow A$, $p : V^{|N|} \rightarrow \mathbb{R}^{|N|}$ with $V := [0, 1]$. Write

$$f_N(v) := \prod_{j \in N} f(v_j), \quad f_{-i}(v_{-i}) := \prod_{j \neq i} f(v_j)$$

for the joint densities of all values and of the values other than i 's.

Define the interim allocations and payments

$$\begin{aligned} X_i(v_i) &= \int_{v_{-i}} x_i(v_i, v_{-i}) f_{-i}(v_{-i}) dv_{-i} \\ P_i(v_i) &= \int_{v_{-i}} p_i(v_i, v_{-i}) f_{-i}(v_{-i}) dv_{-i}. \end{aligned}$$

We know that $X_i(\cdot)$ is nondecreasing.

The seller's problem is to maximize her ex-ante revenue with respect to p_i while selecting

compatible x_i :

$$\begin{aligned}
R &= \max \sum_{i \in N} \int_{(v_i, v_{-i})} p_i(v_i, v_{-i}) f_N(v_i, v_{-i}) d(v_i, v_{-i}) \\
&= \max \int_{(v_i, v_{-i})} \left(\sum_{i \in N} p_i(v_i, v_{-i}) \right) f_N(v_i, v_{-i}) d(v_i, v_{-i}) \\
&= \max \sum_{i \in N} \int_0^1 P_i(v_i) f(v_i) dv_i \quad \text{by Corollary (5.4.3)} \\
&= \max \sum_{i \in N} \left[\int_0^1 v_i X_i(v_i) f(v_i) dv_i - \underbrace{\int_0^1 \int_0^{v_i} X_i(t) dt f(v_i) dv_i}_{(II_i)} \right].
\end{aligned}$$

In this step, we relate the seller's ex-ante expected revenue to an individual's interim expected payment. This is not just because we have the identity for the interim payment from Corollary (5.4.3), but also because interim objects are what agents have strategies over.

We can work on the second term separately:

$$\begin{aligned}
(II_i) &= \int_0^1 f(v_i) \left(\int_0^{v_i} X_i(t) dt \right) dv_i \\
&= F(z) \int_0^z X_i(t) dt \Big|_{z=0}^1 - \int_0^1 F(v_i) X_i(v_i) dv_i \\
&= \int_0^1 X_i(t) dt - \int_0^1 F(v_i) X_i(v_i) dv_i \\
&= \int_0^1 (1 - F(v_i)) X_i(v_i) dv_i
\end{aligned}$$

Substituting (II_i) back in,

$$\begin{aligned}
R &= \max \sum_{i \in N} \int_0^1 \left(\overbrace{v_i - \frac{1 - F(v_i)}{f(v_i)}}^{\text{virtual value}} \right) X_i(v_i) f(v_i) dv_i \\
&= \max \int_{(v_i, v_{-i})} \sum_{i \in N} \left(\underbrace{v_i - \frac{1 - F(v_i)}{f(v_i)}}_{:=c(v_i)} \right) x_i(v_i, v_{-i}) f_N(v_i, v_{-i}) dv.
\end{aligned}$$

This line tells us that for a given value profile of agent i , the expected amount the seller earns from i is the virtual value c_i times the interim assignment probability, which does not condition on others'.

This line tells us the expected amount the seller earns from i is exactly the virtual value c_i times the probability that i gets the good under a given value profile of $(v_1, \dots, v_{|N|})$.

With a fixed value profile v , then R is maximized by setting

$$x_i(v_i, v_{-i}) = \begin{cases} 0, & c(v_i) \leq 0 \text{ or } c(v_i) < c(v_j) \text{ for } j \neq i \\ \frac{1}{m}, & \text{otherwise, where } m := |\{j : v_j = \max_i \{v_i\}\}| \end{cases}. \quad (5.2)$$

We have derived a candidate function for the win probabilities. If an auction is revenue-maximizing, then it must use these ex-post assignment probabilities.²

Recall from Corollary (5.4.3) that there are two more necessary conditions for an auction to be revenue-maximizing and individually rational:

(i) The interim win probabilities X_i are nondecreasing, and

²At least, it must have zero assignment probability for agents with negative virtual values or virtual values that are not the highest.

(ii) The interim payments must satisfy

$$P_i(v_i) = v_i X_i(v_i) - \int_0^{v_i} X_i(t) dt. \quad (5.3)$$

We check the first: because F is regular, we know that $X_i(\cdot)$ is nondecreasing; a higher value leads to a higher virtual value, which $X_i(\cdot)$ is increasing in. In particular, the good is allocated to an agent with the highest value.

Now we need to determine the payments in a way that will satisfy the second condition. We can easily get payments that satisfy the interim payments' characterizations by putting

$$p_i(v_i, v_{-i}) := v_i x_i(v_i, v_{-i}) - \int_0^{v_i} x_i(t, v_{-i}) dt,$$

as integrating both sides recovers Equation (5.3).

The candidate optimum assignment probability in Equation (5.2) is currently written in terms of the virtual values; let's rewrite it in terms of the actual values. Let $\underline{v}$ be the unique value such that $c(\underline{v}) = 0$, or 0 otherwise. Put $s := \max\{\underline{v}, v_j\}_{j \neq i}$.

We have three cases.

Case 1 : If $v_i < s$, then agent i pays

$$v_i \cdot 0 - \underbrace{\int_0^{v_i} x_i(t, v_{-i}) dt}_{=0} = 0.$$

Case 2 : If $v_i > s$, then agent i pays

$$v_i \cdot 1 - \int_0^{v_i} x_i(t, v_{-i}) dt = v_i - v_i + s = \max\{\underline{v}, v_j\}_{j \neq i}.$$

Case 3 : If $v_i = s \neq \underline{v}$ and m people have the highest value, then agent i pays

$$v_i \cdot \frac{1}{m} - \underbrace{\int_0^{v_i} x_i(t, v_{-i}) dt}_{=0} = \frac{v_i}{m}.$$

The ex-post assignment probabilities x_i were the only candidates for an optimal auction, while the payments p_i just needed to satisfy the condition on the interim payments P_i given by Corollary (5.4.3). It is simple to show that the auction defined by $(x_i, p_i)_{i \in N}$ is incentive-compatible, and thus constitutes an equilibrium (in which everyone reports their true values). 

Remark 5.4.8. There are several nice properties of our solution:

- (i) We set out to find a Bayes–Nash equilibrium, but got an auction that also is implementable in dominant strategies. This is a subclass of Bayes–Nash equilibria in which universal truth telling is the specific equilibrium.
- (ii) We just wanted interim individual rationality, but ended with ex-post individual rationality as well.

We know that $\underline{v}$ must be unique because we assumed F is regular, and so c is strictly increasing.

We see that if there is a single winner, as in Case 2, then she pays according to a second-price auction, unless the second-highest value is below the reserve value, in which case she pays the seller's reserve value.

- (iii) If $|N| = 1$, then there is a posted price at $\underline{v}$, which aligns exactly with our finding from Proposition (5.1.2).



5.5 Multiple Goods and a Single Buyer

Now we consider when the seller has two goods to sell and there is a single buyer with values v^1 and v^2 . If the buyer has probability x^1 of getting good 1 and probability x^2 of getting good 2 with payment p , then the interim expected utility is defined to be

$$v^1 x^1 + v^2 x^2 - p.$$

Assume $v^1 \sim F^1$ and $v^2 \sim F^2$ independently.

This structure on the expected utility imposes that the utility function is separable in the two goods.

Proposition 5.5.1.

In a two-good, single buyer auction with independent value distributions $F^1, F^2 \sim \text{Unif}(0, 1)$, the optimal mechanism is one in which the buyer has the options to:

- (i) Buy either good at price $\frac{3}{4}$, or
- (ii) Buy both goods at price $\frac{4-\sqrt{2}}{3}$.

Proof of Proposition 5.5.1. We omit the proof.



More generally, a mechanism can offer a menu of the form

$$\{(z^1, z^2, z^{12}), p\},$$

where p is the payment required for the agent to receive assignment probabilities z^1, z^2, z^{12} .

Proposition 5.5.2.

If both F^1 and F^2 are Beta distributions, the optimal mechanism offers an uncountably large menu.

For intuition on why bundling the two goods together is more profitable for the seller than keeping them separate, we know that if the goods were to be sold individually, we would have two posted price auctions at price

$$p^* = \frac{1 - F(p^*)}{f(p^*)} = \frac{1}{2}.$$

Then the bundle purchased is

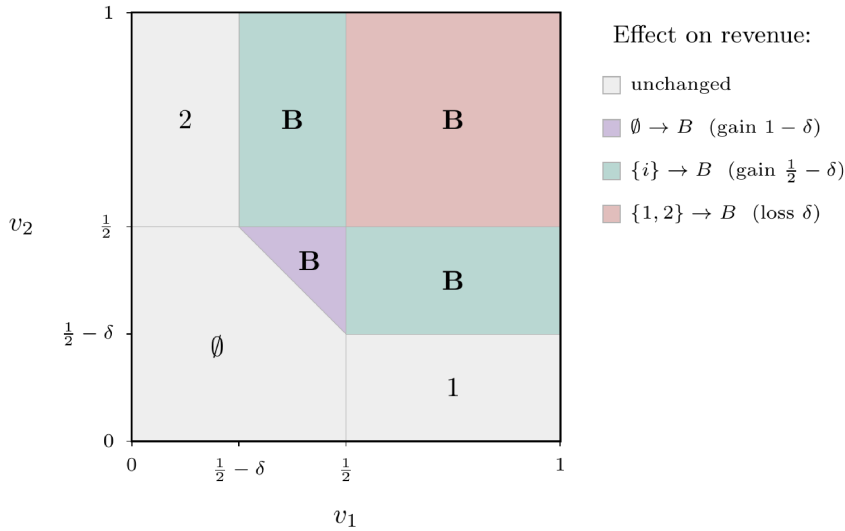
$$\begin{cases} \emptyset, & v_1, v_2 < \frac{1}{2} \\ \{1\}, & v_2 < \frac{1}{2} \leq v_1 \\ \{2\}, & v_1 < \frac{1}{2} \leq v_2 \\ \{1, 2\}, & \frac{1}{2} \leq v_1, v_2 \end{cases}.$$

When bundling with the option to buy $\{1, 2\}$ at discount of δ (so the price of the bundle is $p := 1 - \delta$), an agent who buys the bundle will satisfy:

- (i) $v_1 + v_2 > 1 - \delta$
- (ii) $v_1 + v_2 - p > v_1 - \frac{1}{2} \iff v_2 > p - \frac{1}{2}$
- (iii) $v_1 + v_2 - p > v_2 - \frac{1}{2} \iff v_1 > p - \frac{1}{2}$.

These define regions in the joint value distribution displayed below.

Figure 5.2: The effect of introducing a bundle at price $p := 1 - \delta$ on revenue.



Notes: The original region of both-object-buyers was the red square, while we have now added the other shaded regions.

The total change in revenue from the green regions is $2 \left(\delta \frac{1}{2} \right) \left(1 - \delta - \frac{1}{2} \right) = \delta \left(\frac{1}{2} - \delta \right)$. The change from the purple region is $\frac{1}{2} \delta^2 (1 - \delta)$, and the change from the red region is $-\frac{1}{4} \delta$. Hence, the total change in revenue is

$$\Delta = \delta \left(\frac{1}{2} - \delta \right) + \frac{\delta^2}{2} (1 - \delta) - \frac{1}{4} \delta = \frac{\delta}{2} - \delta^2 + \frac{\delta^2}{2} - \frac{\delta^3}{2} - \frac{1}{4} \delta,$$

which is close to $\frac{\delta}{4}$ when δ is close to zero. Note that when $\delta = 0$, we recover the individual auction case, but $\partial_\delta \Delta|_{\delta=0} \Delta > 0$, so it would be profitable to bundle at a discount.

6.1	Mussa–Rosen (1978)	47
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This section provides a short application of our development of the optimal auction.

6.1 Mussa–Rosen (1978)

Suppose there is a single buyer, single seller (also called a monopolist), and single good. The seller can control the good's quality $q := q(\theta) \in [0, \bar{q}]$, but must pay a cost $C(q)$ for producing quality q . The cost is given to the seller, and we assume that

$$C(0) = 0, \quad C(q) \geq 0, \quad C'(q) > 0, \quad C''(q) > 0, \quad \forall q.$$

Let the buyer have value $\theta \in [0, \bar{\theta}]$ with CDF F and density f .

If the buyer has type θ , buys a good of quality q , and pays p , her utility is defined to be $\theta q - p$. The seller gets $p - C(q)$. The primitive is the buyer's value θ (both true and reported), and the seller chooses both the quality rule $q(\cdot)$ and payment rule $p(\cdot)$. How should she do so to maximize expected revenue?

The seller commits ex ante to a direct mechanism: after the buyer reports a type in $[0, \bar{\theta}]$, the mechanism assigns a quality level $q(\theta)$ and payment $p(\theta)$.

Proposition 6.1.1.

Assume F is regular, so its virtual value

$$c(\theta) := \theta - \frac{1 - F(\theta)}{f(\theta)}$$

is strictly increasing. Then an individually rational revenue-maximizing direct mechanism has quality rule

$$q^*(\theta) = \begin{cases} 0, & c(\theta) \leq C'(0) \\ (C')^{-1}(c(\theta)), & C'(0) < c(\theta) < C'(\bar{q}) \\ \bar{q}, & c(\theta) \geq C'(\bar{q}) \end{cases}$$

and payment rule

$$p^*(\theta) = \theta q^*(\theta) - \int_0^\theta q^*(t) dt.$$

In particular, q^* is nondecreasing.

Proof of Proposition 6.1.1. By the Revelation Principle, we can restrict attention to an incentive-compatible direct mechanism without loss, so the seller chooses functions

$$q : [0, \bar{\theta}] \rightarrow [0, \bar{q}] \quad \text{and} \quad p : [0, \bar{\theta}] \rightarrow \mathbb{R}.$$

The previous section applies with quality playing the role of the one-dimensional allocation

object: instead of a win probability $X(\theta)$, a report θ now determines a deterministic quality level $q(\theta)$, and utility still has the quasilinear form “type times allocation minus payment.” Therefore incentive compatibility and revenue maximization implies that q is nondecreasing and that

$$p(\theta) = \theta q(\theta) - \int_0^\theta q(t) dt.$$

Expected profit is

$$\begin{aligned} \int_0^{\bar{\theta}} [p(\theta) - C(q(\theta))] f(\theta) d\theta &= \int_0^{\bar{\theta}} \left[\theta q(\theta) - \int_0^\theta q(t) dt - C(q(\theta)) \right] f(\theta) d\theta \\ &= \int_0^{\bar{\theta}} [\theta q(\theta) - C(q(\theta))] f(\theta) d\theta - \int_0^{\bar{\theta}} \int_0^\theta q(t) dt f(\theta) d\theta \\ &= \int_0^{\bar{\theta}} [\theta q(\theta) - C(q(\theta))] f(\theta) d\theta - \int_0^{\bar{\theta}} (1 - F(\theta)) q(\theta) d\theta \\ &= \int_0^{\bar{\theta}} \left[\left(\theta - \frac{1 - F(\theta)}{f(\theta)} \right) q(\theta) - C(q(\theta)) \right] f(\theta) d\theta \\ &= \int_0^{\bar{\theta}} [c(\theta) q(\theta) - C(q(\theta))] f(\theta) d\theta \end{aligned}$$

where the third line uses integration by parts, or equivalently Fubini’s theorem.

For each fixed θ , the choice $q(\theta)$ enters the objective only through the bracketed term at that same θ , and $f(\theta)$ is a positive weight. Thus we can maximize pointwise:

$$\max_{q \in [0, \bar{q}]} \{c(\theta)q - C(q)\}.$$

Since C is strictly convex, the maximizer is unique. If $c(\theta) \leq C'(0)$, then the objective is decreasing at $q = 0$, so the maximizer is $q^*(\theta) = 0$. If $C'(0) < c(\theta) < C'(\bar{q})$, the unique interior maximizer satisfies

$$C'(q^*(\theta)) = c(\theta),$$

or equivalently

$$q^*(\theta) = (C')^{-1}(c(\theta)).$$

If instead $c(\theta) \geq C'(\bar{q})$, the objective is still increasing at $q = \bar{q}$, so the constrained maximizer is $q^*(\theta) = \bar{q}$. Because c is increasing and the casewise rule above is itself nondecreasing in $c(\theta)$, the resulting quality schedule q^* is nondecreasing. The payment formula then follows from the envelope formula above. 

How does q^* of the monopolist (seller) compare to the quality that a benevolent social planner picks? The BSP selects q to maximize the sum of utilities $(q\theta - p) + (p - C(q)) = q\theta - C(q)$. The FOC is $C'(q^{FB}(\theta)) = \theta$. It turns out that the seller selects a lower quality for every value $\theta \in [0, \bar{\theta}]$. Indeed, the planner’s quality solves $C'(q^{FB}(\theta)) = \theta$, whereas the monopolist’s quality solves $C'(q^*(\theta)) = c(\theta)$ whenever $q^*(\theta) \in (0, \bar{q})$. Since the virtual value cannot exceed the value, i.e. $c(\theta) \leq \theta$, and C' is increasing, we get $q^*(\theta) \leq q^{FB}(\theta)$. If, in addition,

$$\lim_{\theta \uparrow \bar{\theta}} \frac{1 - F(\theta)}{f(\theta)} = 0,$$

In a more heuristic way, we say that the “no distortion at the top” property holds if f is positive enough near $\bar{\theta}$.

then

$$\lim_{\theta \uparrow \bar{\theta}} \left(\theta - \frac{1 - F(\theta)}{f(\theta)} \right) = \bar{\theta}.$$

In other words, as $\theta \uparrow \bar{\theta}$, the revenue-maximizing quality q^* approaches the benevolent planner's quality. This is the “no distortion at the top” property.

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7.1 Introduction

Example 7.1.1.

Consider nearby cities A, B, and C, each of which needs to build a large public project, e.g. an airport or water treatment facility. They could cooperate or build the project individually.

Define the costs for various cities building it individually or jointly as

$$\begin{aligned} c(A) &= 120, & c(B) &= 140, & c(C) &= 120 \\ c(AB) &= 170, & c(AC) &= 160, & c(BC) &= 190 \\ c(ABC) &= 255. \end{aligned}$$

Clearly, since $c(AB) < c(A) + c(B)$, so the planner would prefer cities A and B to work together than to work separately. How much should she charge each city to make up the total cost of 170? A naive approach is to make each city pay 85. Any payment rule in which both cities pay less than they would have individually is sensible. We might also think that city B should pay more because it has a higher individual cost — we can take a savings-based approach, splitting the total savings of $260 - 170 = 90$ evenly, so A pays 75 and B pays 95.

What if we consider a three-way cooperation? Then $c(ABC) = 255$ must satisfy four constraints:

$$\begin{aligned} c(ABC) &< c(A) + c(B) + c(C) \\ c(ABC) &< c(A) + c(BC) \\ c(ABC) &< c(AB) + c(C) \\ c(ABC) &< c(AC) + c(B), \end{aligned}$$

all of which indeed hold. So the cheapest way to implement the project is to fund it together.

Now, how should all three cities split the cost? Doing so evenly leads to everyone paying 85, but then A and C are collectively paying more than $c(AC)$. The savings-based approach leads to everyone getting savings of $\frac{125}{3}$ over their individual costs. In this case, A pays $78\frac{1}{3}$, B pays $98\frac{1}{3}$, and C pays $78\frac{1}{3}$ — now A and B are collectively paying more than $c(AB) = 170$!



We just had three agents, but the number of comparisons we must make is dizzying.

Let's develop a framework for characterizing solutions in which agents are happy at the three-way joint outcome.

Let x_A, x_B, x_C be the amounts paid by A, B, and C respectively. Then we require the *no subsidy constraints*:

$$\begin{aligned} x_A &\leq 120, & x_B &\leq 140, & x_C &\leq 120, \\ x_A + x_B &\leq 170, & x_A + x_C &\leq 160, & x_B + x_C &\leq 190, \\ x_A + x_B + x_C &= 255. \end{aligned}$$

We can define the no subsidy constraints in a general situation by replacing the numbers with the relevant costs $c(A), c(AB)$, and so on.

Define the savings

$$y_A := c(A) - x_A, \quad y_B := c(B) - x_B, \quad y_C := c(C) - x_C.$$

We can define constraints in terms of the savings as

$$\begin{aligned} y_A &\geq 0, & y_B &\geq 0, & y_C &\geq 0, \\ y_A + y_B &\geq 90, & y_B + y_C &\geq 70, & y_A + y_C &\geq 80, \\ y_A + y_B + y_C &= 125. \end{aligned}$$

Now, the numbers are differences of costs:
 $90 = c(A) + c(B) - c(AB)$.

We can solve this system of linear solutions to get that one solution in the feasible set is

$$(y_A, y_B, y_C) = (55, 35, 35).$$

Is it always possible to find such a solution? If we increase the three-way cost to $c(ABC)' = 265$, the new system has the same one-way and two-way savings restrictions, but a different three-way savings constraint:

$$\begin{aligned} y_A &\geq 0, & y_B &\geq 0, & y_C &\geq 0, \\ y_A + y_B &\geq 90, & y_B + y_C &\geq 70, & y_A + y_C &\geq 80, \\ y_A + y_B + y_C &= 115. \end{aligned}$$

But now summing all three two-way constraints gives

$$(y_A + y_B) + (y_B + y_C) + (y_A + y_C) \geq 90 + 70 + 80,$$

which yields

$$2y_A + 2y_B + 2y_C \geq 240,$$

or

$$y_A + y_B + y_C \geq 120,$$

which is incompatible with the new constraint $y_A + y_B + y_C = 115$. So there is no three-way cooperative outcome that is stable, despite the three-way cooperative outcome being the best (cheapest).

7.2 Cooperative Games with Transferable Utility

Definition 7.2.1 (Cooperative game with transferable utility).

A cooperative game with transferable utility is a pair (N, v) , where N is a set of agents and

$$v : 2^N \setminus \emptyset \rightarrow \mathbb{R},$$

where $v(S)$ is the surplus associated with every nonempty coalition S .

The game has *transferable* utility because there is a single utility for the coalition, not a vector of individual utilities for each member of the coalition.

Definition 7.2.2 (Allocation).

An allocation is a vector $x \in \mathbb{R}^n$ such that

$$\sum_{i \in N} x_i = v(N).$$

The allocation specifies the amount of surplus that each agent gets when the coalition includes all agents.

In Example (7.1.1), the value $v(S)$ was the total savings of coalition S .

Definition 7.2.3 (Core).

The core of (N, v) is the set of all allocations such that the sum of the allocation components within the coalition is larger than the surplus generated from the coalition:

$$\left\{ \text{allocations } x \in \mathbb{R}^n : \text{for all nonempty } S \subsetneq N, \sum_{i \in S} x_i \geq v(S) \right\}.$$

Alternatively and more clearly, it is

$$\left\{ x \in \mathbb{R}^n : \text{for all nonempty } S \subsetneq N, \sum_{i \in S} x_i \geq v(S) \text{ and } \sum_{i \in N} x_i = v(N) \right\}.$$

Consider $N = \{1, 2, 3\}$. Then the core can be written as all $x \in \mathbb{R}^3$ such that

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \geq \begin{pmatrix} v(\{1\}) \\ v(\{2\}) \\ v(\{3\}) \\ v(\{1, 2\}) \\ v(\{1, 3\}) \\ v(\{2, 3\}) \end{pmatrix}, \quad (1 \quad 1 \quad 1) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = v(\{1, 2, 3\}).$$

Before moving to the general case, we need a new definition.

Definition 7.2.4 (Balanced weights).

A set of balanced weights is a function $\delta : 2^N \setminus \{\emptyset, N\} \rightarrow [0, 1]$ such that for every agent $i \in N$,

$$\sum_{S: i \in S} \delta(S) = 1.$$

A set of balanced weights gives a weight to every nonsilly¹ coalition such that when each agent sums the weights of the coalitions she is in, the sum is 1. While a usual coalition structure partitions agents so each agent is in exactly one coalition, a set of balanced weights allows each agent to be in multiple coalitions (excluding the empty and grand coalitions), so long as her coalition assignments have total weight one. Furthermore, because $\delta(S)$ does not depend on i , given a coalition, everyone must have the same weight (spend the same time) in that coalition.

Theorem 7.2.5 (Bonderava–Shapley).

A cooperative game with transferable utility (N, v) has a nonempty core if and only if for every set of balanced weights δ ,

$$\sum_{S \subseteq N | S \neq \emptyset} \delta(S)v(S) \leq v(N).$$

Bonderava–Shapley says that the core is nonempty iff for every artificial coalition arrangement, the total value generated by summing across all coalitions does not exceed the grand coalition's value.

Proof of Theorem 7.2.5. Fix an arbitrary N .

To write the constraints that define the core as a linear system, define $A \in \{0, 1\}^{(2^{|N|}-2) \times |N|}$ by

$$A_{S,i} = \begin{cases} 1, & i \in S \\ 0, & \text{otherwise} \end{cases},$$

where A has one row for each nonempty proper subset of N (i.e. every possible nonsilly coalition) and one column for each agent. Let $\vec{v} \in \mathbb{R}^{2^{|N|}-2}$ have in its coordinate with index S the surplus $v(S)$.

Now we can characterize the core by

$$\{x \in \mathbb{R}^n \mid Ax \geq \vec{v}, \mathbf{1} \cdot x = v(N)\}.$$

Note that $\mathbf{1}$ is a row vector.
So later, $\mathbf{1}'$ is a column vector.

Negating the inequality, Farkas's Lemma (A.3.3) says that the core is nonempty iff there is no row matrix $p \geq 0$ and $q \in \mathbb{R}$ such that

$$\begin{aligned} p(-A) + q\mathbf{1}' &= 0 \\ p(-\vec{v}) + qv(N) &< 0. \end{aligned}$$

If the system has a solution, then $q \neq 0$, as otherwise, we have $p = 0$, which implies from the second equation that $0(-\vec{v}) + 0v(N) = 0 < 0$. Furthermore, $q \geq 0$ because each component of p is nonnegative and each component of A is either 0 or 1. Thus $q > 0$, and

¹Nonempty proper subset.

we can divide both equations by q to get

$$\begin{aligned} \overbrace{\begin{pmatrix} 1 \\ -p \\ q \end{pmatrix}}^{:=\delta} A &= \mathbf{1}' \\ \begin{pmatrix} 1 \\ -p \\ q \end{pmatrix} \vec{v} &> v(N) \\ \begin{pmatrix} 1 \\ -p \\ q \end{pmatrix} &\geq 0. \end{aligned}$$

From the first equation, we see that δ is a set of balanced weights, since for every agent $i \in N$ (corresponding to a column in A and coordinate in the row vector $\mathbf{1}'$),

$$\sum_{S:i \in S} \delta(S) = 1.$$

The second equation translates to the inequality in the proposition statement, and the third requires becomes the definition restriction that weights are nonnegative.

We can think of δ as *artificial coalitions* — time allocations over actual coalitions. 🐘

7.3 Cooperative Games with Non-Transferable Utility

Recall that in games with transferable utility, we had surplus within the coalition of $v(S)$ for every nonempty $S \subseteq N$.

In games with nontransferable utility, we assume that for every nonempty $S \subseteq N$, there is a set of actions A_S that can be taken by the coalition S . Let $A := \cup_S A_S$ be the set of all actions, and assume that each agent i has a weak ranking $\succsim_i$ over her set of actions $\cup_{\{S:i \in S\}} A_S$.

Definition 7.3.1 (Core).

An action $a \in A_N$ is in the core if there is no coalition $S \subseteq N$ and $a' \in A_S$ such that

$$a' \succsim_j a, \forall j \in S \text{ and } a' \succ_k a \text{ for some } k \in S.$$

Answers for when the core exists and on its properties are not very satisfying, so we will look at more specialized examples.

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We now specialize to cases of cooperative games with non-transferable utility. Recall that an action $a \in A_N$ of all agents is in the core if there is no coalition $S \subseteq N$ and $a' \in A_S$ such that a' is a Pareto improvement, i.e.

$$a' \succ_j a, \forall j \in S \text{ and } a' \succ_k a \text{ for some } k \in S.$$

8.1 One-Sided Matching

Consider a set N of agents and a scenario where we want to match the agents into pairs and singletons. Each $i \in N$ has a strict ranking on N .

If $i \succ_i j$, then i would prefer to be unmatched than to match with j .

Definition 8.1.1 (Matching).

A matching is a function $\mu : N \rightarrow N$ such that $\mu \circ \mu = \text{id}_N$. Equivalently, if i is matched with j , then j is matched with i .

Since we want to consider deviations by coalitions S , define A_S to be the set of all matchings among the agents in S , in that

$$A_S = \{\nu : S \rightarrow S \mid \nu \circ \nu = \text{id}_S\}.$$

We can extend agent i 's preference on N to a ranking $\succ_i^*$ on $\cup_{\{S:i \in S\}} A_S$ by letting

$$\nu \succ_i^* \eta \iff \nu(i) \succ_i \eta(i) \text{ or } \nu(i) = \eta(i).$$

There are now indifferences because multiple actions now match agent i with agent j ; the agents i and j are indifferent between these actions.

Definition 8.1.2 (Acceptable, individually rational).

We say that j is acceptable to i if $j \succ_i i$. A matching μ is individually rational if $\mu(i)$ is acceptable for every i such that $\mu(i) \neq i$, i.e. if everyone who is matched weakly prefers his/her partner to him/herself.

Definition 8.1.3 (Stable, blocking pair).

A matching μ is stable if it is individually rational and there is no pair i, j such that $i \neq j$ and

$$i \succ_j \mu(j) \text{ and } j \succ_i \mu(i).$$

A pair (i, j) in which

$$i \succ_j \mu(j) \text{ and } j \succ_i \mu(i)$$

is called a *blocking pair*, as they could form a coalition (blocking the matching μ) and both be better off.

Example 8.1.4.

Consider $N = \{1, 2, 3\}$ with the following preferences:

$$\begin{array}{ccc} \succ_1 & \succ_2 & \succ_3 \\ \hline 2 & 3 & 1 \\ 3 & 1 & 2 \\ 1 & 2 & 3 \end{array} .$$

Then all matchings have a blocking pair.



Proposition 8.1.5.

A matching μ is in the core if and only if it is stable. In particular, to check the core, we just need to check deviations from coalitions of size 2.

Proof of Proposition 8.1.5. The forward direction is immediate from the definition of the core. There is no coalition of any size that can break off and be weakly better off, so there is no coalition of size 2 that can break off and be weakly better off. The agents in any potential coalition of size 2 must be matched to each other in the new matching, so their partners must be different from in the original matching, so if they were weakly better off, they would be strictly better off (by strict preferences on N). μ being in the core gives us that there is no coalition of size 2 where both agents are strictly better off; i.e., there is no blocking pair.

We prove the reverse direction by contraposition. Suppose μ is not in the core, so there exists nonempty $S \subseteq N$ and $\nu \in A_S$ such that for all $j \in S$, $\nu(j) \succ_j^* \mu(j)$, and there exists $i \in S$ such that $\nu(i) \succ_i^* \mu(i)$.

To show nonstability, we need a deviation that makes an agent strictly better off: we must keep i in the pair.

Case 1 : Suppose $\nu(i) = i$. Then $i \succ_i \mu(i)$, so μ is not individually rational and hence not stable.

Case 2 : Suppose $\nu(i) = k \in S$ for $k \neq i$. Since $k \in S$, $\nu(k) \succ_k^* \mu(k)$. But because $\mu(k) \neq i$ (if $\mu(k) = i$, then $\mu(i) = k = \nu(i)$ and i is not actually strictly better off under ν) this preference is strict, so

$$\nu(k) \succ_k^* \mu(k).$$

This implies that (i, k) is a blocking pair at μ , so μ is not stable.



8.2 One-to-One Two-Sided Matching

Suppose $M, W \subseteq N$ are disjoint, nonempty, and $M \cup W = N$. We can view this as a special case of the one-sided matching environment where for all $m \neq m' \in M$ and $w \neq w' \in W$,

$$m' \succ_m m, \quad w' \succ_w w.$$

In other words, each agent would rather be unmatched than be matched to someone from the same group.

Definition 8.2.1 (Gale–Shapley (Deferred Acceptance Algorithm)).

The traditional algorithm is the M -proposing deferred acceptance algorithm. The algorithm proceeds in the following steps:

- (1) Each $m \in M$ proposes to his favorite acceptable $w \in W$. Each $w \in W$ is tentatively matched with her favorite acceptable proposer and rejects everyone else. Call μ_1 this match. If there were no rejections, stop and return the tentative matches.
- (2) In each step $k \geq 2$, each $m \in M$ who was rejected in the previous round proposes to his favorite acceptable w who has not yet rejected him. Each $w \in W$ is tentatively matched with her favorite acceptable m among those who applied in this round and her tentative match from the previous round. All other applicants are rejected. Let μ_k denote this match. If there were no rejections, stop and return the tentative matches.

For any profile of preferences $\succ$, let $DA_M(\succ)$ be the matching returned by the M -proposing deferred acceptance algorithm.

Remark 8.2.2. Note that:

- (i) The welfare of each w is nondecreasing during the course of the algorithm.
- (ii) The algorithm is at most $|M| \cdot (|W| + 1)$ steps.



Example 8.2.3.

Let

$$M := \{m_1, m_2, m_3, m_4\}, W := \{w_1, w_2, w_3, w_4\}.$$

Write out the ordering of *acceptable* matches as

$\succ_{m_1}$	$\succ_{m_2}$	$\succ_{m_3}$	$\succ_{m_4}$	$\succ_{w_1}$	$\succ_{w_2}$	$\succ_{w_3}$	$\succ_{w_4}$
w_1	w_1	w_1	w_1	m_1	m_1	m_3	m_1
	w_2	w_2	w_2		m_4		m_2
		w_3	w_3		m_2		
			w_4		m_3		

In step 1, all m apply to w_1 , who rejects all but m_1 . Then we have the tentative match

$$\mu_1 : (m_1, w_1), m_2, m_3, m_4, w_2, w_3, w_4.$$

In step 2, m_2 , m_3 , and m_4 apply to w_2 , who rejects all but m_4 . Then we have

$$\mu_2 : (m_1, w_1), (m_4, w_2), m_2, m_3, w_3, w_4.$$

In step 3, m_2 would rather be unmatched, so only m_3 proposes to w_3 , and is not rejected, so we have the final matching

$$\mu_3 : (m_1, w_1), (m_4, w_2), (m_3, w_3), m_2, w_4.$$



Proposition 8.2.4.

The M -proposing DA outcome is stable. Hence, the core of the one-to-one two-sided matching problem is nonempty.

Proof of Proposition 8.2.4. Suppose DA results in μ_{DA} . Note that μ_{DA} is individually rational because no agents are ever matched with an unacceptable partner; $m \in M$ only applies to acceptable $w \in W$, and w only accepts m who are acceptable.

For the absence of a blocking pair, suppose that $w \succ_m \mu_{DA}(m)$. Then m proposed to w before he matched with $\mu_{DA}(m)$. Since m and w were not matched, this means that w rejected m at some point. Either w does not find m acceptable, or at the step where w rejected m , w was tentatively matched with an agent m' she prefers to m . Since the welfare of each agent is nondecreasing throughout the algorithm, w ends up with a partner whom she weakly prefers to m' and thus strictly to m , i.e.

$$\mu_{DA}(w) \succeq_w m' \succ_w m.$$

So there is no pair that would prevent μ_{DA} from being stable.



Definition 8.2.5 (Achievable).

We say that $w \in W$ is achievable for $m \in M$ if there exists a stable matching where m and w are matched.

Proposition 8.2.6.

In the M -proposing DA algorithm, each $m \in M$ who is matched is matched with his favorite achievable partner and each $w \in W$ who is matched is matched with her least-favorite achievable partner.

Proof of Proposition 8.2.6. Fix $\succ$ and μ_{DA} . We first show that each m is matched with his favorite achievable partner. Consider the first stage¹ of the DA algorithm at which some $m \in M$ was rejected by an achievable partner. Suppose w rejects m and w is achievable for m with $w \succ_m \mu_{DA}(m)$. Let m' be w 's match at this step of the algorithm. Since this is the first step where any agent in M was rejected by an achievable partner and m' is matched to someone, m' has not yet been rejected by an achievable partner. In particular, m' prefers w to all of his (other²) achievable partners.

w is achievable for m , so let ν be a stable matching in which m and w are matched. Clearly, $m' \succ_w m$, since w rejected m in favor of m' during this stage. But we also have $w \succ_{m'} \nu(m')$. To see why, note that w is matched to m in ν , so $w \neq \nu(m')$. Furthermore,

¹Root calls this the first step, but it is not the literal first step in the DA algorithm.

² w may or may not be achievable.

$\nu(m')$ is an achievable partner for m' by stability, and we saw earlier that m' prefers w to all achievable partners who are not w , including $\nu(m')$. Hence (m', w) is a blocking pair to ν .

Since there is not a first step where an m is rejected by an achievable partner, there is no such step.

It remains to show that every $w \in W$ is matched with her least-favorite achievable partner, i.e. there does not exist $w \in W$ and achievable m such that

$$\mu_{DA}(w) \succ_w m.$$

Suppose that these w and m exist. We know by achievability that there is a stable ν such that $\nu(m) = w$. Denote $m' := \mu_{DA}(w)$. Note $\nu(m') \neq w$, since $\nu(m) = w$ and $m' \neq m$ (as $\mu_{DA}(w) \succ_w m = \nu(w)$ implies $m' := \mu_{DA}(w) \neq \nu(w) = m$). Since m' is matched with w in the M -proposing μ_{DA} , we know that $\mu_{DA}(m') \succ_{m'} \nu(m')$ by the first part of the proof and $\nu(m') \neq w$. But $\mu_{DA}(m') = w$, so $w \succ_{m'} \nu(m')$. Our assumption gives the other side of the blocking pair: $m' = \mu_{DA}(w) \succ_w m = \nu(w)$. Hence (m', w) is a blocking pair for ν . 🐘

Notation 8.2.7.

For each $m \in M$, let P_m be the set of strict preferences on $W \cup \{m\}$. For each $w \in W$, let P_w be the set of strict preferences on $M \cup \{w\}$. Let $\mathcal{P}$ be the set of preference profiles where $\succ_m \in P_m$ and $\succ_w \in P_w$ for every m and w . Denote the set of matchings that never match two agents from the same side as

$$\begin{aligned} \Sigma := \{ \mu : M \cup W \rightarrow M \cup W \mid & \mu \circ \mu = \text{id}_{M \cup W}, \\ & \mu(m) \in W \cup \{m\}, \forall m \in M, \\ & \mu(w) \in M \cup \{w\} \forall w \in W \}. \end{aligned}$$

Definition 8.2.8 (Mechanism).

A one-to-one two-sided matching mechanism is a function $f : \mathcal{P} \rightarrow \Sigma$. It is stable if for every $\succ \in \mathcal{P}$, $f(\succ)$ is stable. In other words, mechanisms are stable if they give stable matchings for every preference profile.

Tentative grades — A: 39+;
A-: [33,39); B+: [27,33); B:
[21,27); B-: [15,21).

Proposition 8.2.9.

There are no stable and strategy-proof one-to-one two-sided matching mechanisms.

Proof of Proposition 8.2.9. Consider the preference profile $\succ$ given by

$$\begin{array}{cccc} \succ_{m_1} & \succ_{m_2} & \succ_{w_1} & \succ_{w_2} \\ w_1 & w_2 & m_2 & m_1 \\ w_2 & w_1 & m_1 & m_2 \end{array}.$$

Then the only stable matchings are $A := \{(m_1, w_1), (m_2, w_2)\}$ and $B := \{(m_1, w_2), (m_2, w_1)\}$. Any matching that leaves agents unpaired is not stable.

Assume that $f(\succ) = A$. If w_1 switches to reporting $\succ'_{w_1}$, which only has m_2 as acceptable, then B is the only stable matching.

If $f(\succ) = B$, then w_2 can misreport only m_1 as acceptable.

In the general case, as long as $|M|, |W| \geq 2$, this problem arises. 🐘

Theorem 8.2.10.

In the mechanism that returns the M -proposing DA outcome for every preference profile, it is a dominant strategy for each $m \in M$ to truthfully report his preference.

The blocking lemma says that when comparing any matching μ to the stable matching of the M -proposing DA mechanism for a given preference, there is at least one $m \in M$ who weakly prefers his match under the DA mechanism. It also says that if $M' \neq \emptyset$, then there is an m outside M' and a w who is matched to someone in M' under μ such that (m, w) blocks μ .

Lemma 8.2.11 (Blocking).

Let μ_{DA} be the outcome of the M -proposing DA mechanism for the preferences $\succ$. Let μ be any other matching (not necessarily stable). Let M' be the set of men who strictly prefer their matches under μ to their matches under μ_{DA} , i.e.

$$M' = \{m \in M : \mu(m) \succ_m \mu_{DA}(m)\}.$$

Then

- (i) $M' \neq M$, and
- (ii) If $M' \neq \emptyset$, then there is an $m \in M \setminus M'$ and a w in $\mu(M')$ such that (m, w) is a blocking pair for μ .

Proof of Lemma 8.2.11. We prove this for when all $m \in M$ and $w \in W$ are mutually acceptable and $|M| = |W|$.

If $M' = \emptyset$, then (i) is immediate and (ii) is vacuous. Thus suppose $M' \neq \emptyset$. It suffices to prove (ii), since the existence of an $m \in M \setminus M'$ will imply $M' \neq M$.

For each $m' \in M'$, $\mu(m')$ must be a woman, since m' strictly prefers $\mu(m')$ to the woman $\mu_{DA}(m')$ and everyone on the opposite side is acceptable. Thus $\mu(M')$ and $\mu_{DA}(M')$ are two sets of women with the same cardinality, namely $|M'|$.

Case 1 : Suppose $\mu(M') \neq \mu_{DA}(M')$. Then we can choose a woman who is matched to someone in M' only under the new matching, with

$$w \in \mu(M') \setminus \mu_{DA}(M').$$

Let $m' := \mu(w)$ be the man in M' who is matched to w under μ , and let $m := \mu_{DA}(w)$ be the man matched to w under μ_{DA} . Since $w \notin \mu_{DA}(M')$, we have $m \notin M'$.

First consider m 's preferences. Since $m \notin M'$, he weakly prefers his DA partner to his partner under μ :

$$\mu_{DA}(m) \succeq_m \mu(m).$$

But $\mu_{DA}(m) = w$. Also, w is matched under μ to $m' \in M'$, so $\mu(m) \neq w$. Preferences are strict, so this weak preference is actually

$$w \succ_m \mu(m).$$

Now consider w 's preferences. If $m' \succ_w m$, then (m', w) would block μ_{DA} : the man $m' \in M'$ strictly prefers $w = \mu(m')$ to $\mu_{DA}(m')$, and w strictly prefers m' to $m = \mu_{DA}(w)$. Since μ_{DA} is stable, this is impossible. Therefore $m \succ_w m' = \mu(w)$. Thus (m, w) is a blocking pair for μ .

The blocking pair will be w and her partner m from the DA matching.

Case 2 : Suppose $\mu(M') = \mu_{DA}(M')$. Then the men in M' are matched to the same set of women under μ and μ_{DA} , but each of them strictly prefers his partner under μ .

Look at the run of the M -proposing DA algorithm that produces μ_{DA} . Let k be the first stage at the end of which every man in M' is tentatively matched to his final partner under μ_{DA} . Choose some $m' \in M'$ who becomes matched to his final partner at stage k , and denote this woman by

$$w := \mu_{DA}(m').$$

Since $\mu(M') = \mu_{DA}(M')$, this same woman w is matched under μ to some man in M' . Call that man

$$m'' := \mu(w).$$

Because $m' \in M'$ strictly prefers $\mu(m')$ to $\mu_{DA}(m') = w$, we have that $\mu(m') \neq w = \mu(m'')$, so $m'' \neq m'$.

Since $m'' \in M'$ and $\mu(m'') = w$, man m'' strictly prefers w to his final DA partner $\mu_{DA}(m'')$. Therefore, during the DA algorithm, m'' must have proposed to w before proposing to $\mu_{DA}(m'')$. He is not matched to w in μ_{DA} , so w rejected him at some point. This rejection happened before step k : if m'' were rejected at stage k , then at the end of step k he would not be tentatively matched to his final DA partner, contradicting the definition of k .

Once a woman has a tentative match in DA, she remains tentatively matched and her partner only improves. Thus, just before step k , woman w was tentatively matched to some man; call him m . At step k , she becomes matched to m' , so she rejects this incumbent m . In particular, $m \notin M'$: otherwise, being rejected at step k would leave m unmatched at the end of step k , again contradicting the definition of k as a particular stage where every man in M' is tentatively matched to his final partner. (The final partner cannot be himself, as all of W is acceptable to m and $|M| = |W|$.)

Now compare preferences. On w 's side, she rejected m'' before she was holding m , and her tentative partner only improves over time, so

$$m \succ_w m'' = \mu(w).$$

On m 's side, he was holding w before being rejected at step k , so he must prefer w to his final DA partner:

$$w \succ_m \mu_{DA}(m).$$

Since $m \notin M'$, he weakly prefers his DA partner to his partner under μ :

$$\mu_{DA}(m) \succeq_m \mu(m).$$

Hence $w \succ_m \mu(m)$. Therefore (m, w) is a blocking pair for μ .



Proof of Theorem (8.2.10). Fix a preference profile $\succ$ and a potential misreport $\succ'_m$. Let μ be the DA- M outcome under $\succ$ and let ν be the DA- M outcome under $(\succ_{m'}, \succ_{-m})$.

Suppose that μ is not strategy-proof for m , so there is another matching where $\nu(m) \succ_m \mu(m)$. m is better off in ν than μ , so by the blocking lemma, there must be a blocking pair (m', w') for ν under $\succ$ such that $M' \not\supseteq m' \neq m \in M'$ and $w' \in \nu(M')$.

We know that ν is stable under $(\succ_{m'}, \succ_{-m})$ because it is the DA- M outcome. But the preferences of m' and w' do not change between $\succ$ and $(\succ'_m, \succ_{-m})$, so (m', w') blocks ν under $(\succ_{m'}, \succ_{-m})$ as well.; ν is not stable under $(\succ_{m'}, \succ_{-m})$, a contradiction. 🐘

8.3 Many-to-One Two-Sided Matching

Suppose each w has an integer-valued capacity $c_w \geq 1$.

Definition 8.3.1 (Matching).

A matching is a function $\mu : M \rightarrow M \cup W$ such that:

- (i) $|\mu^{-1}(w)| \leq c_w$ for every w
- (ii) If $\mu(m) = m' \in M$, then $m = m'$.

Now, each w is matched with a set of agents from M , so we assume that w has a strict preference $\succ_w$ on the set of all subsets of M with cardinality at most c_w : $\succ_w$ is defined on

$$\{M' \subseteq M : |M'| \leq c_w\}.$$

Definition 8.3.2 (Stable, blocking pair).

A matching μ is stable if:

- (i) Each m is matched to an acceptable w or is unmatched, and
- (ii) There is no $M' \subseteq M$ and some w such that $M' \succ_w \{m : \mu(m) = w\}$ and each $m \in M'$ weakly prefers w to his match in μ .

Such a pair (M', w) in (ii) is called a blocking pair.

Example 8.3.3.

Suppose $M = \{m_1, m_2\}$, $W = \{w_1, w_2\}$, $c_{w_1} = 2$, and $c_{w_2} = 1$. Consider

$$\begin{array}{cccc} \tilde{\succ}_{m_1} & \tilde{\succ}_{m_2} & \tilde{\succ}_{w_1} & \succ_{w_2} \\ \hline w_2 & w_1 & \{m_1, m_2\} & \{m_2\} \\ w_1 & w_2 & \{m_1\} & \{m_1\} \\ m_1 & m_2 & \emptyset & \emptyset \end{array}.$$

We investigate stability, where each blocking pair defines the next case.

Case 1 : Consider $\mu(m_1) = m_1$ and $\mu(m_2) = m_2$. There is a blocking pair $(\{m_2\}, w_2)$.

Case 2 : Consider $\mu(m_1) = m_1$ and $\mu(m_2) = w_2$. Then $(\{m_1\}, w_1)$ is a blocking pair.

Case 3 : Consider $\mu(m_1) = w_1$ and $\mu(m_2) = w_2$. Then $(\{m_1, m_2\}, w_1)$ is a blocking pair.

Case 4 : Consider $\mu(m_1) = \mu(m_2) = w_1$. Then $(\{m_1\}, w_2)$ is a blocking pair.

Case 5 : Consider $\mu(m_1) = w_2$ and $\mu(m_2) = w_1$. Then $(\emptyset, w_1)$ is a blocking pair.

Case 6 : Consider $\mu(m_1) = w_2$ and $\mu(m_2) = m_2$. Then $(\{m_2\}, w_2)$ is a blocking pair.

The outside option for the w agents is remaining unmatched. The definition of stability builds in a form of individual rationality for w . If w prefers the empty set to her currently-matched set, then we take $(M' = \emptyset, w)$ as the blocking pair.

Case 7 : $\mu(m_1) = \mu(m_2) = w_2$ is blocked by $(\emptyset, w_2)$.

Case 8 : Consider $\mu(m_1) = m_1$ and $\mu(m_2) = w_1$. Then $(\{m_1\}, w_2)$ is a blocking pair.

Case 9 : Consider $\mu(m_1) = w_1$ and $\mu(m_2) = m_2$. Then $(\{m_2\}, w_2)$ is a blocking pair. 🐼

So it's possible for there to be no stable matching. When exactly is a matching stable?

We want to turn the preference of a school/hospital $\succ_w$ over subsets of M to be defined on a preference over elements of M itself.

Definition 8.3.4 (Responsive).

A preference $\succ_w$ is responsive if there is a strict ranking $\succ_w^*$ on $M \cup \{w\}$ such that for any S with $|S| < c_w$,

- (i) For any $m, m' \notin S$, we have that

$$S \cup \{m\} \succ_w S \cup \{m'\} \iff m \succ_w^* m',$$

- (ii) and for every $m \notin S$,

$$S \cup \{m\} \succ_w S \iff m \succ_w^* w.$$

Responsiveness is much stronger than a not-minimal sufficient condition for stability, which is *rejection monotonicity*. *Observable monotonicity* is the minimal sufficient condition. There are good reasons why we want to allow for complementarity.

Note that $|S| < c_w$ strictly because the responsiveness conditions are about *adding* an agent to w 's matchings, S .

With these preferences, a stable matching always exists. An easy way to do so is to adapt the M -proposing DA algorithm by having, in each step, w reject all m except for those in w 's favorite subset of M including the agents currently matched with w and all new applicants.

Definition 8.3.5 (Deferred acceptance algorithm, many-to-one).

Let $\mu_0(m) = m$ for every m . The algorithm proceeds as follows:

- (i) All $m \in M$ apply to their favorite acceptable w . If for every m , no w is acceptable, return μ_0 . Otherwise, each $w \in W$ considers the set of applicants A_w^1 and rejects all $m \in A_m^1$ except those in

$$\arg \max_{\succ_w} \{M' \subseteq A_w^1 : |M'| \leq c_w\}.$$

Let μ_1 be the matching in which m is matched to w if he applied to w and was not rejected, and where m is otherwise unmatched.

- (ii) For steps $k \geq 2$, *each*^a m applies to his favorite w who has not yet rejected him. If there are not applications, return μ_{k-1} .

Each w gets applicants A_w^k and rejects all m not in

$$\arg \max_{\succ_w} \{M' \subseteq A_w^k \cup \mu_{k-1}^{-1}(w) : |M'| \leq c_w\}.$$

Let μ_k be the matching where each m is matched to his favorite w to whom he applied and was not rejected. If there is no such w , then m is unmatched. If there are no rejections, then return μ_k .

^aContrasts with the one-to-one DA- M algorithm.

Proposition 8.3.6 (Stability of DA- M under responsiveness.).

The many-to-one two-sided DA- M algorithm is stable if each w 's preferences are responsive.

The following proof was left as an exercise.

Proof of Proposition 8.3.6. Suppose $\succ_w$ is responsive for every $w \in W$, and let μ be the DA- M outcome. Suppose for contradiction that μ is not stable. Then at least one of the following is true:

- (i) There is some m who is matched to an unacceptable w , or
- (ii) There is $M' \subseteq M$ and some w^* such that

$$M' \succ_{w^*} \{m : \mu(m) = w^*\}$$

and each $m \in M'$ weakly prefers w^* to his match in μ .

(i) cannot be the case, because the DA- M algorithm only matches $m \in M$ to schools that he applies to, which are schools that he finds acceptable. So fix the specific M' and w^* from (ii). Without loss, we can take M' to only contains partners acceptable to w^* , since removing unacceptable partners would produce a set of matches for w^* that she strictly prefers to M' and hence $\mu(w^*)$ as well.

We claim that by responsiveness of $\succ_{w^*}$, there exists $m^* \in M' \setminus \mu(w^*)$ such that either

Case 1 : $|\mu(w^*)| < c_{w^*}$, or

Case 2 : there is some $m \in \mu(w^*)$ such that $m^* \succ_{w^*} m$.

First, we claim that $M' \setminus \mu(w^*)$ is nonempty. If $M' \subseteq \mu(w^*)$, then either $M' = \mu(w^*)$, contradicting $M' \succ_{w^*} \mu(w^*)$, or M' omits some acceptable member(s) of $\mu(w^*)$. Adding those omitted acceptable members one at a time strictly improves the set by responsiveness, so $\mu(w^*) \succ_{w^*} M'$, again a contradiction.

Now suppose that we are not in Case 1, so $|\mu(w^*)| = c_{w^*}$. If no $m^* \in M' \setminus \mu(w^*)$ is ranked above any $m \in \mu(w^*)$ under $\succ_{w^*}$, then every member of $\mu(w^*)$ is ranked above every member of $M' \setminus \mu(w^*)$. Starting from M' , replace the members of $M' \setminus \mu(w^*)$ one by one with distinct members of $\mu(w^*) \setminus M'$. Each replacement strictly improves the set by responsiveness. Note that since $|\mu(w^*)| = c_{w^*}$, there are more potential replacements than to-be-replaced agents. If any members of $\mu(w^*)$ are still missing, add them one by one; each addition also strictly improves the set, since these members are acceptable (w^* is only matched with acceptable partners in the DA- M algorithm). Hence $\mu(w^*) \succ_{w^*} M'$, contradicting $M' \succ_{w^*} \mu(w^*)$. Therefore some $m^* \in M' \setminus \mu(w^*)$ satisfies $m^* \succ_{w^*} m$ for some $m \in \mu(w^*)$.

Since in each step of the algorithm, each m applies to his favorite acceptable w that has not yet rejected him, he must have applied and been rejected by w^* at one step, since $m^* \in M' \setminus \mu(w^*)$ implies that $w^* \succ_{m^*} \mu(m^*)$ by definition of M' .

We claim that for every $k \geq 1$, if $m' \in \mu_{k-1}(w^*)$ and $m' \notin \mu_k(w^*)$, then each $m \in \mu_k(w^*) \setminus \mu_{k-1}(w^*)$ is weakly preferred by w^* to m' . This is true because if $m' \succ_{w^*} m$ strictly, then by responsiveness,

$$\mu_k(w^*) \setminus \{m\} \cup \underbrace{\{m'\}}_{\notin \mu_k(w^*) \text{ so } \notin \mu_k(w^*) \setminus \{m\}} \succ_{w^*} \underbrace{\mu_k(w^*) \setminus \{m\} \cup \{m\}}_{\notin \mu_k(w^*) \setminus \{m\}} = \mu_k(w^*).$$

Since each m keeps applying to his favorite w who has not rejected him and $m' \in \mu_{k-1}(w^*)$ means that m' was accepted by w^* in step $k-1$, m' still applies to w^* in step k . Hence $\mu_k(w^*) \setminus \{m\} \cup \{m'\}$ is a subset of applicants to w^* in step k , so $\mu_k(w^*)$ cannot be the argmax over the applicant set, a contradiction.

Let ℓ be the first iteration of the algorithm in which m^* applied to w^* but $m^* \notin \mu_\ell(w^*)$. Since m^* is acceptable to w^* (we showed earlier it is without loss to take M' containing only acceptable partners), we know that $|\mu_\ell(w^*)| = c_{w^*}$. Because m^* applied to w^* in round ℓ and was not chosen, each of the c_{w^*} members in $\mu_\ell(w^*)$ is weakly preferred by w^* to m^* . Furthermore, any match added in $\mu_{\ell+1}(w^*)$ is weakly preferred by w^* to a discarded match in $\mu_\ell(w^*) \setminus \mu_{\ell+1}(w^*)$ and hence to m^* as well.

Iterating, we find that all men matched to w^* in the final matching $\mu(w^*)$ are weakly preferred by w^* to m^* , a contradiction. 

8.4 More on the Deferred Acceptance Algorithm

Why is the DA- M algorithm good? Why is it stable? Why is it strategy-proof for all agents on the proposing side?

These properties were all very nice for when the M side was more important than the W side, like in school choice.

In Problem Set 3, we showed that stable matchings are Pareto efficient. But this efficiency considers preferences on the W side. This efficiency breaks down when we stop consider the “preferences” of “agents” in W .

Example 8.4.1.

For intuition, consider a one-to-one matching from the DA algorithm.

$\succ_{m_1}$	$\succ_{m_2}$	$\succ_{m_3}$	$\cdots$	$\succ_{m_n}$	$\succ_{m_{n+1}}$	$\succ_{w_1}$	$\succ_{w_2}$	$\cdots$	$\succ_{w_n}$
w_1	w_2	w_3	$\cdots$	w_n	w_n	m_2	m_3	$\cdots$	m_1
w_2	w_3	w_4	$\cdots$	w_1	w_1	m_3	m_4	$\cdots$	m_2
w_3	w_4	w_5	$\cdots$	w_2	w_2	m_4	m_5	$\cdots$	m_3
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$	$\vdots$	$\vdots$		$\vdots$
w_{n-2}	w_{n-1}	w_n	$\cdots$	w_{n-3}	w_{n-3}	m_{n-1}	m_n	$\cdots$	m_{n-2}
w_{n-1}	w_n	w_1	$\cdots$	w_{n-2}	w_{n-2}	m_n	m_{n+1}	$\cdots$	m_{n-1}
w_n	w_1	w_2	$\cdots$	w_{n-1}	w_{n-1}	m_{n+1}	m_1	$\cdots$	m_n
						m_1	m_2	$\cdots$	m_{n+1}

In the first step, each m_k applies to w_k for $k \leq n$, and m_{n+1} applies to w_n . Then all m are accepted except for m_{n+1} . In the second round, m_{n+1} applies to w_1 , who accepts him in favor of m_1 . In the third round, m_1 applies to w_2 , who accepts him in favor of m_2 . This continues until m_{n+1} is rejected by everyone, and each m_k for $k \leq n$ is matched with his least-favorite partner.

Thus, the M -proposing DA mechanism is not Pareto efficient when we consider just the agents on the M side. 

In practice, when markets are large, the correlation structure that leads to these harmful cycles do not really show up.

Another criticism of DA is that there are two overwhelming factors that students care about:

- (i) School quality.

(ii) Distance.

Priorities are generally decreasing in distance and decreasing in test scores.

Definition 8.4.2 (Aligned).

We say that preferences $(\succ_m)_{m \in M}$ and $(\succ_w)_{w \in W}$ are aligned if there exists $U : M \times W \rightarrow \mathbb{R}$ such that for every $m \in M$ and $w \in W$, $U(m, \cdot)$ represents m 's preferences and $U(\cdot, w)$ represents w 's preferences.

An example of aligned preferences could agree with the underlying U with values

	w_1	w_2	w_3
m_1	7	6	8
m_2	1	2	0
m_3	4	5	7

Since 8 is the highest value, we know that any stable matching must match (m_1, w_3) ; otherwise, the pair would block. Then we consider the lower left submatrix, where (m_3, w_2) match, leaving (m_2, w_1) to be matched.

Welfare maximization implies the diagonal. The leximin ordering also implies the diagonal.

The stable matching is done via a greedy algorithm, which can be quite bad for the people left at the end, as they face a small choice set. It does not agree with welfare maximization or the leximin ordering.

Suppose each m has utility

$$-d(m, w) + q(w)$$

for each w , and that each w has similar priority

$$-d(m, w) + h(m)$$

for each m . Then the preferences are aligned with the function

$$U(m, w) = -d(m, w) + q(w) + h(m).$$

Thus, the preferences in school choice are at least approximately aligned, so the DA- M algorithm may be “too greedy.”

9.1	Deterministic Allocation	67
9.2	Random Allocation	75

In this section, we will consider the problem of allocating a finite number of indivisible objects to a set of agents who have preferences over the objects. Example applications include the distribution of public housing units and the assignment of tasks to workers in a firm.

9.1 Deterministic Allocation

We first focus on deterministic mechanisms.

9.1.1 Svensson (1999)

We have N agents and a set X of houses.¹ We assume for simplicity that $|N| = |X|$.

Suppose each i has a strict preference $\succ_i$ on X .

Definition 9.1.1 (Allocation, mechanism).

An allocation is a bijection $\mu : N \rightarrow X$. We denote the set of allocations as A .
A mechanism is a function $f : P^N \rightarrow A$.

Agents are indifferent between any two allocations that give her the same object, while the planner is to decide on the assignments for everyone.

Definition 9.1.2 ((Group) strategy-proof).

A mechanism f is:

- Strategy-proof if for every i , $\succ \in P^N$, and $\succ'_i$,

$$f(\succ)(i) \succsim_i f(\succ'_i, \succ_{-i})(i).$$

- Group strategy-proof if for all $\succ \in P^N$, there is no nonempty coalition $S \subseteq N$ and joint misreport $\succ'_S = (\succ'_i)_{i \in S}$ such that

$$f(\succ'_S, \succ_{-S})(j) \succsim_j f(\succ)(j) \quad \text{for all } j \in S$$

and

$$f(\succ'_S, \succ_{-S})(i) \succ_i f(\succ)(i) \quad \text{for some } i \in S.$$

¹More generally, objects

- Monotonic if for any $\succ, \succ'$ such that for all i and all $x \in X$, if

$$f(\succ)(i) \succ_i x \implies f(\succ')(i) \succ'_i x,$$

then $f(\succ') = f(\succ)$. If the set of objects dispreferred to $f(\succ)(i)$ weakly grows from $\succ$ to $\succ'$ for agent i , then f is monotone if i maintains the same object.

- Non-bossy if for all $i, \succ_{-i}, \succ_i$, and $\succ'_i$,

$$f(\succ_i, \succ_{-i})(i) = f(\succ'_i, \succ_{-i})(i) \implies f(\succ_i, \succ_{-i}) = f(\succ'_i, \succ_{-i}).$$

For an agent i and preferences for the remaining agents, if i is able to change her preference without affecting her own allocation, then her changing her preference can't change others' allocations.

In house allocation, strategy proofness is actually a very weak condition and doesn't buy us much. Group strategy proofness is a much nicer restriction to work with.

Lemma 9.1.3.

For any mechanism f , the following are equivalent:

- (i) f is group strategy-proof.
- (ii) f is strategy-proof and non-bossy.
- (iii) f is monotonic.

Proof of Lemma 9.1.3. We showed (i) $\iff$ (ii) in Problem Set 4. Now we show (ii) $\iff$ (iii).

($\implies$) Suppose f is strategy-proof and non-bossy. Consider $\succ$ and $\succ'$ such that for all i and $x \in X$,

$$f(\succ)(i) \succ_i x \implies f(\succ')(i) \succ'_i x.$$

We will link $f(\succ)$ to $f(\succ')$ by changing one agent's preference at a time.

Consider agent 1 changing her report from $\succ_1$ to $\succ'_1$ and everyone else maintaining their preferences. By strategy-proofness,

$$f(\succ)(1) \succsim_1 f(\succ'_1, \succ_{-1})(1).$$

If equality did not hold, then we would have $f(\succ)(1) \succ_1 f(\succ'_1, \succ_{-1})(1)$, so by the assumption,

$$f(\succ)(1) \succ'_1 f(\succ'_1, \succ_{-1})(1),$$

and strategy-proofness of f at $(\succ'_1, \succ_{-1})$ would be violated with the profitable deviation of agent 1 to $\succ_1$.

So $f(\succ)(1) = f(\succ'_1, \succ_{-1})(1)$. By non-bossiness, $f(\succ) = f(\succ'_1, \succ_{-1})$.

Now we induct. Suppose $f(\succ) = f(\succ'_1, \dots, \succ'_k, \succ_{k+1}, \dots, \succ_{|N|})$. Strategy-proofness of f at $(\succ'_{\{1, \dots, k\}}, \succ_{\{k+1, \dots, |N|\}})$ gives

$$f(\succ'_{\{1, \dots, k\}}, \succ_{\{k+1, \dots, |N|\}})(k) \succsim_k f(\succ'_{\{1, \dots, k+1\}}, \succ_{\{k+2, \dots, |N|\}}).$$

If equality does not hold, then strategy-proofness of f at $(\succ'_{\{1, \dots, k+1\}}, \succ_{\{k+2, \dots, |N|\}})$ is violated, so k 's object does not change between the two reports. Non-bossiness forces

This proof was left as an exercise.

everyone else to maintain their objects as well, so

$$f(\succ) = f(\succ_{\{1, \dots, k+1\}'}, \succ_{\{k+2, \dots, |N|\}}).$$

Eventually, we reach that $f(\succ) = f(\succ')$. Since $\succ, \succ'$ were arbitrary, f is monotonic.

($\Leftarrow$) Suppose f is monotonic. Fix agent i , $\succ$, and a misreport $\succ'_i$. Let $a := f(\succ)(i)$ and $b := f(\succ'_i, \succ_{-i})(i)$. Suppose for contradiction that $b \succ_i a$. Consider the preference $\succ_i^*$ where i moves b to the top of $\succ_i$ and leaves everything else unchanged. By monotonicity, since the lower contour set of a does not shrink from $\succ_i$ to $\succ_i^*$, and all other agents' preferences are unchanged,

$$f(\succ) = f(\succ_i^*, \succ_{-i}),$$

and in particular

$$a = f(\succ)(i) = f(\succ_i^*, \succ_{-i})(i).$$

Again by monotonicity since b is top-ranked in $\succ_i^*$, the lower contour set of b does not shrink from $\succ'_i$ and $\succ_i^*$ (in fact, it may grow), and all other agents' preferences are unchanged so

$$f(\succ'_i, \succ_{-i}) = f(\succ_i^*, \succ_{-i}).$$

In particular,

$$b = f(\succ'_i, \succ_{-i})(i) = f(\succ_i^*, \succ_{-i})(i) = a.$$

But this contradicts $b \succ_i a$, so $a \succsim_i b$, and f is indeed strategy-proof.

Showing non-bossiness is similar. Suppose $\succ_i, \succ'_i$ are such that

$$a := f(\succ)(i) = f(\succ'_i, \succ_{-i})(i).$$

Move a to the top of $\succ_i$ to create $\succ_i^*$ and move a to the top of $\succ'_i$ to create $\succ''_i$. By monotonicity, since the other agents' preferences do not change,

$$f(\succ) = f(\succ_i^*, \succ_{-i}), \quad f(\succ'_i, \succ_{-i}) = f(\succ''_i, \succ_{-i}),$$

and in all allocations, i gets a . By monotonicity again, since $\succ_i^*$ and $\succ''_i$ have the same lower contour set of a , and all other agents' preferences do not change,

$$f(\succ_i^*, \succ_{-i}) = f(\succ''_i, \succ_{-i}).$$

Combining this display with the previous one yields that

$$f(\succ) = f(\succ'_i, \succ_{-i}),$$

so f is non-bossy. 

Lemma 9.1.4.

If a mechanism is group strategy-proof, then it is Pareto efficient iff it is surjective.

Proof of Lemma 9.1.4. ($\Rightarrow$) Suppose f is group strategy-proof and Pareto efficient. For an outcome $x \in X$, let $\succ^*$ be a preference over X where x is top-ranked. Then by efficiency, $f(\succ^*, \dots, \succ^*) = x$. Since x was arbitrary, f is surjective.

($\Leftarrow$) Suppose f is group strategy-proof and surjective. Assume for contradiction that f is not Pareto efficient, so there exists some $\mu \in A$ such that

$$\mu(i) \succsim_i f(\succ)(i), \quad \mu(j) \succ_j f(\succ')(j).$$

This proof was left as an exercise.

By surjectivity, there exists $\succ'$ such that $\mu = f(\succ')$. Then the grand coalition $S = |N|$ has a profitable group deviation, so f is not group strategy-proof, a contradiction. 

Definition 9.1.5 (Serial dictatorship mechanism).

The serial dictatorship mechanism with order $\rho : \{1, 2, \dots, |N|\} \rightarrow N$ (ρ is a bijection) is the mechanism that gives $\rho(1)$ her top choice, $\rho(2)$ her favorite choice of the remaining objects, and so on.

We call ρ the *picking order* of the mechanism.

Note that since ρ is independent of $\succ$, a single order ρ is used to prioritize the agents regardless of preference.

Definition 9.1.6 (Neutrality).

Let σ be a permutation on X . If $\mu \in A$, define $\sigma\mu$ to be the allocation where each agent i gets $\sigma(\mu(i))$, i.e.

$$(\sigma\mu)(i) := \sigma(\mu(i)).$$

If $\succ_i \in P$, let $\sigma \succ_i$ be the preference such that

$$x \sigma \succ_i y \iff \sigma^{-1}(x) \succ_i \sigma^{-1}(y).$$

A mechanism f is neutral if for every $\succ \in P^{|N|}$ and all permutations σ ,

$$f(\sigma \succ) = \sigma f(\succ).$$

What kinds of mechanisms satisfy neutrality?

Example 9.1.7 (Neutrality).

Consider $N = \{1, 2, 3\}$, $X = \{a, b, c\}$, and $\sigma(a) = b, \sigma(b) = a, \sigma(c) = c$. With the preferences

$$\begin{array}{ccc} \succ_1 & \succ_2 & \succ_3 \\ \hline a & a & b \\ b & c & a \\ c & b & c \end{array} ,$$

we have that

$$\begin{array}{ccc} \sigma \succ_1 & \sigma \succ_2 & \sigma \succ_3 \\ \hline b & b & a \\ a & c & b \\ c & a & c \end{array} .$$

In class, some specific f was given, but I didn't write it down. But an f that implements the Borda count satisfies neutrality, as

$$f(\sigma \succ) = b = \sigma(a) = \sigma f(\succ).$$



Theorem 9.1.8 (Svensson).

A mechanism f is group strategy-proof and neutral iff it is a serial dictatorship.

Proof of Theorem 9.1.8. ($\Leftarrow$). Suppose $f = SD_\rho$. Consider a possible coalition manipulation, and let $\rho(k)$ be the first member of the coalition in the picking order. The agents

It turns out that group strategy-proofness is equivalent to group strategy-proofness limited to coalitions of size 1 and 2. Still, we may think group strategy-proofness is too strong, because the jump from strategy-proofness to group strategy-proofness requires nonbossiness. The second price auction is bossy.

before $\rho(k)$ in the order are not in the coalition, so their reports and assignments do not change. Hence when $\rho(k)$ picks, the same set of objects is available as before. Truthful reporting already gives $\rho(k)$ her favorite object from that set, so she cannot get a strictly better object. If she gets a different object, she is worse off; if she gets the same object, the same argument applies to the next coalition member in the picking order. Therefore no coalition can make all its members weakly better off and one member strictly better off, so serial dictatorships are group strategy-proof.

Serial dictatorships are also neutral. At each step k , the set of remaining objects under $\sigma \succ$ is the image under σ of the set of remaining objects under $\succ$. Therefore the favorite remaining object of agent $\rho(k)$ under $\sigma \succ_{\rho(k)}$ is exactly σ applied to her favorite remaining object under $\succ_{\rho(k)}$. Hence $f(\sigma \succ) = \sigma f(\succ)$.

($\Rightarrow$). Fix a single agent's preference $\succ_* \in P$. Consider the preference profile $(\succ_*, \dots, \succ_*) \in P^{|N|}$. Fix a mechanism f that is group strategy-proof and neutral, so it is also strategy-proof and nonbossy by Lemma (9.1.3).

Without loss, suppose $\succ_*$ ranks x_1 first, x_2 second, and so on. Since $f(\succ_*, \dots, \succ_*)$ is an allocation, each object is assigned to exactly one agent. Define the order ρ by letting $\rho(k)$ be the agent who gets x_k at $(\succ_*, \dots, \succ_*)$.

By neutrality, in any profile where all agents agree, agent $\rho(1)$ gets the common top choice, agent $\rho(2)$ gets the common second choice, and so on. To see this, if the common ranking is $y_1, \dots, y_{|X|}$, take the permutation σ with $\sigma(x_k) = y_k$ for each k and apply neutrality to $(\succ_*, \dots, \succ_*)$. Hence f agrees with the serial dictatorship mechanism SD_ρ on unanimous preference profiles.

Now let $\succ$ be an arbitrary preference profile. Let $\mu = SD_\rho(\succ)$. Let $\succ^\mu$ be the preference where $\mu(\rho(1))$ is top-ranked, $\mu(\rho(2))$ is second-ranked, and so on, and let $\succ' = (\succ^\mu, \dots, \succ^\mu)$ be the corresponding unanimous profile. For each k , by the definition of SD_ρ , $\mu(\rho(k))$ is agent $\rho(k)$'s favorite object among the objects not assigned to agents $\rho(1), \dots, \rho(k-1)$. In particular, agent $\rho(1)$ top-ranks $\mu(\rho(1))$ under $\succ_{\rho(1)}$; more generally, if agent $\rho(k)$ ranks an object above $\mu(\rho(k))$ under $\succ_{\rho(k)}$, that object must be one of $\mu(\rho(1)), \dots, \mu(\rho(k-1))$.

Then $f(\succ')$ agrees with the serial dictatorship with picking order ρ , so agent $\rho(1)$ gets the top-rank in $\succ'$, agent $\rho(2)$ gets the second rank in $\succ'$, and so on. Thus $f(\succ') = \mu$.

We want to show that $f(\succ) = \mu$ and will do so by replacing the reports in $\succ'$ with the reports in $\succ$ one at a time. For each $k = 0, 1, \dots, |N|$, define

$$\succ_{\rho(k)}^k = \begin{cases} \succ_{\rho(k)} & \text{if } \ell \leq k, \\ \succ^\mu & \text{if } \ell > k \end{cases},$$

so $\succ^0 = \succ'$ and $\succ^{|N|} = \succ$. We claim by induction on k that $f(\succ^k) = \mu$. The base case $k = 0$ follows from the previous paragraph.

Now suppose $f(\succ^{k-1}) = \mu$. The only difference between $\succ^{k-1}$ and $\succ^k$ is that agent $\rho(k)$ reports $\succ_{\rho(k)}$ instead of $\succ^\mu$. First apply strategy-proofness at the profile $\succ^k$: if agent $\rho(k)$ instead reported $\succ^\mu$, the outcome would be $f(\succ^{k-1}) = \mu$, so

$$f(\succ^k)(\rho(k)) \succeq_{\rho(k)} \mu(\rho(k)).$$

By the definition of SD_ρ , the only objects that agent $\rho(k)$ ranks above $\mu(\rho(k))$ under $\succ_{\rho(k)}$ are among $\mu(\rho(1)), \dots, \mu(\rho(k-1))$.

Second, apply strategy-proofness at the profile $\succ^{k-1}$: if agent $\rho(k)$ instead reported $\succ_{\rho(k)}$, the outcome would be $f(\succ^k)$, so $\mu(\rho(k))$ is weakly preferred to $f(\succ^k)(\rho(k))$ under $\succ^\mu$. Since $\succ^\mu$ ranks exactly $\mu(\rho(1)), \dots, \mu(\rho(k-1))$ above $\mu(\rho(k))$, this rules out the possibility that $f(\succ^k)(\rho(k))$ is one of those earlier objects. Therefore $f(\succ^k)(\rho(k)) = \mu(\rho(k))$.

Since agent $\rho(k)$'s allocation is unchanged when her report changes from $\succ^\mu$ to $\succ_{\rho(k)}$, non-bossiness implies $f(\succ^k) = f(\succ^{k-1}) = \mu$. By induction, $f(\succ) = \mu = SD_\rho(\succ)$. Since $\succ$ was arbitrary, f is a serial dictatorship with picking order ρ . 🐘

9.1.2 Top Trading Cycles

Suppose we start with an initial allocation μ_0 . How can we determine the best reallocation?

We describe algorithm to compute the *top trading cycle*, which is the mechanism that agrees with the result of this algorithm. Given a preference profile $\succ \in P^{|N|}$, it proceeds as follows:

Step 1 : All agents point at the agent who is assigned her favorite house in μ_0 . Formally, we have a graph of agents as nodes and directed edges with one tail at each agent of an edge with head at the agent who is assigned to her favorite house.

A cycle $i_1 \rightarrow i_2 \rightarrow \dots \rightarrow i_n \rightarrow i_1$ must exist, since in any finite directed graph where every node has exactly one outgoing edge, following edges from any node eventually repeats a node, and hence produces a directed cycle. Trade endowments along all cycles

Note that the cycle may be a degenerate cycle.

$$\begin{aligned} i_1 &\rightarrow \mu_0(i_2), \\ i_2 &\rightarrow \mu_0(i_3), \\ &\vdots \\ i_n &\rightarrow \mu_0(i_1). \end{aligned}$$

Then remove all agents in these cycles and the houses they receive from the problem.

Step k : Repeat with the remaining agents and houses who were not previously in a cycle. Stop when there are no agents left.

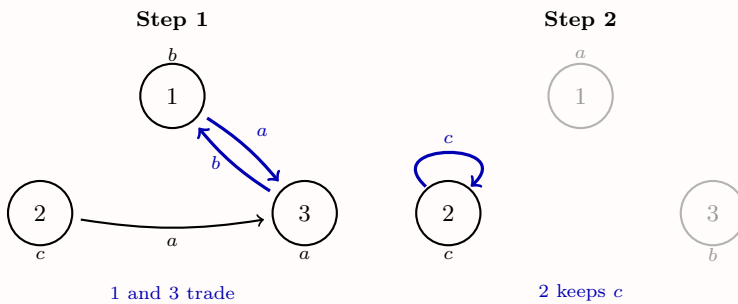
Example 9.1.9.

Again consider

$$\begin{array}{ccc} \succ_1 & \succ_2 & \succ_3 \\ \textcircled{a} & a & \textcircled{b} \\ b & \textcircled{c} & a \\ c & b & c \end{array},$$

and suppose we start with

$$\mu_0(1) = b, \quad \mu_0(2) = c, \quad \mu_0(3) = a.$$



In step 1, agents 1 and 2 point at 3, who points at 1, so agents 1 and 3 trade. In step

2, agent 2 trades with himself, and then we are done.
So in the last allocation,

$$\mu_2(1) = a, \quad \mu_2(2) = c, \quad \mu_2(3) = b.$$

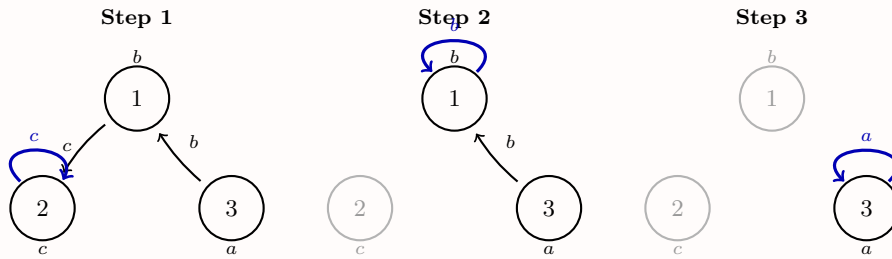
Since TTC is group strategy-proof by Proposition (9.1.10), Theorem (9.1.8) implies that TTC cannot be neutral unless it is a serial dictatorship (it is not). The following permutation shows the failure of neutrality directly. Consider

$$\sigma(c) = a, \quad \sigma(b) = b, \quad \sigma(a) = c.$$

Then the new preference profile is

$$\begin{array}{ccc} \succ_1 & \succ_2 & \succ_3 \\ \hline c & \textcircled{c} & b \\ \textcircled{b} & a & c \\ a & b & \textcircled{a} \end{array}.$$

The endowments are not permuted.



Agent 1 points to 2, 2 points to himself, and 3 points to 1. So 2 gets her top choice, which is already a violation. In step 2, 1 points to herself, while 3 points to 1. Then in step 3, 3 points to herself. Hence

$$\mu_3(1) = b \neq \sigma(a), \quad \mu_3(2) = c = \sigma(a), \quad \mu_3(3) = a = \sigma(c) \neq \sigma(b).$$

While agent 2 ends with the permuted version of her original outcome, agents 1 and 3 fail to have the permuted versions of their original outcomes. 

Proposition 9.1.10.

The top trading cycles mechanism is group strategy-proof and Pareto efficient.

Proof of Proposition 9.1.10. We will first show that TTC is monotonic, since this implies group strategy-proofness (and non-bossiness). Then we will show surjectivity, which in conjunction with group strategy-proofness, is equivalent to Pareto efficiency.

Monotonicity: Suppose $\text{TTC}_{\mu_0}(\succ) = \mu$ and $\succ'$ is such that for all i and x ,

$$\mu(i) \succ_i x \implies \mu(i) \succ'_i x.$$

We want to show that $\text{TTC}_{\mu_0}(\succ') = \mu$. Let $r(i)$ denote the round in which agent i is removed in $\text{TTC}_{\mu_0}(\succ)$. We prove by induction on the rounds of $\text{TTC}_{\mu_0}(\succ')$ that every cycle removed under $\succ'$ is a cycle that was removed under $\succ$, and that every removed

agent receives the same object as under μ .

Base case: Before the first round of $\text{TTC}_{\mu_0}(\succ')$, no agents have been removed, so the induction hypothesis is vacuous.

Inductive step: Suppose that every cycle already removed in TTC under $\succ'$ is a cycle from TTC under $\succ$, and that every already removed agent received the same object as under μ . Consider a cycle $j_1 \rightarrow j_2 \rightarrow \dots \rightarrow j_m \rightarrow j_1$ be a cycle that is about to be removed in TTC under $\succ'$ where j_1 is such that $r(j_1)$ is minimal among the agents in this cycle.

We first show a lemma. In the original run, then any object preferred by i to $\mu(i)$ must be owned by an agent in an earlier original round:

$$x \succ_i \mu(i) \implies r(\mu_0^{-1}(x)) < r(i).$$

By the contrapositive of the monotonicity assumption,

$$x \succ'_i \mu(i) \implies x \succ_i \mu(i) \implies r(\mu_0^{-1}(x)) < r(i).$$

So any object x that an agent i prefers to $\mu(i)$ in $\succ'_i$ is owned by an agent who was removed earlier than i in TTC under $\succ$. (Note that this lemma is consistent with agents who are removed in the first round of TTC under $\succ$ receiving their favorite objects.)

Since $r(j_1)$ is minimal among the cycle, we know that $\{\mu(j_1), \dots, \mu(j_m)\}$ are all still available at this point in TTC under $\succ'$. In this step, if j_1 points to some object x other than $\mu(j_1)$, then

$$x \succ'_{j_1} \mu(j_1).$$

But then by the lemma, x is the initial object of an agent in an earlier original round than j_1 . Since j_1 points to this agent at this time in TTC under $\succ'$, j_1 cannot have the minimal original round of this cycle in TTC under $\succ'$, a contradiction. So j_1 must point to an object not preferred to $\mu(j_1)$. Since $\mu(j_1)$ is still available, she points to $\mu(j_1)$, which is owned by the next agent j_2 in the current cycle under consideration, i.e. $\mu_0(j_2) = \mu(j_1)$. Thus j_2 is the successor of j_1 in the original TTC cycle containing j_1 .

j_1 and j_2 are in the same original TTC cycle, so $r(j_1) = r(j_2)$, and j_2 also has minimal round in the current cycle. We get that j_2 points to $\mu(j_2) = \mu_0(j_3)$ in the TTC cycle under $\succ'$, so j_3 is the successor of j_2 in the original TTC cycle.

For each $\ell = 1, \dots, m$, agent j_ℓ points to $\mu(j_\ell)$ under $\succ'$, so the current cycle under $\succ'$ is an original TTC cycle under $\succ$, and every agent in it receives the same object as under μ .

Nothing in the proof considers agents in another cycle to be removed at the same time as $j_1 \rightarrow \dots \rightarrow j_m \rightarrow j_1$ in TTC under $\succ'$, so all cycles formed in round 1 of TTC under $\succ'$ were formed in TTC under $\succ$ with the same pairings, and so on for future rounds.

Surjectivity: Let $\mu \in A$. Then with $\mu_0 := \mu$ and preferences $\succ$ such that each agent i top-ranks $\mu(i)$, in the first round of TTC, everyone points at themselves, and so $f(\succ) = \mu_0 = \mu$. 

9.2 Random Allocation

For any $\mu \in A$, we can associate a permutation matrix $M(\mu)$ be the $|N| \times |N|$ matrix defined by

$$M(\mu)_{ij} = \begin{cases} 1, & \mu(i) = j \\ 0, & \text{otherwise} \end{cases}.$$

Going forward, we will identify μ and $M(\mu)$ by μ .

Definition 9.2.1 (Bistochastic matrix).

A square matrix M is bistochastic if the rows and columns sum to 1 and all entries are positive, i.e.

$$\sum_j M_{ij} = 1, \forall i, \quad \sum_i M_{ij} = 1, \forall j.$$

Let $\mathcal{B}_{|N|}$ be the set of bistochastic $|N| \times |N|$ matrices.

We think of row i of a bistochastic matrix M as corresponding to a lottery over objects for agent i , while column x corresponds to a lottery over agents for object x .²

Note that $M_1, \dots, M_p$ are permutation matrices and $\alpha_1, \dots, \alpha_p$ are nonnegative numbers that sum to 1, then

$$\alpha_1 M_1 + \dots + \alpha_p M_p$$

is a bistochastic matrix.

Theorem 9.2.2 (Birkhoff Von–Neumann).

Any bistochastic matrix can be written as a convex combination of permutation matrices.

Definition 9.2.3 (Random allocation mechanism).

A random allocation mechanism is a function $f : P^{|N|} \rightarrow \mathcal{B}_{|N|}$. $f(\succ)_{ix}$ is agent i 's probability of being assigned house x when preferences are $\succ$ under f . We write $f(\succ)_i$ for the i th row of the matrix $f(\succ)$, which is agent i 's lottery over houses.

By the Birkhoff Von–Neumann theorem, there exists a lottery over A (the set of all allocations) which yields $f(\succ)$ for every $\succ$. We call this lottery a *social implementation* of $f(\succ)$.

Basically, f is outputting a matrix of probabilities.

Definition 9.2.4 (First-order stochastic dominance).

Let $p, p' \in \Delta(X)$ and let $\succ_i$ be a strict ordering on $X = \{x_1, x_2, \dots, x_{|N|}\}$. We say that p first-order stochastically dominates p' with respect to $\succ_i$ if

- $p(x_1) \geq p'(x_1)$,
- $p(x_1) + p(x_2) \geq p'(x_1) + p'(x_2)$,

Note that FOSD is *not* a binary relation. When neither lottery dominates the other, we say that the lotteries are not *comparable*.

²We don't really consider the second perspective too much.

- and so on, such that for every $x \in X$,

$$\sum_{y: y \succsim_i x} p(y) \geq \sum_{y: y \succsim_i x} p'(y).$$

We denote first order stochastic dominance of p over p' with respect to $\succsim_i$ by

$$p \text{ FOSD}_{\succsim_i} p'.$$

It turns out that $p \text{ FOSD}_{\succsim_i} p'$ iff

$$\mathbb{E}[u(p)] := \sum_{x \in X} p(x)u(x) \geq \sum_{x \in X} p'(x)u(x) =: \mathbb{E}[u(p')]$$

for all $u : X \rightarrow \mathbb{R}$ that agree with $\succsim_i$.

We are overloading notation of u , where we have utility over objects and lotteries over the objects.

Definition 9.2.5 (Strategy-proof, Ex-post efficient, Equal treatment of equals).

Let $f : P^{|N|} \rightarrow \mathcal{B}_{|N|}$ be a random mechanism. f is strategy-proof if for all i , $\succsim_{-i}$, $\succsim_i$, and $\succsim'_i$,

$$f(\succsim_i, \succsim_{-i})_i \text{ FOSD}_{\succsim_i} f(\succsim'_i, \succsim_{-i})_i.$$

Implicitly, if f is strategy-proof, then the lotteries $f(\succsim_i, \succsim_{-i})_i$ and $f(\succsim'_i, \succsim_{-i})_i$ must be comparable.

We say that f is ex-post efficient if for all $\succsim$, $f(\succsim)$ can be written as a convex combination^a

$$\alpha_1 \mu_1 + \cdots + \alpha_p \mu_p$$

where each μ_j is a deterministic Pareto efficient allocation.

f satisfies equal treatment of equals (ETOE) if for any i, j and $\succsim_{-ij}$, if $\succsim_i = \succsim_j$,

$$f(\succsim_i, \succsim_j, \succsim_{-ij})_i = f(\succsim_i, \succsim_j, \succsim_{-ij})_j.$$

^aWe alternatively say that $f(\succsim)$ has a social implementation $\sum_k \alpha_k \mu_k$.

Note that strategy-proofness is not just about non-FOSD of the misreport over the truthful report; the truthful report must FOSD all potential misreports.

Recall that we use $f(\succsim)_i$ to denote the i th row of $f(\succsim)$, i.e. the lottery given to i under f with reported preference profile $\succsim$.

Definition 9.2.6 (Random serial dictatorship, random top trading cycles).

Random serial dictatorship (RSD) is the mechanism

$$\text{RSD}(\succsim) = \frac{1}{|N|!} \sum_{\rho} \text{SD}_{\rho}(\succsim).$$

Random top trading cycles (RTTC) is the mechanism

$$\text{RTTC}(\succsim) = \frac{1}{|N|!} \sum_{\mu_0 \in A} \text{TTC}_{\mu_0}(\succsim).$$

Note that $|N|! = |A|$, where A is the set of *allocations*, not objects.

Both SD and TTC are strategy-proof deterministic mechanisms for the relevant deterministic assignment problems. Randomizing over a collection of such mechanisms, as in RSD or RTTC, preserves strategy-proofness because each realized deterministic mechanism makes truthful reporting weakly optimal, and taking the average preserves the FOSD inequalities.

Proposition 9.2.7.

RSD = RTTC.

Proof of Proposition 9.2.7. There is no intuition. See the course notes for a proof. 🐘

Proposition 9.2.8.

RSD and RTTC are strategy-proof, ex-post efficient, and satisfy ETOE. Furthermore, by Bastek and Ehlers (2024), these three properties uniquely characterize RSD/RTTC up until $|N| = 4$, while there exist strategy-proof, ex-post efficient, and ETOE mechanisms for $|N| \geq 5$.

Proof of Proposition 9.2.8. Of course, we just prove the first part.

The proof was left as an exercise.

Strategy-proofness: Each serial dictatorship that RSD randomizes over is strategy-proof for each agent, since she gets her favorite outcome compatible with her picking order, which her reported preference does not affect. Her reported preference doesn't affect the uniform randomizations over picking orders, so misreporting cannot lead to any gain.

We showed that each instance of TTC given a starting allocation μ_0 is group strategy-proof. Thus, it is individually strategy-proof as well. Since an individual's reported preference does not affect the uniform randomization over initial allocations, RTTC is strategy-proof as well.

Ex-post efficiency: Both RSD and RTTC are lotteries with uniform probabilities $\frac{1}{|N|!}$ and $\frac{1}{|A|}$ respectively over Pareto efficient deterministic allocations, so they are efficient.

Equal treatment of equals: Fix two agents $k \neq k'$. Consider a picking order ρ . There is a paired picking order ρ' where the picking order of everyone but k and k' is maintained but k and k' switch spots. If k and k' have the same preferences, then k gets $SD_\rho(k)$ in the first mechanism, whereas k' gets $SD_{\rho'}(k') = SD_\rho(k)$ in the second mechanism, both with probability $\frac{1}{|N|!}$. No two distinct picking orders can be paired with the same picking order, so every lottery where agent k gets an object $x \in X$ under a picking order ρ has exactly one paired picking order ρ' such that agent k' gets x . Since each picking order is chosen with equal probability, agents k and k' get any object $x \in X$ with the same total probability.

The same logic applies to the starting allocation μ_0 by switching the initial objects k and k' have. Every starting allocation μ_0 has one paired allocation μ'_0 where all other than k, k' have their starting objects preserved and k, k' swap objects. The outcome of μ_0 and μ'_0 are identical, except that k' and k swap their ending objects. Since initial allocations are determined uniformly, the marginal probabilities of each object match between k and k' .



Example 9.2.9.

Suppose $N = \{1, 2, 3, 4\}$, $X = \{a, b, c, d\}$. Let the preferences over objects be:

$\succ_1$	$\succ_2$	$\succ_3$	$\succ_4$
a	a	b	b
b	b	a	a
c	c	c	c
d	d	d	d

Then we get that:

	a	b	c	d
1	$\frac{5}{12}$	$\frac{1}{12}$	$\frac{1}{4}$	$\frac{1}{4}$
2	$\frac{5}{12}$	$\frac{1}{12}$	$\frac{1}{4}$	$\frac{1}{4}$
3	$\frac{1}{12}$	$\frac{5}{12}$	$\frac{1}{4}$	$\frac{1}{4}$
4	$\frac{1}{12}$	$\frac{5}{12}$	$\frac{1}{4}$	$\frac{1}{4}$

It's easy to fill in the columns for c and d . Note that agent 1 gets a if she is the first dictator (probability $1/4$) or if she is the second dictator and 3 or 4 is the first dictator (probability $1/2 \cdot 1/3 = 1/6$), which happens with total probability $5/12$.

Note that while $\text{RSD}(\succ)$ is strategy-proof, ex-post efficient, and has ETOE, there is a simple change that first-order stochastically dominates $\text{RSD}(\succ)$ under $\succ$ for every agent i , as we can use

	a	b	c	d
1	$\frac{1}{2}$	0	$\frac{1}{4}$	$\frac{1}{4}$
2	$\frac{1}{2}$	0	$\frac{1}{4}$	$\frac{1}{4}$
3	0	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$
4	0	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$



The previous example motivates the following.

Definition 9.2.10 (Ordinal efficiency).

We say that f is ordinally efficient if for every $\succ$, there is no $M \neq f(\succ)$ such that

$$M_i \text{ FOSD}_{\succ_i} f(\succ)_i$$

for every i .

Though they both use FOSD as a comparison tool, strategy-proofness is concerned with a single agent's potential profit from misreporting, while ordinal efficiency examines if there is another mechanism where everyone becomes better off.

Definition 9.2.11 (Probabilistic serial mechanism).

The probability serial mechanism is defined by the following:

- (i) Start at $t = 0$.
- (ii) As t increases, let agent "eat" probability shares of her favorite object at unit

speed until one or more objects is fully consumed (i.e. the total probability share eaten of some object is 1). Continue by having every agent “eat” probability shares of her favorite object remaining until $t = 1$.

In Example (9.2.9), agents 1 and 2 begin by eating a , while agents 3 and 4 begin by eating b . We can fill in the first two columns of $f(\succ)$ at this point. At time $t = 1/2$, everyone switches to c , and at time $t = 3/4$ (after we fill in the third column), everyone switches to d .

Proposition 9.2.12.

The probabilistic serial mechanism is ordinally efficient.

Proof of Proposition 9.2.12. Fix an agent i , $\succ$, and potential misreport $\succ'_i$. Let $t_k \in (0, 1]$ be the time at which agent i finished eating her k th favorite object. In the original probabilistic serial mechanism, she eats objects in order of preference in $\succ_i$, so she gets one of her favorite k objects in $\succ_i$ with probability t_k .

If $\succ'_i$ maintains the top k objects of $\succ_i$, then i spends the same t_k time eating her top k objects, again because she eats her most-preferred objects first. If $\succ'_i$ replaces one of the top k objects with a less-preferred object, i spends less than t_k time eating her k favorite objects in $\succ_i$ when she reports $\succ'_i$, so she gets one of her k favorite objects with probability less than t_k by misreporting.

This holds for all $k = 1, \dots, |X|$, so we are done. 

Unfortunately, the probabilistic serial mechanism is not strategy-proof.

Example 9.2.13.

Set $N = \{1, 2, 3\}$ and $X = \{a, b, c\}$:

$\succ_1$	$\succ_2$	$\succ_3$
a	a	b
b	c	a
c	b	c

Then, with time breaks $t = 0, \frac{1}{2}, \frac{3}{4}$,

$$\text{PS}(\succ) = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & 0 & \frac{1}{4} + \frac{1}{4} \\ 0 & \frac{1}{2} + \frac{1}{4} & \frac{1}{4} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & \frac{3}{4} & \frac{1}{4} \end{pmatrix}.$$

But, if agent 3 instead announced $a \succ'_3 b \succ'_3 c$, we would get the allocation

$$\text{PS}(\succ'_3, \succ_{-3}) = \begin{pmatrix} \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \end{pmatrix}.$$

Note that $\text{PS}(\succ)$ and $\text{RSD}(\succ)$ are not comparable under the notion of FOSD because while $\text{PS}(\succ'_3, \succ_{-3})$ gives 3 a higher total probability of her top two choices, it gives a lower probability of her top choice. 

We have shown that PS is not strategy-proof.³ Furthermore, a first-order improvement is not possible, which is a different requirement from strategy-proofness: ordinal efficiency rules out a feasible allocation that FOSD-improves the agents' allocations, while strategy-proofness compares truthful and deviating reports for a single agent and requires truthful reporting to FOSD the deviation. We also see that PS satisfies ETOE.

Theorem 9.2.14 (Bogomolnaia and Moulin (2001)).

There is no strategy-proof, ordinally efficient mechanism that satisfies equal treatment of equals.

³Under the definition above, strategy-proofness requires the truthful outcome to FOSD every deviation outcome. Thus, to show that a random mechanism is not strategy-proof, it is enough to find a deviation outcome that FOSDs the truthful outcome, or to find a truthful/deviation pair that is not comparable by FOSD.

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10.1 Kamenica and Gentzkow (2012)

Consider a prosecutor (called the *sender*) and a judge (called the *receiver*), along with a defendant who both the prosecutor and judge believe to be guilty with probability 0.3.

The prosecutor has utility 1 if the defendant is convicted and 0 otherwise. The judge gets utility 1 if she makes the correct convict/acquit decision and 0 otherwise.

The sender chooses two distributions

$$\pi(\cdot \mid \text{guilty}), \quad \pi(\cdot \mid \text{innocent})$$

over some set of signals S .

If $|S| = 1$, a single signal comes out regardless of innocence or guilt, so the judge gets no information. Then she acquits.

An opposite approach would be to commit to giving full information: $S := \{s_g, s_i\}$ with

$$\begin{aligned} \pi(s_g \mid \text{guilty}) &= 1, & \pi(s_i \mid \text{guilty}) &= 0, \\ \pi(s_g \mid \text{innocent}) &= 0, & \pi(s_i \mid \text{innocent}) &= 1. \end{aligned}$$

Then the judge convicts with ex-ante probability 0.3.

Alternatively with $S = \{s_g, s_i\}$, the sender can distort the signal when the defendant is guilty, putting

$$\begin{aligned} \pi(s_g \mid \text{guilty}) &= 1, & \pi(s_i \mid \text{guilty}) &= 0, \\ \pi(s_g \mid \text{innocent}) &= \alpha, & \pi(s_i \mid \text{innocent}) &= 1 - \alpha. \end{aligned}$$

What is the judge's posterior for the defendant being guilty when the signal received is s_g ? It is

$$\mathbb{P}[\text{guilty} \mid s_g] = \frac{\pi(s_g \mid \text{guilty})\mathbb{P}[\text{guilty}]}{\mathbb{P}[s_g]} = \frac{0.3}{0.3 + 0.7 \cdot \alpha} = \frac{3}{3 + 7\alpha}.$$

We can think of the sender's choice to be one over requesting a blood test. But we will describe the situation in a general form.

Assumption 10.1.1.

Given the utility assigned, the judge is indifferent between convicting and acquitting when the probability of guilt is 0.5. We break this tie in favor of conviction.

Since having larger α increases the chance of sending s_g , we put

$$\frac{3}{3 + 7\alpha} = \frac{1}{2},$$

while implies the optimal α for the sender is $\alpha^* = \frac{3}{7}$.

Then the judge convicts with probability

$$0.3 + 0.7 \cdot \frac{3}{7} = 0.6.$$

10.2 General Model

Consider a sender and receiver, finite set of states Ω , and suppose the sender and receiver share a common prior $\mu_0 \in \Delta(\Omega)$. Assume $\mu_0(\omega) > 0$ for every $\omega \in \Omega$. Let A be a finite set of actions that can be taken by the receiver.

Let $u_S : \Omega \times A \rightarrow \mathbb{R}$ and $u_R : \Omega \times A \rightarrow \mathbb{R}$.

Definition 10.2.1 (Signal structure).

A signal structure is a collection of distributions $\{\pi(\cdot | \omega)\}_{\omega \in \Omega}$ where there is a finite set S such that each $\pi(\cdot | \omega)$ has support in S .

Given the prior belief μ_0 and the signal structure π , the receiver forms the posterior μ_S given each signal $s \in S$ such that

$$\mu_S(\omega) = \frac{\mu_0(\omega)\pi(s | \omega)}{\underbrace{\sum_{\omega'} \mu_0(\omega')\pi(s | \omega')}_{\pi(s)}}.$$

The denominator is the probability of receiving signal s .

If the receiver has a belief μ , she chooses $\hat{a}(\mu) \in A$ to maximize her expected utility

$$\hat{a}(\mu) \in \arg \max_{a \in A} \sum_{\omega} \mu(\omega) u_R(a, \omega).$$

Assume the receiver breaks ties in favor of the sender. This induces the expected utility

$$\hat{u}_S(\mu) = \sum_{\omega} \mu(\omega) u_S(\hat{a}(\mu), \omega).$$

A signal structure $\{\pi\}$ induces a distribution τ over posteriors, i.e.

$$\tau \in \Delta(\Delta(\Omega)).$$

τ has value (the sender's expected utility)

$$\int \hat{u}_S(\mu) d\tau(\mu).$$

It would be easier to work directly with the distribution of posterior beliefs instead of the signal structure. There is a constraint, though. The sender cannot choose any distribution of posterior beliefs that she wants;¹ there must be some signal structure that induces this distribution over posteriors.

Since Ω is finite, $\Delta(\Omega)$ can be thought of a simplex, and so $\Delta(\Delta(\Omega))$ is a probability measure on this simplex.

We can think of the sender's expected utility as

$$\int \hat{u}_S(\mu) d\tau_{\pi}(\mu).$$

¹For example, the prosecutor cannot impose a distribution over posteriors where the judge believes the defendant is guilty with probability 1.

Lemma 10.2.2.

A necessary condition for a distribution over posteriors to have some signal structure that induces it is that the expectation of the posterior beliefs is the prior.

Proof of Lemma 10.2.2. We see that for each $\omega \in \Omega$,

$$\begin{aligned}\sum_{s \in S} \pi(s) \mu_S(\omega) &= \sum_{s \in S} \pi(s) \frac{\pi(s \mid \omega) \mu_0(\omega)}{\pi(s)} \\ &= \sum_{s \in S} \pi(s \mid \omega) \mu_0(\omega) = \mu_0(\omega).\end{aligned}$$

So the expectation of the posteriors something

**Definition 10.2.3 (Bayes-plausible).**

Given a prior μ_0 , $\tau \in \Delta(\Delta(\Omega))$ is Bayes-plausible if

$$\int \mu d\tau(\mu) = \mu_0.$$

The distribution of posteriors must be Bayes-plausible. In fact, this is the only limitation over the distribution of posteriors. Wow!

Proposition 10.2.4.

There is a signal structure with value V^* iff there is a Bayes-plausible distribution over posterior beliefs such that

$$\int \hat{u}_S(\mu) d\tau(\mu) = V^*.$$

We first need the following theorem, which we take as given.

Theorem 10.2.5 (Carathéodory).

Suppose $X \subseteq \mathbb{R}^d$ and y is a point in the convex hull of X . Then there are $d + 1$ points in X such that y is in the convex hull of those points.

Proof of Proposition (10.2.4). ($\implies$) was shown in Lemma (10.2.2).

($\impliedby$) Suppose $\tau \in \Delta(\Delta(\Omega))$ is Bayes-plausible and has value V^* . Consider the set T of all pairs $(\hat{u}_S(\mu), \mu)$ for μ in the support of τ . Note that $T \subseteq \mathbb{R}^{|\Omega|+1}$.

We have that (V^*, μ_0) is in the convex hull of T . By Carathéodory's Theorem,² there is

Why?

2

Carathéodory's theorem

5 languages

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In mathematics, **Carathéodory's theorem** may refer to one of a number of results of Constantin Carathéodory:

- Carathéodory's theorem (conformal mapping), about the extension of conformal mappings to the boundary
- Carathéodory's theorem (convex hull), about the convex hulls of sets in $\mathbb{R}^d$
- Carathéodory's existence theorem, about the existence of solutions to ordinary differential equations
- Carathéodory's extension theorem, about the extension of a measure
- Carathéodory–Jacobi–Liu theorem, a generalization of Darboux's theorem in symplectic topology
- Carathéodory's criterion, a necessary and sufficient condition for a measurable set
- Carathéodory kernel theorem, a geometric criterion for local uniform convergence of univalent functions
- Bourl–Carathéodory theorem, about the boundedness of a complex analytic function
- Vitali–Carathéodory theorem, a result in real analysis

Pick the right theorem!

a distribution τ^* with finite support T° such that

$$\sum_{\mu \in T^\circ} (\hat{u}_S(\mu), \mu) \tau^*(\mu) = (V^*, \mu_0).$$

We can turn this into a signal structure by setting $S = T^\circ$ and put

$$\pi(\mu \mid \omega) = \frac{\mu(\omega) \tau^*(\mu)}{\mu_0(\omega)}.$$

Now we verify that the belief about ω given $\mu \in T^\circ$ is

$$\frac{\pi(\mu \mid \omega) \mu_0(\omega)}{\sum_{\omega'} \mu_0(\omega') \pi(\mu \mid \omega')} = \frac{\mu(\omega) \tau^*(\mu)}{\sum_{\omega'} \mu_0(\omega') \frac{\mu(\omega') \tau^*(\mu)}{\mu_0(\omega')}} = \mu(\omega).$$



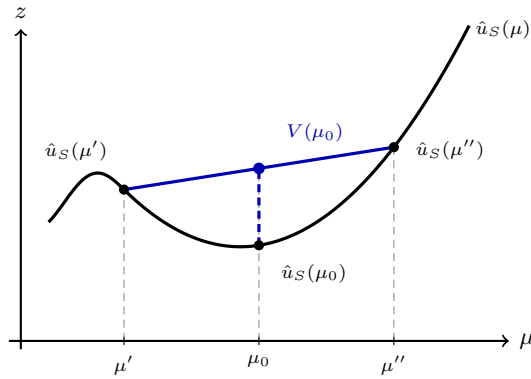
Now let

$$G := \{(\hat{u}_S(\mu), \mu)\}_{\mu \in \Delta(\Omega)},$$

which is the graph of μ under $\hat{u}_S$. Let $\text{CO}(G)$ be the convex hull of G and

$$V(\mu) := \sup \{z \mid (z, \mu) \in \text{CO}(G)\}$$

be the *concavification* of $\hat{u}$. $V(\mu_0)$ is exactly the optimal utility for the sender.



If $\hat{u}_S$ is concave, then $V = \hat{u}_S$, so there is no benefit to information design. But if it is not concave, there is at least one prior where information design is beneficial.

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A.1 Systems of Linear Inequalities

For $A \in \mathbb{R}^{m \times n}$, $x \in \mathbb{R}^n$, and $b \in \mathbb{R}^m$, we can write b linear constraints on the n components of x by writing

$$Ax \leq b$$

if $e'_i Ax \leq b_i$ for every i .

We interpret $\leq$ componentwise: for $u, v \in \mathbb{R}^m$,

$$u \leq v \iff u_i \leq v_i \quad \forall i = 1, \dots, m.$$

The feasible set is

$$x \in \mathbb{R}^n : Ax \leq b.$$

Each inequality cuts off a half-space, as the overall space is split into two parts: one fulfilling the inequality and one not. The feasible set is the intersection of all half-spaces.

Definition A.1.1 (Polyhedron, polytope).

A polyhedron is the solution set to a system of linear inequalities $Ax \leq b$, and a polytope is a bounded polyhedron.

Polyhedra are convex.

A.2 Fourier–Motzkin Elimination

Suppose a linear system involves many variables. If we want to eliminate variable x_n , we can rewrite all inequalities so that they get either upper bounds on x_n or lower bounds on x_n . Then we have

$$x_n \leq u_i(x_1, \dots, x_{n-1}), \quad x_n \geq l_j(x_1, \dots, x_{n-1}).$$

Feasibility requires that

$$l_j(x_1, \dots, x_{n-1}) \leq u_i(x_1, \dots, x_{n-1})$$

for every i, j , which is a system of linear inequalities in $x_1, \dots, x_{n-1}$.

The variable can be chosen iff

$$\max_j l_j(x_{-n}) \leq \min_i u_i(x_{-n}),$$

and so we can eliminate x_n by replacing the original inequalities with $l_j(x_{-n}) \leq u_i(x_{-n})$ for all i, j .

We can continue this process until we only get one variable left, from which it is easy to see if the resulting system of inequalities is contradictory. We then know if there is a solution to the original system.

Definition A.2.1 (Consequence relation).

An inequality

$$a_0 x \leq b_0$$

is a consequence relation of the system $Ax \leq b$ if every $x \in \mathbb{R}^n$ satisfying $Ax \leq b$ also satisfies $a_0 x \leq b_0$.

If $z \in \mathbb{R}^m$ satisfies $z \geq 0$ and $z \neq 0$, then

$$(zA)x \leq z \cdot b$$

is a consequence relation of $Ax \leq b$.

A.3 Farkas' Lemma

If there does not exist an x such that $Ax \leq b$, then can we prove infeasibility in a systematic way? Informally, Farkas' Lemma says that if a linear system has no solution, then there is a linear combination of the constraints that visibly produces a contradiction.

Lemma A.3.1 (Farkas').

Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Either

(i) There exists $x \in \mathbb{R}^n$ such that

$$Ax \leq b.$$

(ii) There exists^a $y \in \mathbb{R}^m$ with $y \geq 0$ such that

$$yA = 0, \quad y \cdot b < 0.$$

but not both.

^aHere, and in the rest of the chapter, we take $y \in \mathbb{R}^m$ to be a row vector. We call y the *multiplier vector* or *certificate of infeasibility*.

Proof of Lemma A.3.1. First, the two alternatives are mutually exclusive. If both held, then multiplying the inequalities by y gives

$$y(Ax) \leq y \cdot b,$$

but $yA = 0$, so

$$0 \leq y \cdot b < 0,$$

a contradiction.

Now suppose there is no x such that $Ax \leq b$. Repeatedly applying Fourier–Motzkin elimination eventually removes all variables and produces an impossible scalar inequality

$$0 \leq c$$

with $c < 0$. Every inequality produced along the way comes from a nonnegative linear combination of previously derived inequalities, and therefore from a nonnegative linear combination of the original inequalities. Hence this final contradiction has the form

$$(yA)x \leq y \cdot b$$

for some $y \in \mathbb{R}^m$ with $y \geq 0$, where the left-hand side is identically zero because all variables have been eliminated. Thus $yA = 0$ and $y \cdot b = c < 0$. 

Farkas is useful when checking whether transfers or prices exist that make a mechanism implementable.

There are some alternative formulations of Farkas’ Lemma.

Lemma A.3.2 (Rational Farkas’).

Let $A \in \mathbb{Q}^{m \times n}$ and $b \in \mathbb{Q}^m$. Either

(i) There exists $x \in \mathbb{Q}^n$ such that

$$Ax \leq b.$$

(ii) There exists $y \in \mathbb{Z}^m$ with $y \geq 0$ such that

$$yA = 0, \quad y \cdot b < 0.$$

but not both.

If all entries of A and b are rational, then Fourier–Motzkin elimination preserves rationality. Hence in Farkas’ Lemma we may take y to be rational, and then clear denominators to take y integer-valued.

What if our system had equalities as well as inequalities? Suppose we want to know if there is an $x \in \mathbb{R}^n$ with

$$Ax \leq b, \quad Bx = c$$

for some matrix $B \in \mathbb{R}^{p \times n}$. We have that

$$\begin{aligned} Ax \leq b, \quad Bx = c \\ \iff Ax \leq b, \quad Bx \leq c, \quad -Bx \leq -c. \end{aligned}$$

So we can rewrite the problem as a system of inequalities and apply Farkas’ Lemma. Writing the resulting multiplier as $[s \ t_1 \ t_2]$ with $s \in \mathbb{R}^m$ and $t_1, t_2 \in \mathbb{R}^p$, and then setting $t = t_1 - t_2$, gives the following variant.

Lemma A.3.3 (Farkas’ II).

Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $B \in \mathbb{R}^{p \times n}$, and $c \in \mathbb{R}^p$. Then either

(i) There exists $x \in \mathbb{R}^n$ such that

$$Ax \leq b, \quad Bx = c.$$

(ii) There exists $s \in \mathbb{R}^m$ and $t \in \mathbb{R}^p$ such that

$$s \geq 0, \quad sA + tB = 0, \quad s \cdot b + t \cdot c < 0.$$

but not both.

Another common situation is to want solutions with $x \geq 0$. A similar argument gives:

Lemma A.3.4 (Farkas' III).

Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $B \in \mathbb{R}^{p \times n}$, and $c \in \mathbb{R}^p$. Then either

(i) There exists $x \in \mathbb{R}^n$ with $x \geq 0$ such that

$$Ax \leq b, \quad Bx = c.$$

(ii) There exists $s \in \mathbb{R}^m$ and $t \in \mathbb{R}^p$ with $s \geq 0$ such that

$$sA + tB \geq 0, \quad s \cdot b + t \cdot c < 0.$$

but not both.

Lemma A.3.5 (Gordan's).

Let $A \in \mathbb{Q}^{m \times d}$. Then either

(i) There exists $y \in \mathbb{R}^d$ such that

$$Ay \gg 0,$$

i.e. every entry of Ay is strictly positive, or

(ii) There exists a nonzero vector of nonnegative integers $z \neq 0$ such that

$$zA = 0.$$

but not both.

A.4 Minimax Theorem

Consider a two-player zero-sum game. Players 1 and 2 can play actions in the sets

$$A_1 = \{1, \dots, q\}, \quad A_2 = \{1, \dots, r\}$$

respectively. The payoffs are captured by a matrix $B \in \mathbb{R}^{q \times r}$ where B_{ij} records the amount player 2 pays to player 1 if player 1 plays i and player 2 plays j . If $B_{ij} < 0$, this means player 1 pays player 2.

Each player can choose a mixed strategy

$$\alpha_1 \in \Delta A_1, \quad \alpha_2 \in \Delta A_2,$$

which gives the expected payoff $\alpha_1 B \alpha_2$ to player 1 and $-\alpha_1 B \alpha_2$ to player 2.

Given a fixed strategy α_1 for player 1, player 2 would choose α_2 to minimize $\alpha_1 B \alpha_2$.

Thus player 1's value is

$$V_1^* = \max_{\alpha_1 \in \Delta A_1} \min_{\alpha_2 \in \Delta A_2} \alpha_1 B \alpha_2.$$

Conversely, player 2's value is

$$V_2^* = \min_{\alpha_2 \in \Delta A_2} \max_{\alpha_1 \in \Delta A_1} \alpha_1 B \alpha_2.$$

If α_1^* solves the maximin problem and α_2^* solves the minimax problem, then

$$\begin{aligned} V_1^* &= \max_{\alpha_1 \in \Delta A_1} \min_{\alpha_2 \in \Delta A_2} \alpha_1 B \alpha_2 \\ &= \min_{\alpha_2 \in \Delta A_2} \alpha_1^* B \alpha_2 \\ &= \min \{ \eta(1) \alpha_1^* B e_1 + \cdots + \eta(r) \alpha_1^* B e_r : (\eta(1), \dots, \eta(r)) \in \Delta A_2 \} \\ &= \min \{ \alpha_1^* B e_j : 1 \leq j \leq r \}. \end{aligned}$$

That is, once player 1 has fixed α_1^* , player 2 can do no better with mixed strategies than with a pure strategy. Hence

$$V_1^* = \min_{1 \leq j \leq r} \alpha_1^* B e_j \leq \alpha_1^* B \alpha_2^* \leq \max_{\alpha_1 \in \Delta A_1} \alpha_1 B \alpha_2^* = V_2^*,$$

so always $V_1^* \leq V_2^*$.

Theorem A.4.1 (Minimax).

In a two-player zero-sum game,

$$\max_{\alpha_1 \in \Delta A_1} \min_{\alpha_2 \in \Delta A_2} \alpha_1 B \alpha_2 = \min_{\alpha_2 \in \Delta A_2} \max_{\alpha_1 \in \Delta A_1} \alpha_1 B \alpha_2.$$

Proof of Theorem A.4.1. Suppose $V_1^* < V_2^*$. Then the following system is infeasible:

$$\begin{aligned} xB &\geq V_2^* \mathbf{1}'_r \\ x &\geq 0 \\ x \cdot \mathbf{1}_q &= 1. \end{aligned}$$

Indeed, if it had a solution, then that strategy x would guarantee player 1 a payoff of at least V_2^* against every pure strategy of player 2, and hence against every mixed strategy of player 2 as well.

We can rewrite the inequality $xB \geq V_2^* \mathbf{1}'_r$ as

$$-B'x' \leq -V_2^* \mathbf{1}_r.$$

By Farkas' Lemma III, there must exist $s \in \mathbb{R}^r$ with $s \geq 0$ and $t \in \mathbb{R}$ such that

$$s(-B') + t \mathbf{1}_q \geq 0, \quad -V_2^* s \cdot \mathbf{1}_r + t < 0.$$

Equivalently,

$$sB' \leq t \mathbf{1}_q, \quad V_2^* s \cdot \mathbf{1}_r > t.$$

Note that $s \neq 0$, since otherwise we would have $0 \leq t \mathbf{1}_q$ and $0 > t$, a contradiction. Let

$$s^* = \frac{s}{1 \cdot s}.$$

Then

$$s^* B' \leq \frac{t}{1 \cdot s} \mathbf{1}_q, \quad \frac{t}{1 \cdot s} < V_2^*.$$

Taking transposes gives

$$B(s^*)' \leq \frac{t}{1 \cdot s} \mathbf{1}_q, \quad \frac{t}{1 \cdot s} < V_2^*.$$

But this says that the strategy s^* for player 2 guarantees a payment of at most $\frac{t}{1 \cdot s}$, which is strictly less than the optimal value V_2^* , a contradiction. 🐼

A.5 The Standard Form

It is often useful to rewrite a linear system in the form

$$Cz = d, \quad z \geq 0.$$

We can convert any system

$$Ax \leq b$$

into this form in two steps.

First, introduce slack variables $s \geq 0$ so that

$$Ax + s = b.$$

Second, rewrite each unrestricted variable as the difference of two nonnegative variables:

$$x_i = y_i - y'_i, \quad y_i, y'_i \geq 0$$

for every i . Then

$$A(y - y') + s = b$$

with $y, y', s \geq 0$. Equivalently,

$$(A \quad -A \quad I) \begin{pmatrix} y \\ y' \\ s \end{pmatrix} = b, \quad \begin{pmatrix} y \\ y' \\ s \end{pmatrix} \geq 0.$$

Proposition A.5.1.

The system $Ax \leq b$ has a solution if and only if

$$(A \quad -A \quad I) \begin{pmatrix} y \\ y' \\ s \end{pmatrix} = b$$

has a solution with $y, y', s \geq 0$.

Proof of Proposition A.5.1. If x satisfies $Ax \leq b$, let

$$s = b - Ax$$

so that $s \geq 0$. Then define

$$y_i = \max\{x_i, 0\}, \quad y'_i = \max\{-x_i, 0\}$$

for each i . Thus $x = y - y'$ and

$$A(y - y') + s = Ax + (b - Ax) = b.$$

Conversely, if $y, y', s \geq 0$ satisfy

$$A(y - y') + s = b,$$

then setting $x = y - y'$ gives

$$Ax = b - s \leq b$$

since $s \geq 0$. 

A.6 Polyhedra

Thus far our approach to systems of the type $Ax \leq b$ has been algebraic. It is also useful to take a geometric point of view. Write

$$P(A, b) = \{x \in \mathbb{R}^n : Ax \leq b\}.$$

Definition A.6.1 (Face).

A face of $P(A, b)$ is a set of the form

$$F = P(A, b) \cap \{x : a_0 x = b_0\}$$

where $a_0 x \leq b_0$ is a consequence relation of $Ax \leq b$.

Taking $a_0 = 0$ and $b_0 = 1$ shows that $\emptyset$ is always a face, while taking $a_0 = 0$ and $b_0 = 0$ shows that $P(A, b)$ itself is a face.

Definition A.6.2 (Vertex).

A face consisting of a single point is called a vertex.

If P is a polytope, let $\text{vert}(P)$ denote the set of vertices of P .

Proposition A.6.3.

Let $v \in P(A, b)$. The following are equivalent:

- (i) v is a vertex of $P(A, b)$.
- (ii) There do not exist distinct $x, y \in P(A, b)$ and $\theta \in (0, 1)$ such that

$$v = \theta x + (1 - \theta)y.$$

- (iii) There is no nonzero vector d such that both $v + d$ and $v - d$ belong to $P(A, b)$.
- (iv) There are n linearly independent rows $A_{i_1}, \dots, A_{i_n}$ of A such that

$$A_{i_1} v = b_{i_1}, \quad \dots, \quad A_{i_n} v = b_{i_n}.$$

Proof of Proposition A.6.3. (i) $\implies$ (ii): Since v is a vertex, there is a consequence relation

$a_0x \leq b_0$ such that

$$\{v\} = P(A, b) \cap \{x : a_0x = b_0\}.$$

If $v = \theta x + (1 - \theta)y$ with $x, y \in P(A, b)$ and $\theta \in (0, 1)$, then

$$b_0 = a_0v = \theta a_0x + (1 - \theta)a_0y.$$

Since $a_0x \leq b_0$ and $a_0y \leq b_0$, this is only possible if $a_0x = b_0 = a_0y$. Hence $x, y \in \{v\}$.

(ii) $\implies$ (iii): If $v + d$ and $v - d$ both lie in $P(A, b)$ for some $d \neq 0$, then

$$v = \frac{1}{2}(v + d) + \frac{1}{2}(v - d)$$

writes v as a nontrivial convex combination of two distinct feasible points.

(iii) $\implies$ (iv): Let

$$T = \{A_i : A_i v = b_i\}$$

be the set of tight rows at v . If $\text{span}(T) \neq \mathbb{R}^n$, then there is a nonzero vector d orthogonal to every row in T . For rows $A_j \notin T$, we have $A_j v < b_j$, so for sufficiently small $\varepsilon > 0$,

$$A_j(v \pm \varepsilon d) < b_j.$$

For rows in T , orthogonality gives

$$A_i(v \pm \varepsilon d) = A_i v = b_i.$$

Hence $v \pm \varepsilon d \in P(A, b)$, contradicting (iii).

(iv) $\implies$ (i): Let $I = \{i_1, \dots, i_n\}$. If $x \in P(A, b)$ satisfies

$$\sum_{i \in I} A_i x = \sum_{i \in I} b_i,$$

then since each $A_i x \leq b_i$, equality of the sums implies $A_i x = b_i$ for every $i \in I$. Because the rows $\{A_i\}_{i \in I}$ are linearly independent, this system has the unique solution $x = v$. Thus

$$\{v\} = P(A, b) \cap \left\{x : \left(\sum_{i \in I} A_i\right) x = \sum_{i \in I} b_i\right\},$$

so v is a face consisting of a single point. 

Definition A.6.4 (Facet).

A facet of $P(A, b)$ is a proper face $F \neq P(A, b)$ that is maximal among proper faces, i.e. there is no proper face $G \neq F$ with $F \subseteq G$.

Proposition A.6.5.

Every polytope is the convex hull of its vertices. Conversely, the convex hull of any finite subset of $\mathbb{R}^n$ is a polytope.

This gives two useful descriptions of a polytope P :

- (i) the *H-representation*, $P = \{x : Ax \leq b\}$,
- (ii) the *V-representation*, $P = \text{co}(\text{vert}(P))$.

The H-representation makes membership easy to check, while the V-representation is convenient for optimization because linear objectives attain their optima at vertices.

Example A.6.6 (Birkhoff–von Neumann).

A matrix $B \in \mathbb{R}^{n \times n}$ is *bistochastic* if

- (i) $\sum_{j=1}^n B_{ij} = 1$ for every row i ,
- (ii) $\sum_{i=1}^n B_{ij} = 1$ for every column j ,
- (iii) $B_{ij} \geq 0$ for all i, j .

The set of bistochastic matrices is a polytope, called the *Birkhoff polytope* and denoted B_n .

A matrix $P \in \{0, 1\}^{n \times n}$ is a *permutation matrix* if each row and each column contains exactly one 1. 

Theorem A.6.7 (Birkhoff–von Neumann).

The vertices of B_n are exactly the permutation matrices.

Proof of Theorem A.6.7. First, every permutation matrix is a vertex. Indeed, let P be a permutation matrix and suppose

$$P = \theta M + (1 - \theta)N$$

with $M, N \in B_n$ and $\theta \in (0, 1)$. If $P_{ij} = 0$, then nonnegativity implies $M_{ij} = N_{ij} = 0$. If $P_{ij} = 1$, then since $M_{ij}, N_{ij} \leq 1$, we must have $M_{ij} = N_{ij} = 1$. Hence $M = N = P$.

Conversely, let M be a vertex of B_n . Suppose some entry satisfies $0 < M_{ij} < 1$. Since row i sums to 1, there is another column $k \neq j$ with $0 < M_{ik} < 1$. Since column k also sums to 1, there is some row $\ell \neq i$ with $0 < M_{\ell k} < 1$. Continuing in this way, and using finiteness, we eventually obtain a cycle of entries of M lying strictly between 0 and 1.

Along that cycle, add δ to every other entry and subtract δ from the alternating entries. For sufficiently small $|\delta|$, all entries remain in $[0, 1]$, and because each row and column in the cycle gains and loses exactly one δ , the row and column sums remain equal to 1. Thus both modified matrices $M^{+\delta}$ and $M^{-\delta}$ belong to B_n , and

$$M = \frac{1}{2}M^{+\delta} + \frac{1}{2}M^{-\delta}$$

with $M^{+\delta} \neq M^{-\delta}$, contradicting that M is a vertex. Therefore every entry of a vertex is either 0 or 1, and the row and column sum conditions force the matrix to be a permutation matrix. 

A.7 Lattices and Tarski's Fixed Point Theorem

Definition A.7.1 (Partial order).

A binary relation $\geq$ on a set L is a partial order if it is reflexive, transitive, and antisymmetric.

Definition A.7.2 (Complete lattice).

A complete lattice $(L, \geq)$ is a set L with a partial order $\geq$ such that:

- (i) for every $A \subseteq L$, there is an element $u \in L$, denoted $\bigvee A$, such that $u \geq x$ for all $x \in A$ and whenever $v \geq x$ for all $x \in A$, we have $v \geq u$;
- (ii) for every $A \subseteq L$, there is an element $u \in L$, denoted $\bigwedge A$, such that $u \leq x$ for all $x \in A$ and whenever $v \leq x$ for all $x \in A$, we have $v \leq u$.

In particular, a complete lattice has a largest element $1 = \bigvee L$ and a smallest element $0 = \bigwedge L$.

Example A.7.3.

For any set X , the power set 2^X ordered by inclusion is a complete lattice. The join is union and the meet is intersection. 

Example A.7.4.

For any set X , the set $2^X \times 2^X$ becomes a complete lattice under

$$(A, B) \geq (A', B') \iff A \supseteq A' \text{ and } B \subseteq B'.$$

Then

$$(A, B) \vee (A', B') = (A \cup A', B \cap B'),$$

while

$$(A, B) \wedge (A', B') = (A \cap A', B \cup B').$$


Definition A.7.5 (Order-preserving).

If $(L, \geq)$ is a complete lattice and $f : L \rightarrow L$, we say f is order-preserving if

$$x \leq y \implies f(x) \leq f(y).$$

Theorem A.7.6 (Tarski).

If $(L, \geq)$ is a complete lattice and $f : L \rightarrow L$ is order-preserving, then the set of fixed points

$$F = \{x \in L : f(x) = x\}$$

is a nonempty complete lattice under $\geq$.

Proof of Theorem A.7.6. Let

$$D = \{x \in L : x \leq f(x)\}.$$

Since $0 \leq f(0)$, the set D is nonempty. Let $u = \bigvee D$. For every $x \in D$, we have $x \leq u$, hence

$$f(x) \leq f(u)$$

by monotonicity. But $x \leq f(x)$ for $x \in D$, so $x \leq f(u)$ for all $x \in D$. Therefore $u \leq f(u)$, which implies $f(u) \in D$. Since u is the least upper bound of D , we also have $f(u) \leq u$. Hence $f(u) = u$, so F is nonempty.

Now let $G \subseteq F$ and put $w = \bigvee G$. For every $x \in G$, we have $x = f(x) \leq f(w)$, so $f(w)$ is an upper bound for G . Hence

$$w \leq f(w).$$

Thus f maps the interval $[w, 1]$ into itself. Applying the first part of the argument to the lattice $[w, 1]$ shows that f has a least fixed point in $[w, 1]$, and that fixed point is exactly the least upper bound of G inside F . A symmetric argument gives greatest lower bounds in F . Therefore F is a complete lattice. 